Patronage Hierarchies, Corruption, and Tax Collection*

Raul Duarte†

October 2026

Click here for the latest version.

Why are developing country bureaucracies ineffective despite repeated modernization efforts? Patronage (hiring on partisanship rather than merit) is often thought to undermine public sector performance, yet it is difficult to measure and usually studied either among frontline agents or supervisors separately, rather than across hierarchical layers. I emphasize patronage hierarchies, instances where layers of the bureaucracy have patronage hires, and measure how they affect state performance. I study this problem using comprehensive data on around 300,000 shipments inspected by Paraguayan customs and develop a method for identifying patronage hires based on political appointment cycles. My findings suggest that patronage inspectors monitored by a patronage port administrator (patronage pairs) exhibit compounded underperformance. They detect less customs fraud and more often fail to detect fraud subsequently identified by headquarters audits, while also deviating more frequently from prescribed random assignment of inspectors to shipments. Additionally, their fraud detection decreases further when dealing with high shipment volumes at their port, as high workloads can mask lower inspection effort. Finally, patronage pairs make smaller tax adjustments than non-patronage pairs, even among comparable products with similar tax evasion risks. Shipments handled by patronage pairs generate around 11% less tax revenue, undermining state capacity.

JEL Codes: D73, H26, H83

*Previously circulated as "Patronage Hierarchies and Tax Collection." I thank my advisors Melissa Dell, Alisha Holland, Anders Jensen, Horacio Larreguy, and Vincent Pons for their guidance and support. I also thank Feyaad Allie, Pierre Bachas, Augustin Bergeron, Sima Biondi, Emily Breza, Peter Buisseret, Peter Calcagno, Daniel Carpenter, Stephen Chaudoin, Juan Dodyk, Rocio Duarte Recalde, Benjamin Enke, Ray Fisman, Frances Hagopian, Shanon Hsu, Callan Hummel, Edward Glaeser, Shao-Yu Jheng, Asim Khwaja, Nicholas Kuipers, Lukas Leucht, Steven Levitsky, Kaixiao Liu, Alexey Makarin, Mashail Malik, John Marshall, Lucy Martin, Benjamin Marx, Konstantinos Matakos, Joaquim Meira, Hector Mendoza, Andrew O'Donohue, Laura Paler, Konstantin Poensgen, Bokai Qi, Pia Raffler, Reed Rasband, Bob Rijkers, Daniel T. Roberts, Tara Slough, Julio Solis, Marco Tabellini, Jorge Tamayo, Felipe Valencia, and Jaya Wen and participants at the Zurich Conference on Public Finance in Developing Countries, the Pacific Conference for Development Economics, APSA 2025, the Northeast Workshop in Empirical Political Science, the Latin American Political Economy Network (REPAL) Annual Meeting, multiple Harvard workshops (Development Economics, Comparative Politics, Political Economy, Latin America Discussion Group, Political Economy of Development, and Political Economics and Culture) and seminars (Development Economics and Alberto Alesina Seminar on Political Economy), and the Institute for Humane Studies Political Economy Workshop and Junior Fellowship Convenings for helpful inputs, comments, and feedback provided. I thank Pierpaolo Fuster for research assistance. I thank the Institute for Humane Studies for research funding under grant no. IHS018614. Refine.ink was used to check the paper for consistency and clarity. This research was approved by the Harvard University-Area Committee on the Use of Human Subjects under Protocol IRB23-0332.

†Cowles Foundation Postdoctoral Fellow, Yale University. Email: raul.duarte@yale.edu.

1 Introduction

Developing country governments adopt new technologies, pass regulations, and restructure agencies, yet weak enforcement and service delivery persist. These persistent shortcomings raise a broader question about the political and organizational roots of state capacity. Why do state agencies in developing countries often underperform despite repeated investments in institutional reform, digital infrastructure, and legal modernization?

Patronage—recruitment based on political connections rather than merit—is frequently linked to state underperformance, yet most research on patronage focuses on a single layer within bureaucracies (Oliveros, 2021; Xu, 2018; Colonnelli et al., 2020). Such single-layer studies can identify performance issues stemming from the selection of less competent or low-effort individuals but misses how patronage across multiple bureaucratic layers facilitates collusion between supervisors and agents. Building on Tirole (1986), I introduce a conceptual framework where shared patronage ties between agents and their immediate supervisors—what I call patronage hierarchies—weaken oversight and enable collusion. This conceptual shift from isolated patronage to vertically-aligned patronage networks reveals how informal loyalties compromise organizational performance even when formal oversight structures are in place. This perspective highlights why traditional reforms, focusing solely on digitization or incomplete anti-corruption enforcement at single organizational layers, may fail by overlooking incentive alignments across bureaucratic hierarchies.

I study this problem within customs, a key revenue source in developing countries and a natural setting for examining patronage hierarchies between supervisors and inspectors. Customs authorities collect an average of 37% of developing countries' tax revenue (WCO, 2020). Many countries have introduced reforms such as evasion risk-based inspection systems, electronic documentation, periodic rotations, and randomized assignments of inspectors to shipments, yet customs authorities still frequently underperform (IMF, 2022), undermining fiscal strength and economic development. Paraguay provides an ideal case for studying this phenomenon given its entrenched patronage system (Schuster, 2021), high reliance on customs revenue (40-45% of tax revenues),

and granular administrative data. I combine detailed trade transaction records with appointment data to examine how patronage affects inspectors' ability and incentives to detect customs fraud, which constitutes misrepresenting information to customs to evade taxes or regulations.

My study draws on extensive fieldwork conducted over several years at Paraguayan customs. These include in-depth interviews with customs brokers, importers, and officials at all organizational levels—from frontline inspectors and port administrators to directors, human resource managers, and IT specialists—as well as direct observation of inspections. This fieldwork revealed specific appointment periods during which patronage hiring surged and mechanisms through which corruption manifests at customs, shaping my theoretical framework and empirical strategies.

A key challenge in studying the effects of patronage is measurement. I identify patronage hires based on appointment dates. Using granular data on thousands of inspected shipments, I examine the effectiveness of customs officials appointed during election cycles or under authoritarian regimes. I operationalize patronage through two distinct appointment cycles: election-cycle hires, referring to individuals appointed to customs shortly after elections and the start of a new administration, and dictatorship hires, referring to those appointed based on political connections during Paraguay's dictatorship (up to 1989). I argue that in contexts where public sector jobs are routinely exchanged for electoral support or allocated without merit-based criteria, appointments during these cycles plausibly reflect patronage. Empirical support for this assumption comes from observing that patronage inspectors have significantly lower educational attainment, while patronage administrators (who supervise inspectors) exhibit significantly greater political involvement based on a comprehensive political engagement index discussed further in Section 4. With this measure, around half of customs inspectors and administrators in my data are patronage hires.

Crucially, I can identify whether both the customs inspector and administrator are patronage hires. I exploit variation from periodic rotations of inspectors and administrators across customs ports, which generates different combinations of inspector–administrator dyads over time. My empirical strategy includes inspector and administrator fixed effects, which absorb all time-invariant

differences in inspection ability and monitoring style at the individual level. As my patronage proxy (appointment timing) is fixed for each official, the fixed effects absorb the individual-level patronage indicators. What remains is the coefficient on the interaction term identifying *patronage pairs*. This captures whether outcomes shift beyond the individual contributions of patronage inspectors and administrators. In additional triple-difference specifications, I separate dyads with a single patronage hire from patronage pairs, showing compounding results.

As the fixed effects absorb time-invariant unobservables at the inspector or administrator level, my results are not driven by unilateral selection of patronage inspectors or administrators into shipments. To bias my estimates, endogeneity would have to operate jointly and in a time-varying way at both levels (e.g., if patronage inspectors and administrators systematically rotated together into specific ports at specific times). A related concern, which I address explicitly in Section 5, is that the patronage pair indicator may proxy for the pair being similar on another dimension, such as age, gender, education, or hire cohort. Following Marx et al. (2021), I show that the results survive controlling for matches on each of these dimensions. To further probe the mechanisms underlying their fraud detection, I sequentially add controls that consider shipment composition. I introduce product chapter (HS 2-digit) fixed effects to ensure comparisons are made across shipments with similar goods. I also control for a comprehensive tax evasion risk index capturing shipment-level heterogeneity in incentives for fraud and enforcement. Comparing baseline estimates to these specifications shows whether patronage pairs underperform due to handling different shipments or because they detect less fraud even for similar goods with comparable evasion risk.

Empirically, I find that patronage pairs consistently underperform across key operational dimensions. They detect less customs fraud, particularly on days when officials' ports receive many shipments concurrently and inspectors have cover for lower detection performance. Fraud omitted by patronage pairs on these high-workload days is significantly more likely to be later uncovered by audits conducted by a separate team at customs headquarters.[1] These pairs are also more likely

[1]These audits provide a credible benchmark for undetected fraud as the team is based at headquarters, operates independently of ports, and has access to information (such as importer tax and banking records) not available to port

to deviate from random assignment (of inspectors to shipments) prescribed by official rules, as administrators handle the inspector assignment to shipments and can distort it. The deviations involving patronage pairs stand out because the same inspectors repeatedly match with specific customs brokers, generating significant "excess inspection shares" (*suspect assignments*).[2] These signal collusion with brokers, leading inspectors to flag less fraud, although some is subsequently uncovered by headquarters audits. Finally, on the intensive margin, patronage pairs impose fraud value adjustments that are 9% lower, even for comparable products with similar evasion risk.

Notably, although shipments handled by patronage officials yield higher average revenues, this is due to their assignment to higher-value shipments rather than superior fraud detection performance. I estimate the tax revenue loss from patronage hires by comparing actual tax yields to counterfactual yields that would have been collected had shipments been handled by non-patronage officials, using a conservative benchmark based on internal reference prices. Losses are largest when both the inspector and administrator are patronage hires, reducing tax revenue on shipments by up to 11%, with a shortfall across all inspected shipments of 5%. Taken together, the evidence points to corruption: patronage pairs appear to gain from misdetection deals with importers.

While prior work investigates patronage within specific bureaucratic layers—either frontline agents (Oliveros, 2021) or high-level administrators (Xu, 2018)—I show how multi-level patronage compounds governance failures by enabling collusion. Thus, their underperformance is not only due to lower human capital. I develop a new appointment-cycle-based approach for identifying patronage hires that is applicable in both one-party dominant regimes and fragmented democracies. I also contribute to the growing evidence on favoritism in public employment, including nepotism (Riaño, 2023), providing a tractable way to measure connected hiring and quantify its governance and fiscal consequences. Finally, by linking these patronage dynamics to revenue outcomes, this study helps explain why customs revenue collection remains low in developing countries, despite

inspectors. They also avoid repeated local interactions with brokers and importers that can foster collusion.

[2]Excess inspection shares measure positive deviations from prescribed random assignment, constructed following the methodology introduced by Chalendard et al. (2023). See Section 4 for details.

customs taxes being theoretically simpler to enforce than domestic taxes (Baunsgaard and Keen, 2010). Patronage hierarchies enable coordination among bureaucrats that erodes organizational performance, helping explain why institutional reforms often fail to improve state capacity.

2 Ineffective Oversight of Patronage Pairs

2.1 Conceptual framework

I propose a framework illustrating how political patronage weakens oversight and performance within hierarchical bureaucracies by affecting bureaucrats' competence and integrity. Following Tirole (1986), consider a three-tiered bureaucracy comprising a principal, supervisor, and agent. The principal delegates oversight to the supervisor, who monitors the agent responsible for policy implementation. When both supervisors and agents are hired based on political ties—forming *patronage pairs*—shared political loyalties compromise organizational performance in two primary ways. First, patronage hires typically exhibit lower competence due to selection based on political loyalty rather than merit (Colonnelli et al., 2020; Dessein and Garicano, 2023; Moreira and Pérez, 2024). Second, shared political backgrounds foster mutual trust and aligned incentives, reducing supervisors' willingness to penalize politically aligned agents and diminishing overall enforcement integrity.[3] Weak oversight arising from these compromised factors risks significant government revenue and service quality losses across diverse contexts with a similar structure, including customs (Moreira and Pérez, 2024), tax collection (Dzansi et al., 2022), agricultural extension (Dal Bó et al., 2021), and public procurement (Celhay et al., 2024).

Mutual political loyalties foster trust and align interests, lowering the costs of bargaining and collusion (see Appendix Section A for a formal treatment). Trust emerges rationally from shared expectations of loyalty (Gambetta, 1988), informed by joint political engagement (Leucht, 2023; Lachi and Rojas-Schaffer, 2018). Thus, patronage pairs face reduced collusion costs compared to non-patronage pairs, enabling easier coordination to circumvent oversight. In contrast, non-patronage hires have less information about each other and face higher bargaining costs.

[3]This dynamic parallels evidence showing that vertically homogeneous worker-supervisor pairs (e.g., in terms of ethnicity) exert less effort due to weaker oversight (Marx et al., 2021).

Patronage supervisors impose less discipline on patronage agents, lowering the probability of punishment ($\underline{p}_S$), whereas non-patronage pairs experience stricter oversight ($\overline{p}_S$). Supervisors and agents face an additional risk of detection by a higher-level principal who periodically conducts audits of the work performed by the pair. This principal-level audit probability, denoted as p_R, captures the likelihood of re-inspection and fraud being identified.[4] I model supervisors' punishment probabilities as endogenous to collusion costs and patronage ties, reflecting their close monitoring and discretion. In contrast, principals conduct lower-frequency, audit-based inspections constrained by distance. These differences justify treating p_S and p_R as distinct in the framework.

Agents decide whether to exert effort or collude with citizens they interact with. Effort incurs personal costs but yields formal rewards from detecting non-compliance. Collusion generates immediate informal benefits through bribes or favors, at the risk of penalties if caught. Lower detection may thus stem from reduced effort or deliberate misreporting in exchange for bribes.

Initially, the organization's human resources division determines whether hires are patronage-based and assigns their posting locations. The game unfolds as follows. In period 0, a citizen interacts with an agent during inspection or service provision. The citizen attempts a violation valued at ϕ, while external conditions (such as workload spikes or weather shocks) introduce a masking factor δ granting agents plausible deniability, which lowers the expected penalty they face if supervisors or principals detect misconduct. In period 1, the agent chooses either honest effort or collusion. Collusion yields an immediate informal benefit (b) but carries potential penalties (μ) if detected by the supervisor or principal. Conversely, honest effort incurs a cost (e) but offers formal rewards $\alpha\phi$, where $\alpha \in (0,1)$, upon successful violation detection with probability p_F. In period 2, supervisors observe the agent's behavior. Patronage supervisors are less likely to impose

[4] In an extended framework considered in the Appendix, the principal chooses an audit intensity a_{ij} for each dyad based on observable risk, so that the detection probability is dyad-specific, $p_R^{ij} = \pi a_{ij}$, where π is the probability that an audit uncovers the violation (see Appendix Section A.6). One could also consider an extension of the framework where patronage pairs face a lower expected penalty from reinspections due to their political connections (such that a lower $\underline{p}_R < p_R$ applies to them). However, in the main settings I consider there is less evidence to suggest that the ineffective oversight extends to the principal.

penalties ($\underline{p}_S$) on patronage agents compared to non-patronage supervisors ($\overline{p}_S$), given their lower bargaining costs of collusion. Finally, in period 3, a higher-level principal conducts audits or re-inspections with probability p_R, penalizing any detected collusion. A formal statement of the game, its extensive form and decision tree, and the payoffs are presented in Appendix Section A.

Patronage pairs face lower collusion costs ($\underline{c}$) than non-patronage pairs ($\overline{c}$), reducing the probability patronage supervisors punish patronage agents ($p_S(\underline{c}) < p_S(\overline{c})$). Consequently, patronage pairs obtain the highest expected payoff from collusion among the four possible pairings of agent and supervisor types. A higher masking factor (δ) lowers the expected penalty from collusion rather than the cost of honest effort, so it raises the payoff from collusion for every pair, but it changes behavior only for pairs close to the margin between collusion and honest effort. Lower collusion costs raise the gain from collusion for patronage pairs relative to any other pairing of the same inspector, and Corollary 2 in Appendix Section A states the condition under which they are the pairs that cross the margin: a higher masking factor then reduces detection within patronage pairs, and not for patronage agents supervised by non-patronage supervisors, whose payoff from collusion stays below their payoff from honest effort. A capacity constraint would instead lower an agent's detection under high workload whoever supervises her. Additionally, as patronage agents have lower detection probabilities ($\underline{p}_F$) compared to non-patronage agents ($\overline{p}_F$), their payoff from honest effort is lower. Thus, for an intermediate range of parameters (Corollary 1), patronage agents supervised by patronage hires gain greater expected utility from collusion than from honest effort, whereas non-patronage agents find honest effort more rewarding, as their higher detection rates yield greater expected benefits than collusion.

This framework highlights a nested accountability problem: aligned interests between patronage supervisors and agents undermine local oversight, while principals face informational constraints that hinder timely intervention. As a result, patronage-driven bureaucracies systematically underperform merit-based structures, perpetuating inefficiency and corruption. This framework generalizes beyond customs, offering insights for regulatory and public service contexts character-

ized by similar oversight hierarchies and patronage dynamics.[5]

2.2 Application to customs

Customs agencies are well suited to studying oversight failures in multi-tiered bureaucracies. They exhibit a clear hierarchy in which a central authority delegates monitoring responsibilities to local supervisors, who in turn oversee frontline agents—mirroring the principal–supervisor–agent framework introduced above. Customs agencies are also high-stakes institutions: they not only enforce trade and tariff policy, but in many developing countries, they account for a large share of government revenue (Besley and Persson, 2009). The high discretion involved in valuation, classification, and inspection decisions makes customs a fertile ground for discretion and collusion.

To clarify how the framework maps to customs, the agent corresponds to the customs inspector, who can collude with their supervisor (the port administrator) against the objective of the principal (customs headquarters), which can send in a post-clearance audit (PCA) team to reinspect shipments. Citizens are importers represented by customs brokers who attempt shipment undervaluation.[6] During the period of my study data, inspectors uncovering fraud can impose fines equal to the attempted fraud value, retaining 50% as a performance bonus, with the remainder collected by the customs authority as a penalty fee (ABC Color, 2025).

Masking occurs when ports receive an unusually high volume of shipments to be inspected compared to their typical levels. Since the arrival timing of ships, planes, and trucks is beyond the control of customs officials, these deviations are exogenous. However, inspectors can exploit high-workload days to strategically reduce effort on certain shipments, potentially lowering fraud detection. Anecdotal evidence suggests that when inspector workloads spike, they justify spending

[5] While this framework emphasizes the negative implications of patronage, a recent literature highlights exceptions where patronage may yield positive outcomes. For example, patronage can leverage private information effectively to identify high-potential candidates, as documented historically in the British Royal Navy (Voth and Xu, 2022). Similarly, when patronage appointees are held accountable for performance outcomes, as in certain Brazilian bureaucracies, patronage may improve monitoring, align incentives, and foster trust within bureaucratic hierarchies (Toral, 2024a).

[6] In Paraguay, importers almost universally rely on customs brokers. These are private sector individuals who manage interactions with the customs authority, including shipment tracking, filing shipment declarations, witnessing customs inspections, and paying duties. In most countries they have to register with the customs authority and pass some examination to earn a license and become part of the profession. Only shipments with very low commercial value and other uncommon exceptions can be exempted from having a customs broker managing the import logistics.

less time per shipment to ensure all inspections are completed on schedule. This allows inspectors to claim plausible deniability if shipments they cleared are later revealed to involve customs fraud.

In applying the framework to my setting, two theoretical predictions naturally emerge. First, patronage pairs should collude more frequently than non-patronage pairs, leading to lower customs fraud detection (**prediction I**). Second, patronage pairs should detect less customs fraud when faced with high workloads, as these offer plausible deniability for reduced effort (**prediction II**).

3 Context

3.1 Modernization and persistent corruption at Paraguayan customs

Paraguayan customs provides a compelling setting for studying how political patronage compromises oversight and performance in hierarchical bureaucracies. It is an important revenue collection institution, generating 40–45% of total tax revenues annually. It has implemented multiple modernization reforms aimed at improving performance and reducing discretion. These include the adoption of a rules-based risk management system, the rollout of a dynamic inspection algorithm proposed by the International Monetary Fund (IMF) in 2020, and the formalization of shipment assignment procedures through customs' software. Customs documentation and clearance have also been digitized and tracked electronically. International organizations, including the IMF, World Bank, and the United States Agency for International Development (USAID), have supported these initiatives through technical assistance and capacity-building programs. For example, Paraguay was the first country in the Western Hemisphere to implement the Millennium Challenge Account Threshold Program, with a focus on combating Paraguay's "systemic corruption problems" (U.S. Embassy of Asunción, 2007). Yet, as my findings show, these reforms have not fully mitigated entrenched informal practices like patronage-based hiring and assignment manipulation, which continue to distort inspection outcomes despite formal oversight mechanisms.

Despite its modernization reforms and low regulatory barriers,[7] customs corruption persists

[7]Paraguay ranks among the top countries in Latin America and the Caribbean for "trade freedom," with low average tariffs and minimal non-tariff barriers (Heritage Foundation, 2024).

(InSight Crime, 2020; U.S. Treasury, 2021).[8] Using the mirror analysis methodology (Fisman and Wei, 2004), I find substantial "evasion gaps" (discrepancies between the value of exports reported by the exporting country and the value of imports reported by the importing country for the same goods and time period), reaching hundreds of millions of US dollars (USD) for particular products. For instance, in 2018, the rest of the world reported exporting 893 million USD of cellphones to Paraguay, while Paraguayan customs reported 611 million USD, revealing a trade gap of 282 million USD. Such discrepancies persist, with cellphone trade gaps exceeding 200 million USD in recent years (2021-2024).[9] This environment benefits colluding importers, customs brokers, and customs officials, while the state loses revenue, and non-colluding brokers and importers are at a disadvantage. The latter face unfair competition, as they may need to charge higher prices for the same goods (due to paying higher taxes), while colluding importers can offer lower prices.

3.2 Patronage and dominant-party politics in Paraguay

Paraguay also illustrates patronage in an understudied context of a dominant party. It has two major clientelistic parties: the Colorado Party and the Liberal Party, with the former being the most influential (Abente Brun, 2007; Lachi and Rojas-Schaffer, 2018; Mandl, 2021). During the dictatorship of Alfredo Stroessner (1954–1989), the Colorado Party solidified its control through extensive patronage networks (Abente Brun, 2007; Painter, 1983; Mandl, 2021). These practices persisted during the country's democratic transition, notably with key state institutions such as Paraguayan customs, which has historically been associated with politically-driven hiring practices.[10] Stroessner infamously claimed that "contraband is the price of peace", justifying the proliferation of contraband and economic privileges for political allies (Última Hora, 2013). Customs positions have been highly sought after, given their association with opportunities for graft and

[8] During the dictatorship, customs corruption in Paraguay was top-down and centralized, but more recent corruption has been decentralized: linked to specific officials acting independently (Franks et al., 2005; U.S. Treasury, 2021), which is consistent with the literature on political modernization (Huntington, 1968; Shleifer and Vishny, 1993).

[9] Other products with significant trade gaps in recent years include vehicles, clothing, electronic cigarettes, cigarettes, and beer. The top five products represent an estimated annual revenue loss of about 50 million USD.

[10] This echoes what Schwartz (2023) describes in Guatemala, where entrenched informal networks from an authoritarian past continued to undermine state institutions such as the customs administration under democracy.

illicit enrichment schemes (Última Hora, 2013; ABC Color, 2018). This system of patronage has perpetuated the party's dominance, allowing it to stay at the center of Paraguayan politics despite occasional challenges from the Liberal Party and other opposition forces (Abente Brun, 2007).

The Colorado Party has maintained near-continuous control since 1947, with a brief interruption from 2008 to 2013 when a center-left coalition, which included the Liberal Party, won the presidential election.[11] Its dominance relies on its historical and ongoing ability to leverage clientelism, vote-buying (Finan and Schechter, 2012; Duarte et al., 2025; New York Times, 2023), and patronage (Painter, 1983; Setrini, 2025). Yet even when the dominant Colorado Party wins consecutive elections, internal factional conflicts persist (Rizova, 2007).[12]

Scholars such as Schuster (2021) describe patronage as "the defining feature of Paraguay's public personnel management," and customs is no exception (Mandl, 2021). In practice, patronage hires tend to be drawn from the ruling party and the ruling faction within it. After elections, the party rewards members that contributed to its victory while demoting—or, in rare cases, dismissing—those aligned with its losing party faction.[13] During my study period, the Colorado Party's most recent vote share was 49%, yet over 95% of inspectors and administrators belonged to it, underscoring the patronage-driven character of customs. While most customs inspectors and administrators could be considered patronage hires, my operationalization of patronage can be viewed as a dosage measure, where the most politically connected hires (those hired at the start of new administrations and those hired during the dictatorship) receive the highest hiring prioritization.[14]

[11] This period of opposition rule was marked by instability, culminating in a 2012 impeachment of President Fernando Lugo, orchestrated by the Liberal and Colorado Party. This maneuver allowed the Liberal Party to hold the presidency from 2012 to 2013 before the Colorado Party swiftly regained control.

[12] Although Colorado Party primaries are often fiercely contested, factions typically reconcile before general elections through a process known as the "republican hug" (Goroso, 2023; Última Hora, 2023b), named after the party's official name: National Republican Association. Factional alignment is fluid and shifts with changes in political power (Última Hora, 2022), and notably, no faction has won multiple elections up to the period covered by this study.

[13] Firings are rare due to stringent public employment laws. Most employees gain tenure after a few years of continuous service through a process known as "deprecarization," in which temporary hires transition to tenured status. With tenure, employees cannot be dismissed without "just cause" and, in the event of dismissal, are entitled to compensation based on seniority. In my dataset, over 90% of inspectors and administrators already hold tenure.

[14] In Paraguay, hiring later in an executive's term is more likely to reflect institutional staffing needs and allows for more thorough oversight from the Ministry of Economy, reducing the influence of political favoritism.

3.3 Customs clearance process and inspections

Paraguay is a member of the South American Mercosur trade bloc alongside Brazil, Argentina, and Uruguay. Mercosur eliminates import tariffs for goods with Mercosur origin and applies a Common External Tariff (CET) to others, which averages 11.5% but can reach 35% for some products. Paraguay, however, applies lower tariffs than its neighbors—exempting nearly 650 items from the CET—resulting in an average import tariff of around 3.5% (Mandl, 2021).

Despite being landlocked, Paraguay receives imports through 31 customs ports via the Paraguay and Paraná Rivers (which drain into the Atlantic), as well as two main airports and several land crossings. These ports which are part of my data, include both public and privately managed terminals, though the customs authority maintains its own infrastructure and personnel at all locations.

Figure 1: Customs Clearance Process



Next, I elaborate on the main steps that make up the import clearance process as shown above:

1. **Registration:** first the customs broker acting on behalf of the importer submits an import declaration online on customs' web platform. This declaration includes all the relevant documentation: the invoice provided by the exporter, the bill of lading, a document authorizing the broker to act on behalf of the importer, a certificate of origin, a value declaration document, and any authorization documents for products that are regulated (e.g., agrochemicals require a permit from the Ministry of Agriculture).
2. **Inspection assignment by risk management unit:** shipments are assigned to either a green (no inspection), orange (documentary inspection), or red (documentary and physical inspection) channel.[15] All shipments start as "green," but customs' risk management unit can assign a higher level of inspection for three main reasons. They are sequentially: (i) mandatory inspection laws for certain products (e.g., weapons, used cars, agrochemicals), (ii) risk scores generated by a rules-based system considering shipment characteristics (called

[15]The frequency of each has hovered around a third for each channel during the time period of my data.

the SBR score) and a dynamic importer-based algorithm proposed by the IMF (called the SAS/SGIR score), and (iii) random assignment.[16] The highest risk level from these criteria determines the channel. Inspectors are supposed to be assigned randomly to shipments conditional on having the same workload. That is, if all inspectors at a port have the same number of pending shipments to inspect, then a random number generator within customs' software randomly picks the inspector for the next shipment. While this is how the assignment system is intended to work at many customs authorities, I show it can be manipulated.

3. **Assessment by inspectors:** is based on the documentation submitted by the broker. For orange channel shipments, visual inspectors review the documentation for accuracy. Red channel shipments additionally involve physical inspections to verify shipment contents.[17] In addition, a valuation inspector assesses red channel shipments on whether their values declared show irregularities. The visual and valuation inspectors are always different individuals and there is no rotation between these roles. Based on the information they gather, the inspectors make any necessary adjustments to the declared import value, quality, product classification, and/or country of origin. Finally, inspectors may make tax adjustments and recommend fines if fraud is uncovered (with inspectors receiving 50% of the fine). Port administrators are the most important authority at each port, among their duties they manage inspectors and determine whether fines are applied or exempted after fraud detection.
4. **Clearance:** on this last step, the broker pays relevant taxes, duties, and fines on behalf of the importer (who then compensates the broker), and the products are released for the importer.
5. **Reinspection or post-clearance audit (PCA):** after clearance, some shipments may still be subject to a reinspection or PCA by a small, specialized team (called *fiscalización*) based at customs headquarters. This team can leverage importers' tax and banking data for cross-checks, giving them an informational advantage. Partly due to this advantage, the PCA

[16] Random assignment is determined based on a random number generator on customs' risk management software.
[17] See Figure C1 for an illustration of a physical inspection.

team uncovers customs fraud not flagged by port inspectors, although some omissions by port inspectors arise from collusion with brokers. At the same time, the team's limited size (10–20 employees) restricts its audits to just 1–2% of all imports annually.

3.4 Customs fraud detection

Inspectors are tasked with detecting customs fraud, primarily focusing on four methods that importers or their brokers can commit: product misclassification (towards lower-tariff goods), underreporting quantities, incorrect product origin declaration (usually towards tariff-exempt Mercosur countries), and undervaluation. Misclassification, underreporting, and origin fraud can typically be detected by visual inspectors, whereas undervaluation is harder to identify. The World Trade Organization (WTO) stipulates that the values submitted on invoices constitute the primary basis for customs valuation. However, anecdotal evidence from customs officials and the press indicates that invoices submitted are often falsified to show lower values (Torres Romero, 2019; Última Hora, 2023a). Exporters abroad may even abet this evasion by not responding to inquiries from Paraguayan customs and even providing altered invoices (U.S. Treasury, 2021).

In equilibrium, importers and their brokers should aim to commit fraud through less detectable methods. The fraud handled by visual inspectors is more easily detectable, often requiring collusion and informal payments. Hence, they prefer undervaluation due to its complexity and lower detectability, particularly if invoices are forged. In addition, if their scheme is uncovered they can still attempt to bargain with the valuation inspector on the extent (intensive margin) of undervaluation. Figure 2 is consisten with undervaluation as the predominant fraud method, as valuation inspectors detect fraud in 27.1% of assessed shipments compared to 0.6% by visual inspectors.[18] Given the paucity of customs fraud flagged by visual inspectors, I focus on valuation inspectors.[19]

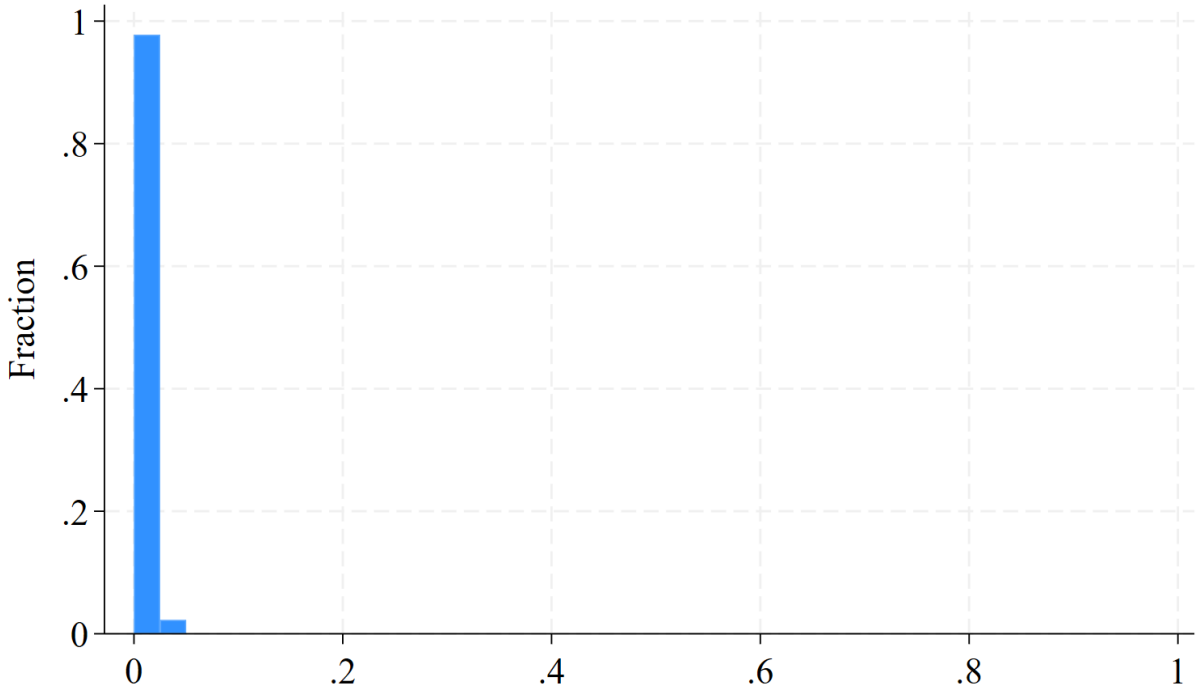
There is an inherent tension at Paraguayan customs between the patronage bureaucracy and the

[18] As documented in the customs evasion literature, undervaluation is the dominant form of customs evasion (Javorcik and Narciso, 2008; Rijkers et al., 2017; Leone et al., 2022; Anne et al., 2023; Madsen et al., 2025).

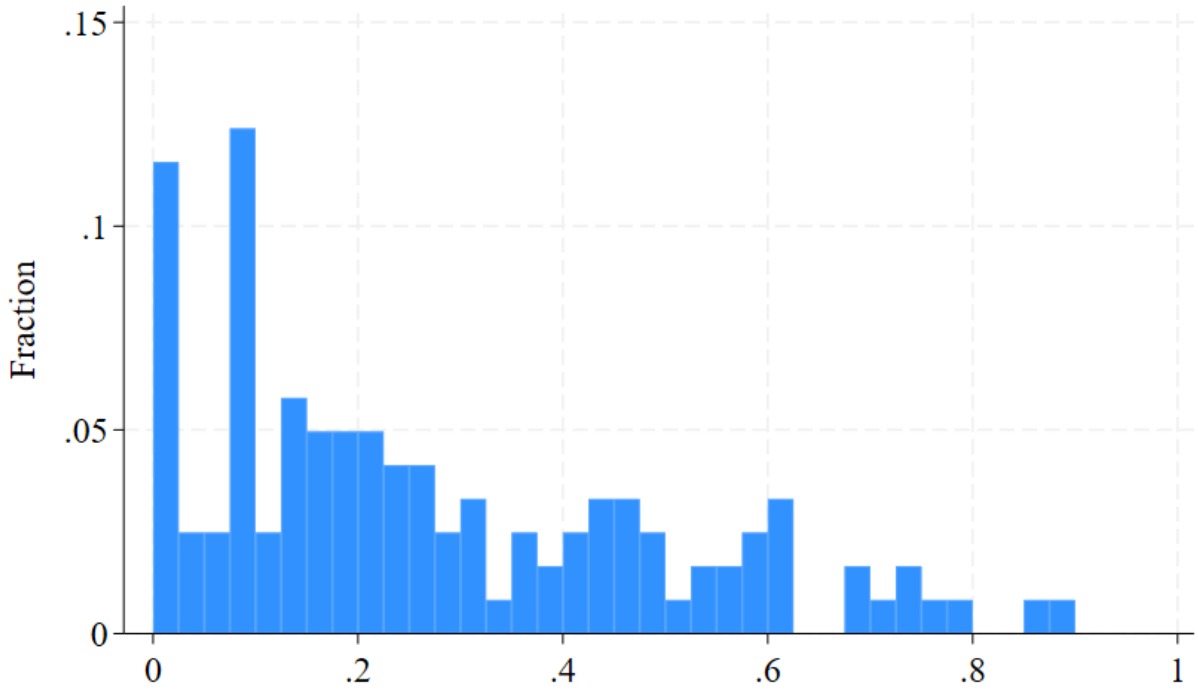
[19] Appendix Figure C2 shows the fraud detection rate of visual inspectors on red channel shipments, confirming similarly low detection rates (1.1%). This suggests that the disparity in detection rates between visual and valuation inspectors is not simply due to valuation inspectors handling only higher-risk (red channel) shipments.

Figure 2: Fraud detection rate for visual and valuation inspectors

(a) Visual inspectors



(b) Valuation inspectors



Notes: Each histogram shows the distribution of fraud detection rates across visual and valuation inspectors, calculated as the number of shipments with detected fraud divided by the total number of shipments inspected by each inspector. Hence, the unit of observation is the inspector. The width of the bars is 0.025 (2.5%).

executive's goal of efficient revenue collection, creating a Janus-faced institution with some "pockets of effectiveness" (Geddes, 1994; Brierley, 2021). The latter occurs as Paraguayan customs collects a substantial share of annual tax revenues (40–45% over the past decade), attracting oversight from the relatively technocratic Ministry of Economy and Finance. International organizations including the IMF, World Bank, and Inter-American Development Bank support capacity-building initiatives at customs. The U.S. government also supported customs via USAID. Diplomatic cables from the U.S. Embassy in Paraguay show their efforts at building the customs authority's capacity, including vetting officials for training (U.S. Embassy of Asunción, 2006), monitoring the perception of customs and other institutions through surveys (U.S. Embassy of Asunción, 2005), and highlighting successes (U.S. Embassy of Asunción, 2008b,a).[20]

Port inspectors are more vulnerable to capture than PCA team members as inspectors typically live near their assigned ports, fostering close relationships with customs brokers—a phenomenon observed in similar settings (Blundo et al., 2008; Ndonga, 2013; Olimpieva, 2010). PCA team members are instead based in the capital and are more highly compensated, so they face lower monitoring costs and reduced capture risks (Mastrorocco and Teso, 2023).[21]

A more technical PCA team coexists with potentially corrupt inspectors because, although PCA audits accurately uncover fraud (approximately 88% of audited shipments in 2023), their findings rarely convert fully into additional revenue. Importers can appeal, and due to the weak rule of law they often prevail in court against customs. Among PCA team audits that uncovered fraud in 2023, only about 12% resulted in additional revenue by 2024. Anecdotally, some port inspectors deliberately omit fraud detection due to collusion with importers and brokers.[22] PCA findings thus

[20]Their interest partly stems from concerns that customs tax evasion in Paraguay facilitates trade-based money laundering linked to financing terrorist groups (U.S. Treasury, 2021; Madureira, 2022).

[21]Proximity may increase state performance in other settings: Toral (2024b) finds that Brazilian prosecutors embedded locally engage more effectively against corruption. However, Brazilian prosecutors differ from officials in my setting as they are hired through competitive examinations, are promoted based on merit, and have greater autonomy.

[22]While cases where customs fraud is identified by the PCA team after being omitted by local port inspectors may result in sanctions, firings are rare (ABC Color, 2019b). Inspectors may have plausible deniability in failing to detect customs fraud without collusion or ill intent. Common sanctions include reassignment to desk duties at customs headquarters ("being sent to the freezer"), un-

serve as an informative benchmark to measure undetected fraud by local inspectors.

4 Empirical Strategy and Hypotheses

4.1 Characteristics of patronage hires based on appointment cycles

My definition of patronage hires attempts to measure the hiring prioritization of patronage-driven governments (Schuster, 2021). This logic is reflected in how Paraguayan customs fills positions: hiring remains deeply embedded in partisan networks and largely bypasses formal meritocratic procedures. Qualitative accounts and news reports link customs corruption to patronage appointments (ABC Color, 2023; Última Hora, 2014a). Mandl (2021) describes Paraguayan customs as a "prize for individuals with political connections," noting that winning parties "dispensed government jobs in return for political favors."[23] Among the roles that turn over with new executive administrations is the head of human resources, who can make new appointments.[24]

The recent literature has operationalized patronage through various proxies: high-level bureaucratic appointments (Xu, 2018; Voth and Xu, 2022; Toral, 2024a), temporary hires (Robinson and Verdier, 2013; Oliveros, 2021), political donors and candidates (Colonnelli et al., 2020), and bureaucrats connected to higher-level officials via promotion patterns (Jiang, 2018; Jiang and Zhang, 2020). My measure defines patronage hires as customs officials appointed in the first year of new executive administrations or during the dictatorship period. The focus on the first year is motivated by the fact that hiring budgets are set on an annual basis (ABC Color, 2024), making this discrete cutoff empirically salient. Election-cycle hires reflect observed patronage hiring spikes shortly

paid suspensions, and port rotations. An exception occurred in 2023 when several officials were fired and legally charged for a collusion scheme at the country's main airport (https://www.observador.com.py/subadministrador-y-secretario-de-aduana-imputados-por-importacion-de-celulares-declarados-como-autopartes/).

[23]These include political brokers of the incumbent Colorado party (Última Hora, 2014b), who later become customs officials and continue to work as brokers during subsequent elections.

[24]The majority of public sector hires in Paraguay are politically appointed without competitive exams (Schuster, 2021). Although tenured roles technically require public contests, many institutions bypass this by hiring on temporary contracts later converted to tenure through deprecarization (Última Hora, 2024). The civil service ministry (now part of the Ministry of Economy) posts competitions for public sector jobs online (https://www.paraguayconcursa.gov.py/sicca/seleccion/buscarConcurso/adjudicados/adjudicados.seam) and customs held only eight public "merit contests" and two "public opposition competitions" since 2014, yet none were for customs inspectors or port administrators.

after elections as political debts are paid (Geddes, 1994; Remmer, 2007; Akhtari et al., 2022).[25] In addition, historical accounts show that dictatorship-era (1954–1989) appointments consistently served as political rewards within a long-standing patronage regime (Abente Brun, 2007; Mandl, 2021). Although this proxy likely underestimates the true prevalence of patronage—biasing results toward zero—it remains particularly suited to dominant-party regimes. Appendix Figure C3 shows a concentration of appointments in the first year of executive terms during democracy.[26]

Table 1 compares characteristics of patronage and non-patronage hires. Patronage valuation inspectors typically have about one education level lower, averaging below a college degree. Indeed, patronage inspectors are 31 percentage points less likely to have a college degree than non-patronage inspectors. This difference highlights a potential selection channel for lower performance among patronage hires. Both tables indicate that patronage hires tend to be older, partly due to the inclusion of dictatorship hires.[27] The relatively high age of customs officials reflects their frequent refusal to retire, with some appealing to the Supreme Court to retain their positions (ABC Color, 2019a). These legal actions underscore the desirability of customs positions.

I construct a "political engagement index" adding indicators for various types of political engagement by customs officials through matching on their national IDs or names. These are: affiliation with the incumbent party, roles as poll workers, poll watchers or party representatives in recent elections,[28] leadership roles in the incumbent Colorado Party,[29] local party headquarters (*seccional*) leadership,[30] local representatives to incumbent party conventions, voting participa-

[25] As a scope condition, conditioning on copartisanship may be needed in competitive democracies. I emphasize post-election hires, as pre-election hires could be easily fired before acquiring tenure if a differing administration takes power (Robinson and Verdier, 2013; Oliveros, 2021).

[26] Although the first year is a salient cutoff, in Appendix Section C.1 I show results have similar or just slightly smaller coefficient magnitudes when identifying the election-cycle hires as those hired during the first year and a half. Similarly, Appendix Figure C4 plots kernel density estimates of the first appointment date during the dictatorship.

[27] Excluding dictatorship hires reduces, but does not eliminate, significant differences in age among valuation inspectors (see Appendix Table C1).

[28] Digitized data on these electoral campaign roles are available from the election authority starting from 2013. Duarte and Carrizosa (2023) explore the effect of partisan poll workers in elections.

[29] These are elected positions published in the incumbent party's website or books (Delvalle Castillo, 2008).

[30] Previous research on Paraguayan politics has emphasized the role of local party headquarters leaders as political brokers and local politicians (Dosek, 2019; Bandiera et al., 2025). Data on these leaders is available on Delvalle

tion at general elections, candidacy for office, and family connections to politicians.[31] This index aims to capture formal political service provided to the incumbent party and the recency of some of these services highlights ongoing political engagement even after appointment to customs positions. This electoral work likely enables patronage hires to identify one another in the workplace.

Patronage administrators are significantly more politically involved than their non-patronage counterparts, consistent with their older age and higher likelihood of attaining party positions. By contrast, inspectors show no significant difference in measured political activity, though the appointment cycles I use likely capture latent informal ties not observed directly. The index captures only formal political activity and omits informal roles such as lower-level political brokering (Duarte et al., 2025). While local *seccional* leaders are political brokers, they hold limited elected posts higher in the party hierarchy, and no systematic data exist for lower-level brokers nationwide.

Although nearly all customs officials are affiliated with the incumbent party, making simple partisanship uninformative, the political engagement index still differentiates hiring priorities based on connections and services provided for administrators (Mandl, 2021; Schuster, 2021). Given significant differences in age and education among patronage inspectors, my regression specifications control for interactions with these variables.

Castillo (2008) and archived versions of the incumbent party's website.

[31] Individual-level digitized turnout data from the election authority for general elections is available from 2008 onward. Data on officials that were political candidates and politicians' relatives were manually collected from media mentions by newspapers of record in Paraguay which cover customs extensively: ABC Color and *Última Hora*.

Table 1: Comparison of patronage and non-patronage officials

(a) Valuation inspectors

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Education level	54	3.41	62	4.47	116	-1.06***
		(0.22)		(0.19)		
College degree	54	0.44	62	0.76	116	-0.31***
		(0.07)		(0.05)		
Age	54	54.24	62	45.08	116	9.16***
		(1.27)		(1.00)		
Political engagement index	54	3.81	62	3.76	116	0.06
		(0.13)		(0.13)		

(b) Port administrators

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Education level	43	4.67	32	4.53	75	0.14
		(0.21)		(0.28)		
College degree	43	0.81	32	0.75	75	0.06
		(0.06)		(0.08)		
Age	43	56.95	32	50.84	75	6.11***
		(0.93)		(1.51)		
Political engagement index	43	4.67	32	4.12	75	0.55*
		(0.23)		(0.19)		

Notes: The education level variable goes from no education mentioned (0), primary (1), secondary (2), some professional/vocational/technician (non-academic) training (3), some (academic) college courses (4), college degree (5), master's degree (6), and doctorate (7). The "political engagement index" aggregates indicators for various types of political involvement that the customs official has taken part in, which are: (i) being affiliated with the incumbent party, (ii) having been a poll worker in recent elections (2013 or 2018), (iii) having been a poll watcher or party representative in recent elections (2013 or 2018), (iv) having a leadership role in the incumbent Colorado party since 2006, (v) being a local party headquarter (*seccional*) leader (Dosek, 2019; Bandiera et al., 2025) sometime between 2006 and 2022, (vi) being a local representative to party authorities sometime between 2006 and 2022, and (vii) having voted in the most recent general elections for my data's time period (2008, 2013, or 2018), (viii) having been a political candidate, and (ix) being a politician's relative.

Besides demographics on officials, I conducted a tailored survey in 2023–2024 with 149 visual and valuation inspectors across 21 customs ports, achieving an 80% response rate.[32] Responses were obtained from 80 visual and 69 valuation inspectors, of which 65 and 63, respectively, overlap with inspectors from my 2017–2021 data period. Among valuation inspectors (the focus of this study), 29 were patronage hires and 34 were non-patronage. Most responses to the 56-question survey were statistically similar between these groups (see Appendix Tables C2 to C6).[33] Still, patronage inspectors differed significantly in four ways: they were 20% more likely to have relatives working at customs, they said receiving detailed shipment-risk information (which could enhance their fraud detection and make them more accountable) would make them feel more monitored, and reported they more often need additional time with inspections and more often receive threats.

4.2 Hypotheses and methods for measuring performance in customs

Having described the patronage hires, I now outline my empirical methods based on my general hypotheses outlined in Section 2. I aim to demonstrate how, despite the standardized inspection process at Paraguayan customs, informal practices among customs officials may undermine state effectiveness. My methods are informed by the literature and validated through qualitative interviews, site visits, and media reports.

Throughout, the null hypothesis is that the patronage pair coefficient is zero: a patronage inspector supervised by a patronage administrator performs as the two officials' individual fixed effects predict, so that patronage would matter only through who is hired and not through who supervises whom. The framework in Section 2 predicts a negative coefficient, as shared political loyalties lower the cost of collusion. In the model, the pair coefficient is the detection a patronage inspector loses by colluding on shipments she would inspect honestly under a non-patronage administrator. It is nonzero only through the oversight channel, the lower bargaining cost within

[32] The survey covered all active ports except four small, remote locations with fewer than three inspectors each (Pilar, Saltos del Guaira, Pedro Juan Caballero, and Santa Helena).

[33] The questions spanned: inspector characteristics, effort, motivation, promotion prospects, institutional environment, their thoughts on customs' evasion risk management system, and varied questions on their time preferences, altruism, and importance of the different communities they are part of.

patronage pairs, while her own detection ability enters her fixed effect and multiplies the loss, since each shipment she stops inspecting honestly costs $\underline{p}_F$ expected detections (see Appendix Section A, Corollary 1 and Remark 1). The opposite sign is also conceivable: patronage ties can improve monitoring and align incentives (Toral, 2024a), or let principals use private information about candidates (Voth and Xu, 2022), in which case patronage pairs would detect *more* fraud. The tests below are accordingly two-sided.

Despite rules mandating random assignment of inspectors to shipments, I uncover plausible evidence that this system is manipulated at Paraguayan customs. This allows certain inspectors to frequently inspect particular customs brokers, where collusion can arise through repeated interactions (Leone et al., 2022; Lameke et al., 2023). From what I could gather through qualitative interviews, administrators can override the random assignment algorithm by temporarily marking certain inspectors as absent, given that the algorithm works by randomly assigning shipments to inspectors among those marked as "present." Therefore, to direct a shipment to preferred inspectors, the port administrator can set as "present" specific inspectors and mark all other inspectors as absent momentarily. After a relevant shipment is matched, administrators can change who is present again to all inspectors (patronage and non-patronage) that are truly present. This deviation from prescribed random assignment has no legitimate justification, as inspectors do not specialize in certain products, which would provide an innocuous explanation for non-random matching.[34]

I follow the statistical simulation methodology developed by Chalendard et al. (2023) to identify excess inspection shares between inspectors and brokers, which is further explained in Appendix Section B. To gauge its validity, I examine the extreme tail (inspector–broker and inspector–importer pairs with the highest cumulative excess shares) because persistent matchings are precisely where collusive ties are most likely to form. The inspector-broker pair with the high-

[34]Customs headquarters was previously unaware of these deviations as they do not keep records of assignment patterns and lacked tools to analyze them statistically. I reconstructed shipment assignments and implemented the methodology from Chalendard et al. (2023), after obtaining their R code to detect significant deviations. In addition, the paper introducing this methodology was only published in 2023, while my data spans imports from 2017 to 2021.

est cumulative excess interactions over several semesters operated in Ciudad del Este, involving a customs broker implicated in a $570 million fraud scheme sanctioned by the U.S. Treasury (U.S. Treasury, 2021; El Trueno, 2021; El Observador, 2022). Additionally, the inspector in this pair had previously been implicated in a contraband tires scheme.[35] Moreover, the importer with the greatest cumulative excess interactions across all inspectors and semesters is a used car importer connected to invoice forgery for undervaluation (Torres Romero, 2019). Similar patterns emerge among several customs brokers, inspectors, and importers with the most suspect assignments.

I hypothesize that these suspect assignments (deriving from significant excess inspection shares) are more frequent with patronage pairs due to their lower collusion costs. In addition, I expect lower fraud detection on shipments with patronage pairs and inspectors with suspect assignments towards the customs broker on the relevant shipment.[36] Thus, prediction I above translates to:

> **Hypothesis 1:** significant excess inspection shares (*suspect assignments*) occur more frequently with patronage pairs.
>
> **Hypothesis 2:** patronage pairs with suspect assignments detect less customs fraud.

As mentioned above, after shipments clear customs they can still be re-inspected or audited up to five years after they clear customs. These post-clearance audits and re-inspections often uncover fraud omitted by local port inspectors, partly due to inspectors' corruption rather than solely the PCA team's informational advantage. Thus, I expect patronage pairs to be disproportionately involved in cases where PCA teams detect fraud missed at the local level.

Building on insights from the literature on workload and performance (Ponticelli and Alencar, 2016; Alquezar-Yus, 2023; Freedman et al., 2021; Shurtz et al., 2022), I examine the impact of workload fluctuations on fraud detection. Specifically, I calculate the daily number of red channel shipments (requiring valuation inspection) at each customs port, divided by the number of valu-

[35] Per my data use agreement with Paraguayan customs, I cannot name individual customs officials in this paper.

[36] As excess inspections are caused intentionally, lower fraud detection combined with excess inspections suggests corruption rather than merely lower capacity.

ation inspectors who handled at least one red channel shipment at that port on that day.[37] This adjustment accounts for differences in port capacities, which I then standardize to measure deviations from inspectors' typical workload. That is:

$$shipments_on_port_per_day_{pd} = \frac{\text{Total red channel shipments at port } p \text{ on day } d}{\text{Number of valuation inspectors at port } p \text{ on day } d}$$

I expect patronage inspectors to justify and mask lower inspection effort when many shipments arrive at their port, which could be leveraged strategically to detect lower fraud on declarations that are actually fraudulent.[38] Therefore, prediction II becomes:

Hypothesis 3: patronage pairs detect less customs fraud under high workloads.

A useful feature of leveraging shipment volumes is that while inspectors may influence *which* shipments they inspect (as evidenced by the excess inspection shares), they appear to have no control over *how many* shipments they are assigned, particularly during periods of high volume. Port administrators must assign shipments to all inspectors present, and inspectors cannot control the timing of ships, planes, or trucks arriving at customs ports.[39] Thus, the number of shipments an inspector gets assigned per day is exogenous for the inspector.

In the Appendix, I further validate this approach by using exogenous variation in the water level of the Paraguay River—the country's main waterway—as an alternative proxy for workload fluctuations. Water level reductions affect the size of vessels that can navigate the river, which are regulated by the General Naval Prefecture, thus influencing shipment volumes at river ports.[40] Following Ignatenko (2024), I exploit unexpected deviations from average monthly water levels and vessel drafts (vessel sizes) to capture changes in shipment volumes. Appendix Tables C87

[37] Only red channel shipments are assigned a valuation inspector, so the count is the number of distinct valuation inspectors with a red channel shipment at the port on the day.

[38] Alternatively, lower fraud detection may be due to cognitive overload as inspectors may struggle to process an unusually high volume of shipments, suggesting capacity constraints beyond corruption. Section 5 and Appendix C.11 distinguish the two: a capacity constraint should lower a patronage inspector's detection under high workload regardless of who supervises her, whereas masking should do so only within patronage pairs.

[39] Appendix Table C18 shows that total shipments per inspector each day at each port predicts inspectors' workload tally (the number of shipments they have pending to inspect). Hence, when more shipments arrive at a customs port, all inspectors present are expected to have a higher workload. Patronage inspectors have slightly lower workloads overall, even after controlling for administrator fixed effects (Appendix Tables C19 and C20).

[40] Appendix Table C82 confirms these effects.

and C88 demonstrate that higher workload (proxied by water level and vessel draft deviations) correlate with lower fraud detection among patronage pairs.

In summary, suspect assignments—where specific inspectors repeatedly assess shipments from certain brokers—should occur more frequently with patronage pairs. This is because administrators manage the inspector assignment to shipments and can prevent the default random assignment from taking place. As argued in Section 2, given that port administrators monitor the inspectors, incentives are altered if both the supervisor and agent are aligned, which can lead to ineffective oversight and consequently lower effort by the agent (Vannutelli, 2022; Raffler, 2022). Furthermore, fraud detection should decrease when inspectors accumulate suspect assignments towards customs brokers, as excess inspections are linked to collusion in customs (Chalendard et al., 2023; Leone et al., 2022). I also hypothesize that when workloads spike, detection falls further for patronage pairs than for non-patronage pairs. Finally, post-clearance audits and reinspections should reveal more undetected fraud for shipments handled by patronage pairs.

Additionally, I hypothesize that patronage pairs make smaller fraud value and tax adjustments, even on shipments with similar products based on Harmonized System (HS) product code chapters, which is consistent with bargaining leading to partial undervaluation.[41] To test this, I examine valuation adjustments across all shipments in which fraud is flagged, controlling for product fixed effects at the HS two-digit (chapter) level to ensure comparisons are made within similar product categories. Hence, applying prediction I from Section 2 at the intensive margin turns to:

Hypothesis 4: when patronage pairs flag fraud, they make lower value adjustments.

4.3 Regression specifications

To test my hypotheses outlined above, I leverage highly detailed shipment imports data from Paraguayan customs from 2017 to 2021.[42] These include the valuation and product code (HS codes) of shipment items, the dates and times of registration, inspection, and clearance of shipments, shipments' inspection channel (green, orange, or red), IDs of agents associated to each

[41] The HS is a standardized system for classifying goods developed by the World Customs Organization.

[42] The years correspond to when my data use agreement was signed and for which data was obtained and cleaned.

shipment (importer, broker, inspector, and administrator), port inspection results by inspectors, and re-inspection and audit results from the post-clearance audit (PCA) team at customs.

My baseline regression specification is:

$$\begin{aligned} y_{sij} = &\alpha_i + \rho_j + \beta_1 PatronInsp_i \times PatronAdm_j \\ &+ \gamma_1 PatronAdm_j \times InspEduc_i + \gamma_2 PatronAdm_j \times InspAge_i + \varepsilon_{sij}, \end{aligned} \tag{1}$$

where y_{sij} are my dependent variables which indicate fraud detection for shipment s by either the port inspector or by the PCA team (when the inspector does not flag fraud) respectively,[43] α_i and ρ_j are inspector and administrator fixed effects (FEs), $PatronInsp_i$ and $PatronAdm_j$ are indicator variables for patronage inspectors and administrators, and $PatronInsp_i \times PatronAdm_j$ identifies patronage pairs.[44] Patronage pairs are found on around 20.8% of all red channel shipments in my data.[45] Standard errors are clustered at the inspector-administrator pair level, as the treatment of interest is at this level (Abadie et al., 2023). The coefficient of interest, β_1, tests my prediction that patronage pairs detect *less* fraud but are associated with *more* PCA-detected fraud (that they omit).

As patronage inspectors differ significantly in age and education, I control for these traits by interacting with inspector education and age ($InspEduc_i$, $InspAge_i$). I use inspector and administrator fixed effects, which absorb all time-invariant individual characteristics (including each official's patronage status) and leaves identified the interaction term for patronage pairs.[46] The fixed effects thus absorb unobserved heterogeneity in inspectors' competence, integrity, or propensity to detect fraud, as well as administrators' discretion in shipment assignment or enforcement. Iden-

[43] While the PCA team may reinspect shipments on which fraud was detected (as they may expect to uncover greater fraud), I focus on PCA-detected fraud on which inspectors did not flag fraud to make this distinction clear.

[44] Each inspector-administrator pair processed an average of 297 shipments in my data, providing a substantial number of observations. While one could consider using a "patronage index" aggregating whether the inspector and administrator are patronage hires to test if having both or just one leads to different results, the inspector and administrator fixed effects absorb values of 1 in this index, such that only the coefficient for patronage pairs remains.

[45] This proportion happens to match what we would expect under independent matching on shipments, as patronage administrators and inspectors are on 55% and 37.9% of red channel shipments respectively.

[46] Given my theoretical arguments that both electoral cycle and dictatorship hires are patronage-based, my main results focus on both types. In the Appendix I present results isolating electoral cycle and dictatorship pairs by excluding the other type of patronage inspector. Results tend to go in my hypothesized direction for patronage hires overall, yet they sometimes lack statistical power due to the lower number of observations.

tification comes from officials' rotations: the same inspector (or administrator) is observed with different counterparts over time, so coefficients are identified within inspector and within administrator.[47] These fixed effects matter because patronage inspectors tend to inspect shipments with higher value and tax liability (Appendix Table C12), which can lead to higher incentives for tax evasion.[48] On the other hand, Table C13 demonstrates that once we run specification (1), patronage pairs are not associated with shipments having significantly higher value or tax liability.[49]

A distinct concern is that β_1 may capture homogeneity of the pair on a dimension other than patronage. As in other studies of dyadic matches (Marx et al., 2021; Bertrand and Duflo, 2017), equation (1) cannot by itself distinguish the effect of two officials sharing patronage status from the effect of other traits on which patronage pairs happen to coincide: patronage hires are older than other officials and enter customs in similar post-election waves, so two patronage officials are more likely to be of similar age, to have been hired under the same executive administration, and possibly to share gender or education. The fixed effects are the first line of defense, since they absorb each official's own gender, age, education, and hire cohort, leaving only pair-level variation to identify β_1. To address the pair-level concern directly, I follow Marx et al. (2021) and define match indicators for the same four dimensions (same gender, similar age, same education level, and "concurrent hires" appointed under the same executive administration), plus a "low-capacity pair" indicator for pairs in which neither official holds a college degree, the median education level. The last match targets a different worry than homophily: because patronage inspectors are less educated, β_1 could reflect two low-human-capital officials working together rather than a patronage tie. Appendix Table C51 first checks, within specification (1), which of these matches patronage pairs are more

[47] The recent panel literature on interactive, grouped and non-separable fixed effects (Bai, 2009; Bonhomme and Manresa, 2015; Ditzen and Karavias, 2025) shows that additivity can be restrictive when a unit's unobserved characteristics are "priced" differently in different contexts. The estimators in that literature (iterative principal components, common correlated effects, grouped and discretized heterogeneity) are designed for unit-by-time panels with a common latent factor structure and a long time dimension, whereas the identifying variation here comes from the same official being observed under counterparts of both types.

[48] This finding aligns with Chalendard et al. (2023), who show inspectors deemed the most corrupt (based on excess inspections) are disproportionately assigned to the highest-value shipments with the greatest expected tax revenue.

[49] Appendix Tables C14 to C17 also show the main results are robust to controlling for "high-value" ports.

likely to exhibit. Appendix Tables C52–C56 then re-estimate each main specification adding the alternative matches one at a time and jointly, so that β_1 is identified only from patronage pairs that differ from otherwise similar pairs on patronage alone. Traits I do not observe for both officials (such as hometown) cannot be checked this way, and party affiliation is uninformative because over 95% of officials belong to the Colorado party. The remaining identifying assumption is thus that, conditional on the fixed effects and on these observable matches, the timing of a patronage inspector's assignment to a patronage administrator is unrelated to shipment-level determinants of fraud.[50]

We can then ask: (i) do patronage pairs underperform because they face a different mix of shipments, or (ii) conditional on similar shipments, do they still detect less fraud? To separate these, I hold product mix constant with product chapter fixed effects (θ_c) and control for shipment-level evasion incentives using a tax "evasion risk index." The index is the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items (Javorcik and Narciso, 2008), shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products (Hummels and Schaur, 2013), and an indicator for registration outside normal operating hours at the port.[51] The unit price gap is computed at the item level (declared FOB value divided by quantity minus the median unit price for the same eight-digit product code and country of origin), then aggregated at the shipment level with product weights. The time-sensitive share captures the proportion of items classified as such, which anecdotally motivate customs brokers to speed

[50] The balance results in Tables C13 and C29 support this assumption.

[51] Besides using this index to more comprehensively control for shipments' evasion risk, Appendix Table C29 shows how patronage pairs correlate with these covariates in my main regression. Four of the ten variables are significantly related to patronage pairs, and in Appendix Section C.8 I show all my main results are also robust to just controlling for these variables that significantly correlate with patronage pairs.

up their shipment clearance through informal payments. Finally, the after-hours indicator captures anecdotal and empirical evidence that reduced staffing outside normal operating hours facilitates collusion, as reflected in its positive correlation with subsequent PCA fraud detection.

To align signs across components so that higher values reflect greater evasion risk, I reverse the sign of the unit price gap, since undervaluation (declaring below the median price) signals higher evasion. I then standardize all variables into z-scores. Following Marx et al. (2024), the index for each shipment is the average of available z-scores, dividing by the number of non-missing components. This adjustment accounts for customs risk scores that are missing when shipments are mandatorily inspected due to laws and regulations (so risk scores are not calculated). Finally, I re-standardize the resulting index so it can be interpreted in terms of standard deviations.

To analyze if suspect assignments (significant excess inspector-broker interactions) occur more often with patronage pairs, I run:

$$y_{ptbi} = \alpha_i + \eta_b + \rho_j + \beta_1 PatronInsp_i \times PatronAdm_j + \varepsilon_{ptbi}, \tag{2}$$

where y_{ptbi} is a binary suspect assignments share measure (for significant excess inspection shares on 90% or more of the simulations) for the inspector i and broker b pair at port p and semester t,[52] and η_b indicates customs broker FEs. If multiple administrators were posted at the same port in the same semester due to rotations, I consider the administrator linked to most shipments for that inspector-broker pair in that semester-port. I expect my coefficient of interest (β_1) to be positive.

Next, to examine how patronage pairs handle suspect assignments,

[52] For each port-semester combination, I (i) construct 99% confidence intervals for inspectors' productivities (shipments handled) using the method from Sison and Glaz (1995); (ii) draw productivity distributions from those intervals; and for each draw (iii) simulate random allocations of shipments to inspectors under multinomial assignment, holding the total number of shipments fixed. For each inspector-broker pair, I compare the observed share to the simulated distribution and set $y_{ptbi} = 1$ if the observed share exceeds the 99th percentile in at least 90% of the productivity draws. Full details are in Appendix Section B.

$$
\begin{aligned}
y_{sij} = &\alpha_i + \rho_j + \eta_b + \beta_1 SES_{ptbi} + \beta_2 PatronInsp_i \times PatronAdm_j \\
&+ \beta_3 SES_{ptbi} \times PatronInsp_i + \beta_4 SES_{ptbi} \times PatronAdm_j \\
&+ \beta_5 SES_{ptbi} \times PatronInsp_i \times PatronAdm_j \\
&+ \gamma_1 PatronAdm_j \times InspEduc_i + \gamma_2 PatronAdm_j \times InspAge_i \\
&+ \nu_1 SES_{ptbi} \times InspEduc_i + \nu_2 SES_{ptbi} \times InspAge_i + \varepsilon_{sij},
\end{aligned} \tag{3}
$$

where y_{sij} is fraud detection for shipment s (by the inspector or PCA team), and SES_{ptbi} is the share of significant excess inspection shares in simulations (ranging from 0 to 1) for inspector i and broker b pair at port p in semester t. I expect patronage hires and suspect assignments lead to *lower* fraud detection by inspectors, which is offset by *higher* fraud detection by the PCA team.[53]

The regression testing if higher workload deviations lower fraud detection is similar to equation (3) except I remove the broker fixed effects and interact with $WorkDev_{pd}$ instead, which is the standardized total number of red channel shipments arriving at port p on day d (divided by the number of valuation inspectors present at port p on day d). I expect patronage hires to detect *less* fraud locally but trigger *more* PCA-detected fraud under heavier workloads.

Finally, I study the *intensive margin*: conditional on fraud detection, do patronage pairs impose smaller fraud value and tax adjustments? I estimate a Poisson pseudo-maximum likelihood (PPML) regression (Santos Silva and Tenreyro, 2006) on the subsample of shipments with detected fraud (one observation per shipment s):

$$
\begin{aligned}
y_{sij} = &\alpha_i + \rho_j + \theta_c + \beta_1 PatronInsp_i \times PatronAdm_j \\
&+ \gamma_1 PatronAdm_j \times InspEduc_i + \gamma_2 PatronAdm_j \times InspAge_i + \varepsilon_{sij},
\end{aligned} \tag{4}
$$

where y_{sij} is fraud value detected (penalties and additional taxes paid by the broker on the importer's behalf) and θ_c are product chapter fixed effects.[54] I expect $\beta_1 < 0$, indicating that

[53] In the Appendix I show my main results are robust to excluding inspector-broker pairs with suspect assignments. The coefficients maintain their significance and direction.

[54] Conditioning on detection helps separate the intensive from the extensive margin, with the caveat it is endogenous as it depends on officials' behavior. I therefore treat these estimates as more suggestive. As a robustness check, Appendix Table C34 re-estimates equation (4) on *all* shipments (setting $y_{sij} = 0$ when no fraud is detected), showing

patronage pairs make lower fraud value adjustments. Coefficients in PPML are interpreted as semi-elasticities: β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of y.

In the Appendix, I run several robustness checks, including testing whether my results reflect low capacity or generic homophily rather than patronage collusion. Appendix Section C.7 adds calendar quarter or calendar month fixed effects to the main specifications, as trade flows and staffing vary over the year. No coefficient moves by more than a tenth of a standard error. On Tables C35 to C38 I show my results hold when controlling for interactions with officials having lower capacity, proxied by having a below-median education (a college degree). Appendix Section C.13 examines the alternative affinities introduced above, both on their own (replacing the patronage pair indicator, Tables C40–C49 and, for low-capacity pairs, Table C50) and jointly with it (Tables C52–C56).[55] If the mechanism were simply "similar people collude," generic matches should reproduce the main effects and absorb the patronage-pair coefficient. On their own, same-gender pairs detect *more* fraud, while similar-age pairs and concurrent hires, like patronage pairs, detect less, though with coefficients of roughly half the size. Entering the low-capacity pair next to the patronage pair leaves the patronage-pair coefficient unchanged (-0.129), while the low-capacity pair itself is smaller and insignificant (-0.051), so the patronage effect is not a low-education effect in disguise. When all five matches enter jointly, the patronage-pair coefficient on fraud detection keeps about three-quarters of its size and its significance, and similar-age and concurrent-hire matches retain independent negative effects. Since age and hire cohort are themselves markers of the post-election hiring waves that define patronage, I view these results as showing that patronage ties have explanatory power beyond generic homophily, with the caveat that part of the effect is shared with the cohort and age structure that patronage hiring creates.

the result holds on all shipments after controlling for product chapter FEs and the evasion risk index.

[55] These affinity matches are common: same-gender pairs cover 59.1%, similar-age pairs (an age gap of at most 10 years) 52%, same-education pairs 35%, and concurrent hires 18% of red-channel shipments. Low-capacity pairs, in which neither official holds a college degree, are rarer at 6.6%.

5 Analysis

Table 2 assesses patronage pairs' fraud detection at the extensive margin. Columns (1) and (5) report the specification without covariates for completeness, while columns (2) and (6) report equation (1).[56] Column (2) indicates patronage pairs detect fraud 47% less often than the sample mean, and columns (3)-(4) shows the gap remains at 13% even within product chapters and controlling for shipments' evasion risk. The two sets of columns answer different questions. Column (2) is the total effect: it combines a *compositional* channel, in which patronage pairs handle shipments whose product mix makes fraud less likely to be detected, with a *treatment* channel, in which a patronage pair is less likely to detect fraud on a given shipment. Columns (3)-(4) compare patronage and non-patronage pairs handling shipments in the same product chapters with the same evasion risk, and so isolate the treatment channel. Product chapter fixed effects absorb roughly 70% of the column (2) coefficient, so composition accounts for a large part, but not all, of the total effect.[57] Following Oster (2019), we can assess how strong selection on unobserved variables would need to be to explain away this estimated effect. Using the Stata command proposed by Oster, I calculate the implied selection ratio (δ) required for the coefficient on patronage pairs to be reduced to zero. I find $\delta = 1.18$, meaning unobserved confounders would need to be 18% stronger than the included covariates (excluding the inspector and administrator fixed effects used for identification) in predicting both treatment assignment and the outcome to eliminate the estimated effect of joint patronage. This exceeds the standard ($\delta = 1$) threshold proposed by Oster, supporting this main result's robustness to omitted variable bias under the proportional selection assumption. Columns (5)–(8) report coefficients in the expected direction (greater detection by the PCA team) but lack statistical precision.[58] The extensive-margin results for fraud detection by the

[56] Appendix Tables C60 to C64 additionally report tables without the inspector and administrator FEs instead.

[57] Patronage pairs handle slightly fewer vehicle shipments (Appendix Table C21), the product group in which most fraud is detected, and within product chapters they still detect less fraud. Coarser product controls give the same answer: Appendix Table C22 replaces the 96 product chapter fixed effects with 15 product sector fixed effects and yields the same coefficient (-0.036) as column (3).

[58] The PCA indicator's relatively high mean reflects high hit rates (see Section 3), even though only $\sim$1–2% of shipments receive audits. Appendix Tables C65 to C68 display the main results with covariate coefficients. Appendix Table C57 has similar results for pairings of distinct patronage types (i.e., a dictatorship administrator paired with an

port inspector establish the baseline: patronage pairs flag less fraud overall.

Appendix Section C.6 shows that the gap is concentrated where there is tax to evade. Splitting shipments by the statutory tax rate implied by their declared tariff classification and origin, patronage pairs detect less fraud only in the top tercile of rates (-0.036, the same as column (4), on a detection rate of 61%), while the pair coefficient is zero on shipments in the bottom tercile, on shipments whose items carry no customs duty, and on shipments of Mercosur origin (Tables C23 and C24). A capacity shortfall would lower detection regardless of the tax at stake, lower effort where evasion pays would not.

Next, I examine intentional manipulation of the shipment assignment process. Patronage administrators can override random assignment rules, leading inspectors to repeatedly inspect the same brokers ("suspect assignments"). If patronage pairs are more likely to generate such patterns, it provides evidence of collusion rather than mere capacity limits. Table 3 shows patronage pairs are 38% more likely than average to have suspect assignments, indicating that vertical patronage alignment substantially increases manipulation of the assignment system. To link suspect assignments to performance, I examine whether manipulated matches reduce inspectors' fraud detection and increase PCA detections.[59] Column (2) of Table 4 shows suspect assignments reduce detection by 43% overall, with an additional 25% reduction for patronage pairs (sum of coefficients for patronage pairs and the interaction of patronage inspector with suspect assignments relative to the sample mean).[60] In columns (3)-(4), the patronage hire indicators marginally lose significance when we control for product chapters, which may be due to the correlation between suspect as-

electoral-cycle inspector, and vice versa). Appendix Table C58 examines how fraud detection varies with inspectors' tenure, showing patronage pairs with longer-serving inspectors have slightly more undetected fraud flagged by the PCA team. Appendix Table C59 adds *concurrent hires*, inspector-administrator pairs appointed during the same executive administration (a proxy for shared factional alignment), to the baseline specification: the patronage pair coefficient is unchanged, and concurrent hires themselves show a larger but imprecisely estimated negative effect on detection.

[59] As suspect assignments correlate with the treatment, the estimates should not be read as causal effects however.

[60] Appendix Table C66 displays all covariates from Table 4, showing that inspectors with an additional educational level forming part of inspector-broker pairs with fully deviant excess inspections are expected to detect more fraud, yet the coefficient is only about 8% of the coefficient for inspector-broker pairs with fully deviant excess inspections. Thus, inspectors with higher education are still expected to detect less fraud if they have significant excess inspections.

Table 2: Patronage hires and customs fraud detected

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector × Patronage administrator	-0.0705**	-0.1294***	-0.0356**	-0.0356**	0.0036	0.0045	0.0028	0.0028
	(0.0317)	(0.0349)	(0.0155)	(0.0153)	(0.0038)	(0.0056)	(0.0056)	(0.0055)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2608	0.2628	0.5751	0.5766	0.0217	0.0218	0.0316	0.0317

Notes: All regressions include inspector and administrator fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

signments and patronage pairs shown on Table 3. Still, suspect assignments (which are driven by patronage pairs) lead to around 31% lower detection even after controlling for product chapters and evasion risk. Conversely, column (6) shows fraud later uncovered by the PCA team is far higher, even when controlling for product chapters: about 5.6 times the average (3.5 times higher when an inspector-broker pair has significant excess interactions and an additional 2.1 times higher for patronage pairs with an inspector-broker pair with significant excess interactions).[61] As suspect assignments reflect intentional efforts to circumvent prescribed random assignment, the results point to corruption rather than merely lower capacity as a driver of patronage underperformance.

Workload shocks offer another way to probe patronage pairs' underperformance relative to non-patronage pairs. If they strategically reduce effort when workloads spike, this would strengthen the interpretation from Section 2 that lower detection reflects collusion or *masking*. Table 5 builds up the results, first showing specifications without covariates in columns (1) and (5). In column (2), a one standard deviation (SD) increase in workload lowers fraud detection by 4.5 percentage points (17% of the sample mean) for patronage pairs, over and above a common workload effect of 6 percentage points that is not statistically significant. A patronage pair facing a workload one SD above its typical level therefore detects fraud 67% less often than the sample mean.[62] With product chapter FEs and the evasion risk index (column (4)), the additional decline for patronage pairs is 1.9 percentage points (7% of the mean). Column (6) indicates that a one SD increase in workload also makes patronage pairs about twice as likely to have fraud later uncovered by the PCA team, and this interaction does not change when we control for product chapter FEs and shipments' evasion risk. Hence, workload shocks appear to provide cover for inspectors' strategic

[61] Appendix Table C66 shows that inspectors with an additional educational level forming part of inspector-broker pairs with fully deviant excess inspections are expected to have less undetected fraud uncovered by the PCA team, yet the coefficient size is a tenth of the coefficient for inspector-broker pairs with fully deviant excess inspections and 16-18% of the coefficient for the triple interaction.

[62] Appendix Table C67, which shows all covariates from Table 5, shows that inspectors with workload deviations and an additional educational level are expected to detect more customs fraud, but the coefficient is one twenty-second the coefficient for patronage pairs and a seventh of the triple interaction coefficient. Thus, inspectors with higher education (a proxy for inspectors' capacity) are still expected to detect less fraud when they are part of a patronage pair.

Table 3: Patronage pairs and suspect assignments

	Binary significant excess inspection share (1)
Patronage inspector	0.0031*
× Patronage administrator	(0.0018)
Mean of Dependent Variable	0.0082
Observations	77,955
R^2	0.0394

Notes: The regression includes inspector, administrator, and customs broker fixed effects. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector combinations which are significantly excessive for 90% or more of the simulations. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. The sample covers all possible non-singleton pairings of valuation inspectors and customs brokers at all semesters and ports.

effort reduction. Appendix C.12 shows that the reallocation of detection from the pair to the PCA team survives when the two outcomes are aggregated to the inspector-port-day.

Could the additional decline instead reflect a capacity constraint, with less-educated patronage inspectors suffering steeper cognitive overload on busy days? Two features of the results argue against this reading, and Appendix C.11 examines it in detail. First, a capacity constraint is a property of the inspector, whereas the decline is a property of the pair: the workload slope of a patronage inspector supervised by a non-patronage administrator is close to zero (0.010, Appendix Table C67), and patronage administrators are not less educated than other administrators (Table 1). Second, pairs of two officials without college education (the median) do not detect less fraud as workload rises: their interaction with workload deviations is positive and not statistically significant (Appendix Table C37). With the low-capacity interactions included, the patronage pair interaction with workload remains negative and significant at the 5% level in every specification, including the most stringent one with product chapter FEs and the evasion risk index.

Table 4: Customs fraud detected, patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Significant excess	0.0149	-0.1179**	-0.0833*	-0.0833*	0.0121**	0.0525**	0.0530**	0.0529**
insp. share	(0.0143)	(0.0488)	(0.0442)	(0.0442)	(0.0059)	(0.0238)	(0.0236)	(0.0236)
Patronage inspector ×	-0.0079	-0.0245**	-0.0106	-0.0107	0.0009	0.0006	0.0005	0.0004
Patronage administrator	(0.0097)	(0.0115)	(0.0091)	(0.0091)	(0.0039)	(0.0057)	(0.0057)	(0.0057)
Patronage inspector ×	-0.0291	-0.0429*	-0.0301	-0.0300	-0.0079	-0.0124	-0.0126	-0.0124
Significant excess insp. share	(0.0214)	(0.0237)	(0.0185)	(0.0185)	(0.0099)	(0.0107)	(0.0108)	(0.0108)
Patronage administrator ×	0.0036	-0.0092	-0.0050	-0.0049	-0.0026	-0.0048	-0.0045	-0.0043
Significant excess insp. share	(0.0181)	(0.0185)	(0.0172)	(0.0172)	(0.0075)	(0.0071)	(0.0071)	(0.0070)
Patronage inspector × Patronage	-0.0026	0.0123	0.0048	0.0048	0.0276*	0.0301**	0.0293*	0.0293*
administrator × Significant excess insp. share	(0.0280)	(0.0286)	(0.0246)	(0.0246)	(0.0155)	(0.0151)	(0.0150)	(0.0151)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,041	292,033	292,033	292,033	294,041	292,033	292,033	292,033
R^2	0.6667	0.6677	0.7118	0.7118	0.0671	0.0677	0.0745	0.0749

Notes: All regressions include inspector, administrator, and customs broker fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table 5: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Workload deviations	0.0070	-0.0608	-0.0146	-0.0144	-0.0029**	0.0049	0.0022	0.0023
	(0.0095)	(0.0385)	(0.0186)	(0.0184)	(0.0012)	(0.0058)	(0.0058)	(0.0058)
Patronage inspector ×	-0.0798**	-0.1365***	-0.0384**	-0.0385**	0.0073**	0.0080	0.0064	0.0064
Patronage administrator	(0.0316)	(0.0351)	(0.0158)	(0.0156)	(0.0037)	(0.0055)	(0.0055)	(0.0054)
Patronage inspector ×	0.0077	0.0096	0.0049	0.0055	-0.0054*	-0.0080**	-0.0078**	-0.0077**
Workload deviations	(0.0136)	(0.0139)	(0.0069)	(0.0068)	(0.0029)	(0.0036)	(0.0035)	(0.0035)
Patronage administrator ×	0.0193	0.0160	0.0089	0.0095	0.0009	0.0008	0.0013	0.0014
Workload deviations	(0.0145)	(0.0139)	(0.0062)	(0.0061)	(0.0014)	(0.0013)	(0.0013)	(0.0013)
Patronage inspector × Patronage	-0.0548***	-0.0451***	-0.0184**	-0.0191**	0.0207***	0.0206***	0.0198***	0.0197***
administrator × Workload deviations	(0.0180)	(0.0173)	(0.0079)	(0.0079)	(0.0045)	(0.0043)	(0.0042)	(0.0042)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2623	0.2645	0.5754	0.5768	0.0266	0.0269	0.0364	0.0366

Notes: All regressions include inspector and administrator fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

So far, the results show patronage pairs underperform on the extensive margin—detecting fewer fraudulent shipments, especially under high workloads and in manipulated assignments. I now turn to the intensive margin: even when patronage pairs flag fraud, they impose smaller tax adjustments, meaning the additional tax payments required of importers are systematically lower. As shown in Table 6, these adjustments are about 10% lower, even after controlling for inspector covariates and product chapter fixed effects to ensure comparisons across similar goods. Even after we control for shipments' evasion risk (which almost doubles the model fit as measured by the pseudo R^2), patronage pairs' adjustments are about 9% lower.[63]

Together, the results show that patronage pairs detect less fraud, manipulate shipment assignment of inspectors more often, omit more fraud later uncovered by audits, and impose smaller tax adjustments—even when handling similar goods. The findings suggest that the underperformance of patronage pairs may not simply reflect administrative failure, but rather a broader political calculus by executive administrations. While patronage hiring undermines the customs authority's revenue collection goals, it aligns with the executive's political objective of rewarding loyal supporters and consolidating electoral coalitions. As a result, administrations face competing priorities: securing political support through appointments while still maintaining enough revenue to govern effectively.[64] As the next section quantifies, the fiscal costs of patronage are substantial, but potentially politically rational under current institutional incentives.

6 Tax revenue loss from patronage hires

This section estimates potential tax revenue losses due to patronage using a measure of hypothetical tax revenue losses analogous to that proposed by Chalendard et al. (2023). We can estimate

[63] While this specification is estimated using Poisson pseudo-maximum likelihood (`ppmlhdfe`), Oster's method applies to linear models. To approximate the robustness of this result to omitted variable bias, I estimate an analogous log-linear specification using `reghdfe`, where the dependent variable is the log of the outcome. In this specification, I find that the implied selection strength is $\delta = 2.46$. That is, unobserved confounders would need to be more than twice as predictive as the included covariates (excluding fixed effects) to fully explain away patronage pairs' negative effect. This comfortably exceeds Oster's robustness threshold ($\delta = 1$), further supporting the credibility of this finding.

[64] As in other developing contexts, political leaders may rationally weaken the state's economic performance to maintain political coalitions and control (Bates, 2014). A similar dynamic is presented by Albertus et al. (2016) with the authoritarian survival of the PRI in Mexico during the past century.

Table 6: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected		
	(1)	(2)	(3)
Patronage inspector ×	-0.1112***	-0.1080*	-0.0976*
Patronage administrator	(0.0425)	(0.0618)	(0.0576)
Mean of Dependent Variable	13,511,141	13,505,530	13,505,530
HS2-product FE	X	X	X
Interactions w/ insp. char.		X	X
Evasion risk index			X
Observations	79,709	79,589	79,589
Pseudo R^2	0.1659	0.1662	0.3099

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects. Regressions in columns (2)-(3) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

hypothetical tax revenue loss for each shipment as the difference between (the natural log of) hypothetical and actual tax yield. As the hypothetical tax yield is unobserved, it is difficult to measure. I take as a reference price for each shipment item their median unit price (value-to-quantity ratio) reported by Paraguayan importers with the same eight-digit product code, country of origin, and year.[65] This reference price is multiplied by the item's quantity and its tax rate (based on the item's product code at the eight-digit level) to estimate an item-level hypothetical yield. Summing across all items within shipments provides a shipment-level hypothetical yield. This approach is conservative, as it assumes median unit prices are not underreported.[66]

To ensure an unbiased estimate of the overall impact of patronage on tax revenue, Chalendard et al. (2023) exclude controls that are potentially endogenous to corruption, which in our case correspond to the inspector and administrator fixed effects.[67] Thus, I estimate the following equation controlling only for variables plausibly exogenous to corruption as indicated by Chalendard et al. (2023): the tax rate, a mixed-shipment indicator (whether it has products from multiple six-digit product codes), the value share of differentiated products, origin country fixed effects, product chapter (HS 2-digit product code) fixed effects, and month-year fixed effects:

$$\begin{aligned} y_{sij} =& \beta_1 PatronInsp_i + \beta_2 PatronAdm_j + \beta_3 PatronInsp_i \times PatronAdm_j \\ &+ \beta_X X_s + \theta_c + \omega_o + \tau_m + \varepsilon_{sij}, \end{aligned} \tag{5}$$

where y_{sij} is the hypothetical tax revenue loss on shipment s, X_s is the vector of exogenous shipment characteristics, while θ_c, ω_o and τ_m represent the product chapter, origin country, and month-year fixed effects respectively. The core inputs for computing this back-of-the-envelope calculation are β_1, β_2, and β_3. Without the fixed effects, the interaction is not significant, but

[65] Chalendard et al. (2023) calculate internal reference prices using median kilogram prices (value-to-weight ratios). However, as not all goods are measured in kilograms, a unit price-based measure is likely more accurate. Unit prices adjust to the appropriate quantity metric for each good, yielding a more precise reference price.

[66] Chalendard et al. (2023) also use external reference prices based on kilogram prices from UN COMTRADE data on exports to the country they study (Madagascar). However, given that COMTRADE data only disaggregates to six-digit product codes, they do not allow us to calculate as similar of a counterfactual.

[67] For their regressions they exclude inspector and customs broker FEs, which are used in the rest of their analyses.

we still observe an additive effect from each layer. Hence, after exponentiating the significant coefficient estimates, we find that shipments handled by patronage inspectors and administrators are linked to tax revenue losses of roughly 7.2% and 3.4%, respectively. In contrast, the interaction is not significant here. Still, patronage pairs account for an estimated 10.8% increase in revenue losses, showing how patronage pairs compound the effect additively here.

Table 7: Patronage hires and tax revenue losses

	Hypothetical tax revenue losses based on internal reference prices using importer unit prices
	(1)
Patronage inspector	0.0696*
	(0.0381)
Patronage administrator	0.0339*
	(0.0180)
Patronage inspector × Patronage administrator	-0.0607
	(0.0497)
Mean of Dependent Variable	-0.0075
Exogenous shipment characteristics	X
Month-year FE	X
HS2-product FE	X
Origin country FE	X
Observations	294,039
R^2	0.0369

Notes: The regression includes fixed effects for: origin country, product chapter (HS 2-digit product code), and month-year. It also controls for the following exogenous shipment characteristics: the tax rate, a mixed-shipment dummy (whether the shipment has products from multiple product subheadings), and the value share of differentiated products within the shipment. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is a hypothetical tax revenue loss variable generated by subtracting a shipment's actual tax yield from their hypothetical tax yield if we use internal reference prices based on unit prices for other imports of the same product, country of origin, and year. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to non-singleton shipment observations. The sample covers all red channel shipments.

Using the coefficient estimates from Table 7, I estimate the counterfactual tax revenue in the absence of patronage hires for each shipment as: $T^{NP} = T \times \exp(\widehat{\beta}_1 \cdot PatronInsp + \widehat{\beta}_2 \cdot PatronAdm + \widehat{\beta}_3 \cdot PatronInsp \times PatronAdm)$, where T denotes the actual tax yield. This expression follows from the log-linear regression model in equation (5), which estimates the log difference between hypothetical and actual tax yields. Exponentiating the predicted difference from the significant coefficients provides a multiplicative adjustment factor to scale the actual yield up to its estimated counterfactual value in the absence of patronage. That is, we are asking how much more tax revenue would have been obtained if shipments were handled by non-patronage hires instead. Further details on the derivation can be found in Appendix Section D.

I then compute the additional tax revenue that would have been collected on shipments if they were handled by non-patronage hires. Table 8 reports average values for red channel shipments across four types: those involving patronage inspectors, patronage administrators, patronage pairs, and all shipments. Table 8 summarizes the results: shipments handled by patronage inspectors, administrators, or both yield more revenue on average—\$8,934, \$9,133, and \$8,794, respectively—than the overall average of \$8,530. However, had these shipments been processed by non-patronage officials, tax revenue would have increased by 7.2%, 3.4%, and 10.9%, respectively—amounting to \$644, \$315, and \$959 per shipment. Two results can be highlighted from this exercise. First, while patronage hires may appear to collect more revenue, this reflects their assignment to higher-value shipments rather than superior fraud detection. Second, restricting attention to patronage inspectors alone would miss around 7.2 percentage points in underperformance, roughly two-thirds of the total loss when both tiers are patronage, while focusing solely on administrators would miss approximately 3.4 percentage points (about one-third). Overall, tax revenue across all red channel shipments would have been 5% higher.[68] This estimate is likely a lower bound, as not all shipments handled by patronage hires are necessarily subject to evasion and my patronage hire measure is an imperfect proxy.

[68] If we expect the average loss to hold for each shipment, we get a total estimated loss of \$122 million USD.

Table 8: Cost of Patronage

	Patronage Inspector		Patronage Administrator		Patronage Pair		All shipments	
	Tax yield	% Increase	Tax yield	% Increase	Tax yield	% Increase	Tax yield	% Increase
Actual tax yield	\$8,934	–	\$9,133	–	\$8,794	–	\$8,530	–
Additional counterfactual yield:	\$644	7.2%	\$315	3.4%	\$959	10.9%	\$417	4.9%

Notes: Tax losses are defined as the difference between the counterfactual tax yield—estimated in the absence of patronage hires—and the actual tax yield. The estimates are converted from the local currency (Paraguayan Guaranis, PYG) to US dollars (USD) using daily exchange rate data. The USD-PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. Counterfactual yields are derived using a hypothetical tax loss measure based on internal reference prices. It is generated by subtracting a shipment's actual tax yield from their hypothetical tax yield if we use internal reference prices based on unit prices for other imports of the same product, country of origin, and year. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. The first two columns report red channel shipments handled solely by patronage inspectors, the next two by patronage administrators only, the following two by both patronage inspectors and administrators, and the final two include all red channel shipments.

7 Conclusion

This study investigates and highlights how patronage alignment between multiple bureaucratic layers magnifies inefficiencies and weakens organizational oversight. In contrast to single-layer analyses, my approach additionally captures how vertically-aligned patronage facilitates collusion between supervisors and agents. I develop a method to identify patronage hires based on bureaucrats' appointment dates—capturing politically motivated hiring both shortly after elections and during authoritarian rule. This approach allows me to trace the long-run consequences of short-term political incentives on public sector performance (Geddes, 1994; Akhtari et al., 2022; Barbosa and Ferreira, 2023), as these bureaucrats outlast the administration that originally hired them.

By examining the Paraguayan customs authority where patronage is prevalent, I find results suggesting that customs officials hired through patronage are less effective in terms of competence and integrity. My empirical evidence indicates that patronage pairs, where both the customs inspector and administrator are patronage hires, are 38% more likely to deviate from the prescribed random assignment of inspectors to shipments. Overall, patronage pairs detect about 47% less fraud, and 13% less when comparing shipments within the same product code chapters and among shipments with similar evasion risk. At the same time, some of this undetected fraud is later uncovered by a separate team at customs headquarters, and this team's fraud detection on shipments handled by patronage pairs is 5.6 times higher when inspectors have suspect assignments from excess interaction with a customs broker. As deviations from random assignment result from intentional decisions facilitated by the port administrator, they likely constitute corruption.

In addition, patronage pairs detect less fraud when an exogenously high number of shipments arrive at the officials' customs port. On days with a standard deviation higher of shipments per inspector, fraud detection by patronage pairs is 29% lower and twice as likely for the PCA team, where the latter uncover undetected fraud by patronage pairs. These results align with anecdotal evidence suggesting that high-workload days offer inspectors greater plausible deniability, allowing them to justify reduced scrutiny, while also creating opportunities for discretionary selection

of shipments for inspection. Patronage pairs impose fraud value adjustments that are 10% lower than those applied by non-patronage pairs on shipments with similar products, and this gap persists at 9% even after accounting for a comprehensive tax evasion risk index. Using internal reference prices, I find that patronage hiring significantly depresses tax collection, especially when both the inspector and administrator are patronage hires—leading to revenue losses of 11% per shipment. About two-thirds of the 11% revenue gap is attributable to patronage inspectors and one-third to administrators. Overall, replacing patronage officials with non-patronage staff could have increased total tax revenue from inspected shipments by 5%, highlighting the fiscal cost of patronage hires.

These results reflect a breakdown of nested accountability: when both an agent and their supervisor are patronage hires, oversight weakens and collusion becomes more feasible. Supervisors fail to discipline agents, and higher-level principals may lack the information needed to intervene effectively on all instances. This aligns with recent work such as Sánchez de la Sierra et al. (2024), which shows the role of state officials' managers in corruption. They also concur with findings from Moreira and Pérez (2024) on the U.S. customs service after the Pendleton Act, where patronage persisted through exempted supervisory roles for port administrators (district collectors).

More broadly, the effectiveness of tariff policy depends not only on formal rules but on how bureaucrats enforce them. When officials are embedded in patronage hierarchies, they may selectively enforce regulations in ways that serve private interests. Such discretionary enforcement can distort competition by granting certain firms preferential treatment unrelated to their productivity (Atkin et al., 2022), thus making bureaucratic discretion, not policy design, the binding constraint.

The dynamics uncovered in this study are not unique to Paraguay, as patronage hierarchies in customs administrations have historically undermined oversight and enabled corruption in other contexts as well. In the 1800s, the New York Custom House—then the largest U.S. federal office—was rife with the "spoils system" (patronage). Samuel Swartwout, a patronage appointee of President Andrew Jackson, embezzled over $1.2 million (about $35 million today), as "checks and balances designed to account for all funds failed" (U.S. Customs and Border Protection, 2019).

Later investigations revealed a system of bribery and political contributions tied to appointments (Hartman, 1953; Rao, 2019; Hoogenboom, 1959). More recently, in Guatemala, the 2015 *La Línea* scandal exposed a patronage network that facilitated large-scale customs fraud and implicated top government officials (Schwartz, 2023). Similar dynamics are documented in Bolivia, where customs roles were used to finance political parties through siphoned state resources (Gingerich, 2013). These cases echo that patronage hierarchies weaken accountability and enable collusion.

My findings suggest reducing political patronage could enhance tax collection efficiency. My framework additionally suggests that principals should allocate oversight resources strategically by targeting high-risk dyads, especially vertically-aligned patronage pairs. Over the medium term, selective task assignment could reduce workload-driven discretion, while longer-run improvements require shifting toward merit-based recruitment to address structural incentives for collusion. Beyond other revenue collection contexts, the dynamics examined here may also extend to other public services featuring a principal-supervisor-agent framework: agricultural extension (Dal Bó et al., 2021), public procurement (Celhay et al., 2024), irrigation systems (Wade, 1982), among others. Future research could investigate patronage hierarchies across these and other sectors.

Policymakers should actively work toward reforms aimed at merit-based hiring and more carefully weigh the adverse effects of patronage, even where it offers electoral advantages. Still, these efforts often confront deep institutional resistance. Moreira and Pérez (2024) show how the 1883 Pendleton Act, which mandated exams for recruiting certain employees in the largest US customs districts, did not improve performance as it led to hiring in exam-exempted positions. Similar challenges are expected in contexts where public employees benefit from the status quo (Klitgaard, 1988). In addition, the transition from political discretion to bureaucratic autonomy in hiring introduces its own challenges, as agencies often need to build reputations and support networks to sustain reforms (Carpenter, 2002; Rao, 2019) and bureaucratic managers may introduce their own hiring biases (Hassan et al., 2025). The literature would benefit from further work on civil service reforms that tackles the countervailing and entrenched practices of patronage bureaucracies.

References

Abadie, Alberto, Susan Athey, Guido Imbens, and Jeffrey Wooldridge, "When should you adjust standard errors for clustering?," *Quarterly Journal of Economics*, 2023, *138* (1), 1–35.

ABC Color, "En Aduanas ya tienen identificados a al menos 127 "magos" de las finanzas," *www.abc.com.py*, 2018.

_ , "Aduaneros que no quieren jubilarse, enviados a sus casas a "trabajar"," *www.abc.com.py*, 2019.

_ , "Las aduanas son usinas de corrupción," *www.abc.com.py*, 2019.

_ , "Política y prebendarismo," *www.abc.com.py*, 2023.

_ , "Cargos públicos: Hay más de 319.000 asignados en el presupuesto," *https://www.abc.com.py/*, 2024.

_ , "Congreso eliminó la repartija de multas en la DNIT," *www.abc.com.py*, 2025.

Abente Brun, Diego, "The quality of democracy in small South American countries: The case of Paraguay," *Kellogg Institute, University of Notre Dame, Working Paper*, 2007, *343*.

Akhtari, Mitra, Diana Moreira, and Laura Trucco, "Political turnover, bureaucratic turnover, and the quality of public services," *American Economic Review*, 2022, *112* (2), 442–493.

Albertus, Michael, Alberto Diaz-Cayeros, Beatriz Magaloni, and Barry R. Weingast, "Authoritarian survival and poverty traps: Land reform in Mexico," *World Development*, 2016, *77*, 154–170.

Alquezar-Yus, Miguel, "Time Constraints and the Quality of Physician Care," Technical Report, HEDG, Department of Economics, University of York 2023.

Anne, Clement, Cyril Chalendard, Ana M. Fernandes, Bob Rijkers, and Vincent Vicard, "Containing Tariff Evasion," 2023.

Atkin, David, Amit Khandelwal, Laura Boudreau, Rafael Dix-Carneiro, Isabela Manelici, Pamela Medina, Brian McCaig, Ameet Morjaria, Luigi Pascali, Bob Rijkers, and Meredith Startz, "International Trade," *VoxDevLit*, October 2022, *4* (1), 1–45.

Bai, Jushan, "Panel data models with interactive fixed effects," *Econometrica*, 2009, *77* (4), 1229–1279.

Bandiera, Antonella, Horacio Larreguy, and Jorge Mangonnet, "Family ties, social control, and authoritarian distribution to elites," *American Political Science Review*, 2025, pp. 1–19.

Bandiera, Oriana, Iwan Barankay, and Imran Rasul, "Social connections and incentives in the workplace: Evidence from personnel data," *Econometrica*, 2009, *77* (4), 1047–1094.

Barbosa, Klenio and Fernando Ferreira, "Occupy government: Democracy and the dynamics of personnel decisions and public finances," *Journal of Public Economics*, 2023, *221*, 104856.

Bates, Robert H., *Markets and states in tropical Africa: The political basis of agricultural policies*, University of California Press, 2014.

Baunsgaard, Thomas and Michael Keen, "Tax revenue and (or?) trade liberalization," *Journal of Public Economics*, 2010, *94* (9-10), 563–577.

Bertrand, Marianne and Esther Duflo, "Field experiments on discrimination," in Abhijit Vinayak Banerjee and Esther Duflo, eds., *Handbook of Economic Field Experiments*, Vol. 1, North-Holland, 2017, pp. 309–393.

Besley, Timothy and Torsten Persson, "The origins of state capacity: Property rights, taxation, and politics," *American Economic Review*, 2009, *99* (4), 1218–1244.

Blundo, Giorgio, Jean-Pierre Olivier de Sardan, N. Bako Arifari, and M. Tidjani Alou, *Everyday corruption and the state: Citizens and public officials in Africa*, Bloomsbury Publishing, 2008.

Bonhomme, Stéphane and Elena Manresa, "Grouped patterns of heterogeneity in panel data," *Econometrica*, 2015, *83* (3), 1147–1184.

Brierley, Sarah, "Combining patronage and merit in public sector recruitment," *Journal of Politics*, 2021, *83* (1), 182–197.

Carpenter, Daniel, *The forging of bureaucratic autonomy: Reputations, networks, and policy innovation in executive agencies, 1862-1928*, Princeton University Press, 2002.

Celhay, Pablo, Paul Gertler, Marcelo Olivares, and Raimundo Undurraga, "How managers can use purchaser performance information to improve procurement efficiency," Working Paper 2024.

Chalendard, Cyril, Ana M. Fernandes, Gael Raballand, and Bob Rijkers, "Corruption in Customs," *Quarterly Journal of Economics*, 2023, *138* (1), 575–636.

Colonnelli, Emanuele, Mounu Prem, and Edoardo Teso, "Patronage and selection in public sector organizations," *American Economic Review*, 2020, *110* (10), 3071–3099.

Dal Bó, Ernesto, Frederico Finan, Nicholas Y Li, and Laura Schechter, "Information technology and government decentralization: Experimental evidence from Paraguay," *Econometrica*, 2021, *89* (2), 677–701.

Delvalle Castillo, Miguel Ángel, *Cronologia historica de la A.N.R. y de las Seccionales Coloradas y todas las autoridades partidarias* 2008.

Dessein, Wouter and Luis Garicano, "La Cordata: Loyalty in Tournaments," Technical Report, Working Paper 2023.

Ditzen, Jan and Yiannis Karavias, "Interactive, grouped and non-separable fixed effects: A practitioner's guide to the new panel data econometrics," 2025. arXiv preprint arXiv:2507.19099.

Dosek, Tomás, "La Persistencia de Caudillos Locales en América Latina," 2019. Doctoral Thesis. Santiago de Chile: Pontificia Universidad Católica del Chile.

Duarte, Raul, Frederico Finan, Horacio Larreguy, and Laura Schechter, "Brokering Votes with Information Spread via Social Networks," *The Review of Economic Studies*, 08 2025, p. rdaf075.

Duarte, Raúl and Andres Carrizosa, "Getting a seat at the (electoral) table: Partisan poll workers and electoral bias," *Available at SSRN 4335982*, 2023.

Dzansi, James, Anders Jensen, David Lagakos, and Henry Telli, "Technology and tax capacity: Evidence from local governments in Ghana," Technical Report, National Bureau of Economic Research 2022.

El Observador, "Rechazan sobreseimiento para Ibrahim Khalil Hijazi," *El Observador Paraguay*, 2022.

El Trueno, "Sancionados por EE.UU. habrían manejado US$ 570 millones," *El Trueno*, 2021.

Finan, Frederico and Laura Schechter, "Vote-buying and reciprocity," *Econometrica*, 2012, *80* (2), 863–881.

Fisman, Raymond and Shang-Jin Wei, "Tax rates and tax evasion: evidence from "missing imports" in China," *Journal of Political Economy*, 2004, *112* (2), 471–496.

Franks, Jeffrey, Randa Sab, Valerie Mercer-Blackman, and Roberto Benelli, "Has Corruption in Paraguay Contributed to Slow Economic Growth?," in "Paraguay," International Monetary Fund, 2005.

Freedman, Seth, Ezra Golberstein, Tsan-Yao Huang, David J. Satin, and Laura Barrie Smith, "Docs with their eyes on the clock? The effect of time pressures on primary care productivity," *Journal of Health Economics*, 2021, *77*, 102442.

Gambetta, Diego, "Can we trust trust?," *Trust: Making and breaking cooperative relations*, 1988, *13* (1), 213–237.

Geddes, Barbara, *Politician's Dilemma: Building State Capacity in Latin America*, Vol. 25, University of California Press, 1994.

Gingerich, Daniel W., *Political institutions and party-directed corruption in South America: Stealing for the team*, Cambridge University Press, 2013.

Goroso, Felpie, "Escándalo: el abrazo republicano," *La Nacion*, 2023.

Hartman, William, "The New York Custom House: Seat of Spoils Politics," *New York History*, 1953, *34* (2), 149–163.

Hassan, Mai, Horacio Larreguy, and Stuart Russell, "Who Gets Hired? Political Patronage and Bureaucratic Favoritism," *American Political Science Review*, 2025, pp. 1–18.

Heritage Foundation, "The Heritage Foundation Index of Economic Freedom," https://www.heritage.org/index/explore 2024.

Hoogenboom, Ari, "The Pendleton Act and the civil service," *The American Historical Review*, 1959, *64* (2), 301–318.

Hummels, David and Georg Schaur, "Time as a Trade Barrier," *American Economic Review*, 2013, *103* (7), 2935–2959.

Huntington, Samuel P., *Political Order in Changing Societies*, Yale University Press, 1968.

Ignatenko, Anna, "Competition and Price Discrimination in International Transportation," 2024.

IMF, *Customs Matters: Strengthening Customs Administration in a Changing World*, Washington, DC: International Monetary Fund, 2022.

InSight Crime, "Paraguay Bungling Chance to Root Out Corruption in Customs," *https://insightcrime.org/*, 2020.

Javorcik, Beata and Gaia Narciso, "Differentiated products and evasion of import tariffs," *Journal of International Economics*, 2008, *76* (2), 208–222.

Jiang, Junyan, "Making bureaucracy work: Patronage networks, performance incentives, and economic development in China," *American Journal of Political Science*, 2018, *62* (4), 982–999.

_ **and Muyang Zhang**, "Friends with benefits: Patronage networks and distributive politics in China," *Journal of Public Economics*, 2020, *184*, 104143.

Klitgaard, Robert, *Controlling Corruption*, University of California Press, 1988.

Lachi, Marcello and Raquel Rojas-Schaffer, *Correligionarios: Actitudes y Prácticas Políticas del Electorado Paraguayo*, Asunción: Arandura, 2018.

Lameke, Aimable Amani, Albert Malukisa Nkuku, Raúl Sánchez de la Sierra, Vincent Tanutama, and Kristof Titeca, "Social Relationships and Joint Profit in Kinshasa," Technical Report, National Bureau of Economic Research 2023.

Leone, Samuel, Nate Grubman, and Jawaher Mbarek, "Corruption Dynamics in International Trade: Evidence on Bribery and Tax Evasion from Tunisian Customs Transactions," 2022.

Leucht, Lukas, "Jobs for Votes: Patronage and Performance in Tammany Hall's NYPD," 2023.

Madsen, Erik, Martin Rotemberg, Sharon Traiberman, and Shizhuo Wang, "Tariffs and State Capacity: A Specific Example," *NBER Working Paper*, 2025, (w34143).

Madureira, Vinicius, "Paraguay Extradites Brazilian Money Launderer to the US," *Organized Crime and Corruption Reporting Project (OCCRP)*, 2022.

Mandl, Tomás, *Modern Paraguay: Uncovering South America's Best Kept Secret*, McFarland, 2021.

Marx, Benjamin, Vincent Pons, and Tavneet Suri, "Diversity and team performance in a Kenyan organization," *Journal of Public Economics*, 2021, *197*, 104332.

_ , _ , and Vincent Rollet, "Electoral Turnovers," *Review of Economic Studies*, 2024, p. rdae108.

Mastrorocco, Nicola and Edoardo Teso, "State Capacity as an Organizational Problem: Evidence from the Growth of the US State Over 100 Years," Technical Report, National Bureau of Economic Research 2023.

Milgrom, Paul and Chris Shannon, "Monotone Comparative Statics," *Econometrica*, 1994, *62* (1), 157–180.

Moreira, Diana and Santiago Pérez, "Civil service exams and organizational performance: Evidence from the Pendleton Act," *American Economic Journal: Applied Economics*, 2024, *16* (3), 250–291.

Ndonga, Dennis, "Managing the risk of corruption in Customs through single window systems," *World Customs Journal*, 2013, *7* (2), 23–37.

New York Times, "One Secret to a Latin American Party's Dominance: Buying Votes," *New York Times*, 2023.

Olimpieva, Irina, "The changing contours of corruption in Russia: Informal intermediaries in state-business relations," *East-West Journal of Economics and Business*, 2010, *13* (2), 61–82.

Oliveros, Virginia, *Patronage at work: Public jobs and political services in Argentina*, Cambridge University Press, 2021.

Oster, Emily, "Unobservable selection and coefficient stability: Theory and evidence," *Journal of Business & Economic Statistics*, 2019, *37* (2), 187–204.

Painter, James, "Paraguay in the 1970s: Continuity and Change in the Political Process," *ISA Occasional Papers*, 1983, pp. 1–38.

Ponticelli, Jacopo and Leonardo Alencar, "Court enforcement, bank loans, and firm investment: evidence from a bankruptcy reform in Brazil," *The Quarterly Journal of Economics*, 2016, *131* (3), 1365–1413.

Raffler, Pia, "Does political oversight of the bureaucracy increase accountability? Field experimental evidence from a dominant party regime," *American Political Science Review*, 2022, *116* (4), 1443–1459.

Rao, Gautham, *National Duties: Custom Houses and the Making of the American State*, University of Chicago Press, 2019.

Remmer, Karen L, "The political economy of patronage: Expenditure patterns in the Argentine provinces, 1983–2003," *Journal of Politics*, 2007, *69* (2), 363–377.

Riaño, Juan Felipe, "Bureaucratic Nepotism," 2023. Working paper, Georgetown University.

Rijkers, Bob, Leila Baghdadi, and Gael Raballand, "Political connections and tariff evasion evidence from Tunisia," *The World Bank Economic Review*, 2017, *31* (2), 459–482.

Rizova, Tatiana Petrova, "The Origins of Paraguayan Post-Authoritarian Electoral Institutions and Their Effects on Internal Party Organization," 2007. Unpublished Manuscript, UCLA.

Robinson, James A. and Thierry Verdier, "The Political Economy of Clientelism," *Scandinavian Journal of Economics*, 2013, *115* (2), 260–291.

Sánchez de la Sierra, Raúl, Kristof Titeca, Haoyang (Stan) Xie, Aimable Amani Lameke, and Albert Malukisa Nkuku, "The Real State: Inside the Congo's Traffic Police Agency," *American Economic Review*, December 2024, *114* (12), 3976–4014.

Santos Silva, J. M. C. and Silvana Tenreyro, "The Log of gravity," *The Review of Economics and Statistics*, 2006, pp. 641–658.

Schuster, Christian, "Paraguay: The Supremacy of Informality in Public Administration," in "The Emerald Handbook of Public Administration in Latin America," Emerald Publishing Limited, 2021, pp. 203–228.

Schwartz, Rachel, *Undermining the State from Within: The Institutional Legacies of Civil War in Central America*, Cambridge University Press, 2023.

Setrini, Gustavo, "Regime Change and Alternancia Clientelar: From Monopolistic to Pluralistic Clientelism in Post-Stroessner Paraguay," *Latin American Perspectives*, 2025.

Shleifer, Andrei and Robert Vishny, "Corruption," *Quarterly Journal of Economics*, 1993, *108* (3), 599–617.

Shurtz, Ity, Alon Eizenberg, Adi Alkalay, and Amnon Lahad, "Physician workload and treatment choice: The case of primary care," *The RAND Journal of Economics*, 2022, *53* (4), 763–791.

Sison, Cristina P. and Joseph Glaz, "Simultaneous confidence intervals and sample size determination for multinomial proportions," *Journal of the American Statistical Association*, 1995, *90* (429), 366–369.

Tirole, Jean, "Hierarchies and bureaucracies: On the role of collusion in organizations," *The Journal of Law, Economics, and Organization*, 1986, *2* (2), 181–214.

Toral, Guillermo, "How patronage delivers: Political appointments, bureaucratic accountability, and service delivery in Brazil," *American Journal of Political Science*, 2024, *68* (2), 797–815.

_ , "Street-Level Rule of Law: Prosecutor Presence and the Fight against Corruption," 2024.

Torres Romero, Jorge, "Facturas mau para justificar la evasión en importación de usados," *La Nacion*, 2019.

U.S. Customs and Border Protection, "Did You Know... Samuel Swartwout Skimmed Staggering Sums?," https://www.cbp.gov/about/history/did-you-know/samuel-swartwout 2019. Last modified December 20, 2019; accessed August 27, 2025.

U.S. Embassy of Asunción, "Paraguay: Survey results send mixed message on corruption," Wikileaks Cable: 05ASUNCION1400_a 2005.

_ , "Paraguay: Human rights vetting request for the MCC Threshold Country Program," Wikileaks Cable: 06ASUNCION1148_a 2006.

_ , "Status report on Paraguay MCA Threshold Program," Wikileaks Cable: 07ASUNCION978_a 2007.

_ , ""Got Viagra?" Paraguay rises to the occasion," Wikileaks Cable: 08ASUNCION148_a 2008.

_ , "Lugo's first major attack on corruption," Wikileaks Cable: 08ASUNCION672_a 2008.

U.S. Treasury, "Treasury Targets Corruption Networks in Paraguay," *https://home.treasury.gov/*, 2021.

Vannutelli, Silvia, "From lapdogs to watchdogs: Random auditor assignment and municipal fiscal performance," Technical Report, National Bureau of Economic Research 2022.

Voth, Joachim and Guo Xu, "Discretion and destruction: Promotions, performance, and patronage in the Royal Navy," Technical Report, CEPR Discussion Paper No. DP13963 2022.

Wade, Robert, "The system of administrative and political corruption: Canal irrigation in South India," *Journal of Development Studies*, 1982, *18* (3), 287–328.

WCO, *WCO Annual Report: 2019-2020*, Brussels, Belgium: World Customs Organization, 2020.

Xu, Guo, "The costs of patronage: Evidence from the british empire," *American Economic Review*, 2018, *108* (11), 3170–98.

Última Hora, "Castiglioni ya tiene movimiento," *Última Hora*, 2007.

_ , "Contrabando: Un abordaje conceptual," *Última Hora*, 2013.

_ , “Con el control en la mano, por ahora,” *Última Hora*, 2014.

_ , “Seccionaleros copan Aduanas y ANNP,” *Última Hora*, 2014.

_ , “Hugo Velázquez está casi fuera de Honor Colorado,” *Última Hora*, 2017.

_ , “Dirigentes migran de bandos en el ajetreo de las internas coloradas,” *Última Hora*, 2022.

_ , “Aduanas recibe reportes de EEUU sobre facturas falsas en importaciones,” *Última Hora*, 2023.

_ , “Santiago Peña y Arnoldo Wiens sellan el tradicional “abrazo republicano” en la ANR,” *Última Hora*, 2023.

_ , “¿Nepotismo o meritocracia?,” *Última Hora*, 2024.

Appendix

A Model

This appendix formalizes the framework of Section 2. A hierarchical bureaucracy comprises a principal, a supervisor, and an agent. The agent chooses between exerting effort to perform her duties effectively—such as a rigorous inspection or diligent assistance—and colluding with the citizen (e.g., a taxpayer, an importer, or a program beneficiary) subject to government oversight. Effort costs e and detects the citizen's violation with probability p_F. Patronage agents have lower human capital (Colonnelli et al., 2020; Dessein and Garicano, 2023; Moreira and Pérez, 2024) and thus a lower detection probability ($\underline{p}_F$) than non-patronage agents ($\overline{p}_F$), so $p_F \in \{\underline{p}_F, \overline{p}_F\}$. When the agent detects a violation of value ϕ, she imposes a fine equal to ϕ and keeps a share α of it as a performance bonus. Collusion yields an informal payoff b but exposes the agent to a sanction μ from the supervisor or the principal. The supervisor observes the agent's conduct and can either sanction her or be bought into the collusion. Being bought in requires bargaining, whose cost is lower within patronage pairs. The principal audits the agent's work with probability p_R. The model is a sequential game of incomplete information between the agent and the supervisor, with the principal's audit and the outcome of effort modeled as chance moves. I characterize its perfect Bayesian equilibrium and derive Predictions I and II of Section 2.

A.1 Setup

The primitives are:

- $e \in (0,1)$: the agent's effort cost.
- $b \in (0,1)$: the informal payoff from collusion, received by each colluding official (the agent, and the supervisor if she is bought in).

- $\mu \in (0,1)$: the sanction if caught, for the agent and for a bought-in supervisor caught by the principal.

- $\phi \in (0,1)$: the value of the violation attempted by the citizen.

- $\alpha \in (0,1)$: the share of ϕ paid to the agent as a bonus for a detected violation.

- $p_F \in \{\underline{p}_F, \overline{p}_F\}$, with $0 < \underline{p}_F < \overline{p}_F < 1$: the detection probability of a patronage and of a non-patronage agent, conditional on effort.

- $c \in \mathbb{R}_+$: the bargaining (collusion) cost that the supervisor bears to be bought in. It equals $\underline{c}$ when both the agent and the supervisor are patronage hires (a *patronage pair*) and $\overline{c} > \underline{c}$ otherwise.

- $w \in \mathbb{R}$: the supervisor's payoff from sanctioning the agent (a career or reputational return, net of the cost of confronting the agent). It is the supervisor's private information, drawn from a continuously differentiable distribution function F with density $f > 0$ on $\mathbb{R}$.

- $p_R \in (0,1)$: the probability that the principal audits the shipment and uncovers the collusion.

- $\delta \in (0,1)$: the masking factor. When misconduct is detected, a sanction follows only if the official's plausible deniability fails, which happens with probability $1-\delta$, so a detected official expects a sanction of $(1-\delta)\mu$.

Masking thus scales the sanction conditional on detection rather than the probability of detection: on a high-workload day, an official whose cleared shipment is later found to be fraudulent can claim an honest mistake. The supervisor's sanction probability $p_S(c)$ is derived in Section A.3, and I write $\underline{p}_S \equiv p_S(\underline{c})$ and $\overline{p}_S \equiv p_S(\overline{c})$. In this section I take p_R as given. Section A.6 lets the principal choose a dyad-specific audit intensity.

Information and conventions. Nature draws w before Period 1 and only the supervisor observes it. F and the other primitives are common knowledge, and the citizen's violation ϕ and the bribe b are exogenous. An indifferent supervisor sanctions and an indifferent agent colludes. Supervisor indifference has probability zero under a continuous F, and the agent's convention only fixes which side of the cutoff c^* is included. Sanctioning by the supervisor and detection by the principal both forfeit the agent's bribe, and the sanction μ is imposed only when plausible deniability fails.

A.2 Timing

Period 0: Patronage versus non-patronage hires and their postings are predetermined by the organization's human resources division. A citizen and the agent interact during an inspection or service provision on a day with masking factor δ, and the citizen attempts a violation worth ϕ.

Period 1: The agent chooses between honest effort and collusion. Collusion yields the bribe b. No effort is exerted, so the violation goes undetected locally. Honest effort costs e and detects the violation with probability p_F. Upon detection the agent levies a fine equal to ϕ, keeps $\alpha\phi$, and the principal collects the remaining $(1-\alpha)\phi$ as a penalty fee. The expected payoff from honest effort is

$$\Pi_A^H = \underbrace{p_F(\alpha\phi - e)}_{\text{successful inspection}} + \underbrace{(1-p_F)(0-e)}_{\text{unsuccessful inspection}} = p_F\alpha\phi - e. \tag{A.1}$$

Period 2: The supervisor observes the agent's conduct. If the agent colluded, the supervisor either sanctions her—the agent forfeits b and expects a sanction of $(1-\delta)\mu$—or is bought in, receiving b and bearing the bargaining cost c. Sanctioning disciplines the agent. It does not record a fraud finding or recover the evaded duties, which only the agent's inspection (Period 1) or the principal's audit (Period 3) can do.[69] The supervisor's choice depends on her private sanctioning

[69] If a sanction also corrected the shipment's assessment with probability ρ, the agent's and the supervisor's payoffs, and with them Lemmas 1 and 2, Proposition 1, and the behavioral content of Corollaries 1 and 2, would be unchanged, while local detection would gain a term equal to the probability of collusion times $p_S(c)\,\rho$, and the detection interaction of Section A.5 would remain nonpositive provided $\rho\, p_S(\underline{c}) \leq \underline{p}_F$, that is, if a patronage supervisor's intervention produces a recorded detection less often than the agent's own effort does. The audit extension of Section A.6 would

payoff w, so from the agent's viewpoint the supervisor sanctions with a probability $p_S(c)$ derived below.

Period 3: The principal audits the shipment and uncovers the collusion with probability p_R. If the supervisor was bought in, each official caught expects a sanction of $(1-\delta)\mu$ and forfeits the bribe. After supervisory discipline, an audit may still recover the duties but has no further effect on the agent's payoff, including when plausible deniability spared her the supervisor's sanction.

Figure A1 shows the game in extensive form: Nature draws the supervisor's sanctioning payoff w at the root, the agent chooses without observing it, and the supervisor, who observes both the agent's conduct and her own w, chooses between sanctioning the agent and being bought in. Figure A2 displays the same game from the agent's viewpoint once the supervisor's choice is summarized by the sanction probability $p_S(c)$ derived below: the agent's choice at the root, followed by the supervisor's sanction decision, the principal's audit, and the outcome of effort, the last three as chance moves.

A.3 The supervisor's sanction decision

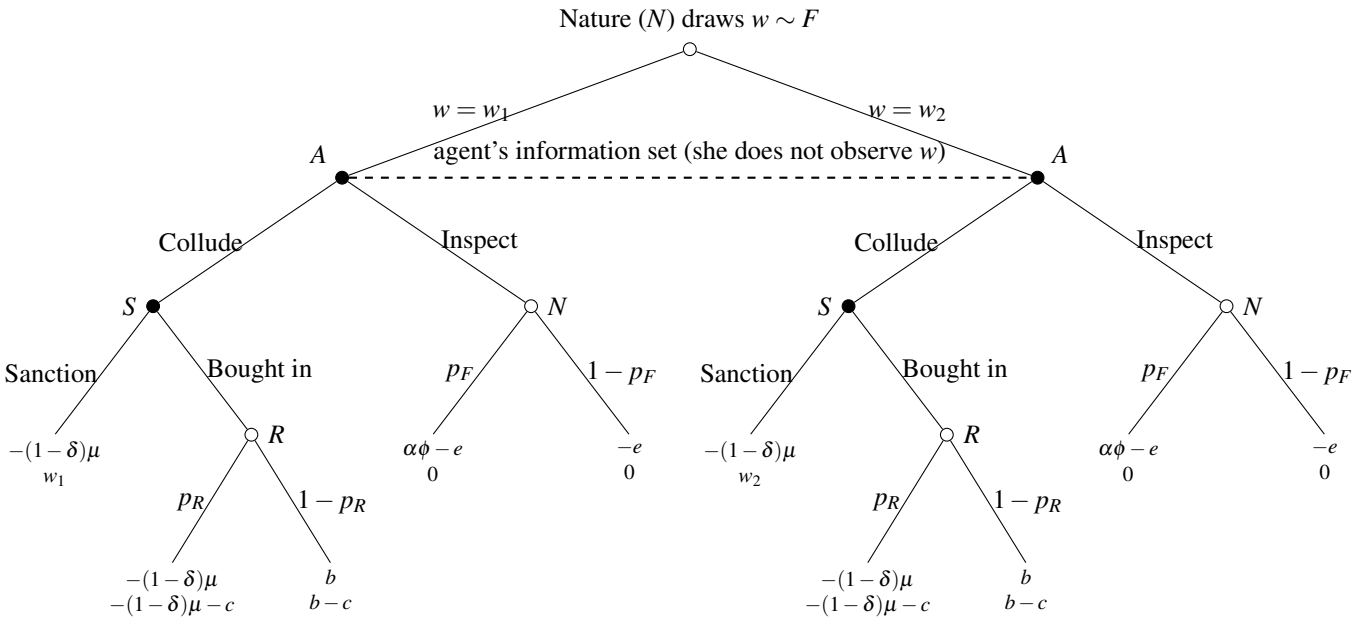
Consider a supervisor who has observed collusion. If she is bought in, she receives b, bears the bargaining cost c, and is caught by the principal with probability p_R, in which case she forfeits b and expects the sanction $(1-\delta)\mu$:

$$\Pi_S^C(c) = \underbrace{(1-p_R)\,b}_{\text{supervisor not caught}} - \underbrace{p_R(1-\delta)\,\mu}_{\text{supervisor caught by principal}} - c. \tag{A.2}$$

If she sanctions the agent, she receives w. Shared political loyalties lower the cost of reaching and enforcing an illicit agreement, because trust emerges from shared expectations of loyalty (Gambetta, 1988). This is what $\underline{c} < \overline{c}$ captures. A patronage supervisor's payoff from being bought in is

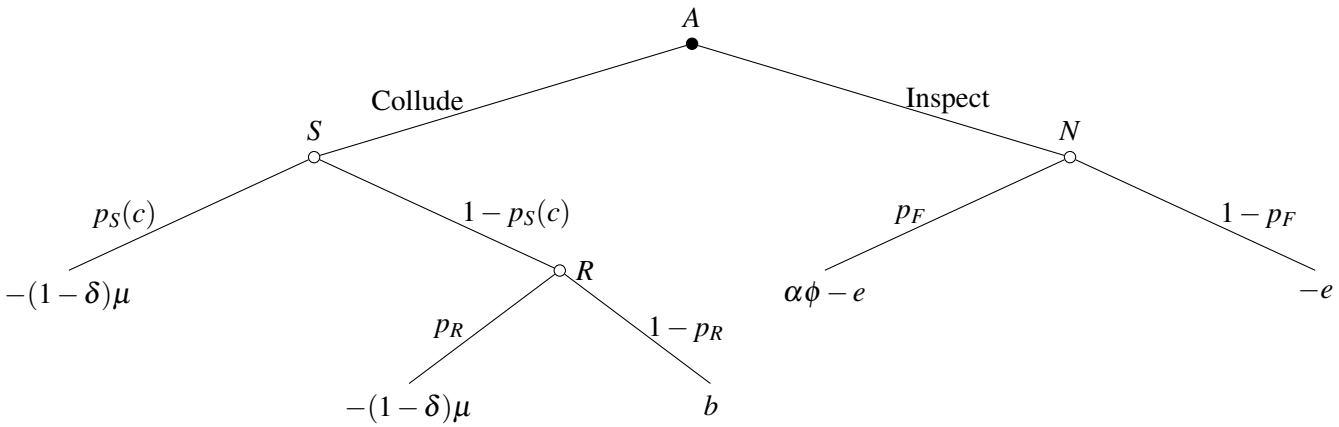
gain a third revenue channel and the stock of cases left for audits would shrink by the same term, so Proposition 2 would have to be restated.

Figure A1: Extensive form of the ineffective-oversight game



Notes: Filled nodes are decision nodes of the two strategic players, the agent (A) and the supervisor (S). Hollow nodes are chance moves: Nature (N) draws the supervisor's private sanctioning payoff w from F at the root (a continuum, shown here by two representative draws, $w_1 < \Pi_S^C(c) \leq w_2$) and decides whether honest effort detects the violation, with probability p_F, and the principal's audit (R) uncovers the collusion with probability p_R. The dashed line joins the agent's decision nodes into one information set, because she knows F but not the realized w. The supervisor's nodes are singletons, since she observes both the agent's conduct and her own w. Each end node lists the agent's payoff above the supervisor's. The supervisor's payoff after honest effort is normalized to zero, and after a sanction the collusion ends, so the principal's audit no longer affects either official's payoff. In equilibrium the supervisor sanctions if and only if $w \geq \Pi_S^C(c)$ (Lemma 1): the type w_2 sanctions and the type w_1 is bought in. Integrating the agent's payoff over F turns the supervisor's node into a chance move with probability $p_S(c)$, which gives the decision tree in Figure A2.

Figure A2: Decision tree of the ineffective-oversight game



Notes: The filled node is the agent's (A) decision. Hollow nodes are chance moves from the agent's viewpoint: the supervisor (S) sanctions with the probability $p_S(c)$ derived in Lemma 1, the principal (R) uncovers the collusion with probability p_R, and Nature (N) determines whether effort detects the violation. The agent's expected payoff is written below each terminal node. The entry $-(1-\delta)\mu$ is the expected sanction of a detected agent, since plausible deniability succeeds with probability δ.

therefore higher with a patronage agent than with a non-patronage agent, $\Pi_S^C(\underline{c}) > \Pi_S^C(\overline{c})$.

Lemma 1 (Supervisor's sanction probability)**.** *A supervisor who observes collusion sanctions the agent if and only if $w \geq \Pi_S^C(c)$. From the agent's viewpoint, the probability of being sanctioned by the supervisor is*

$$p_S(c) \;=\; 1 - F\big((1-p_R)\,b - p_R(1-\delta)\,\mu - c\big), \tag{A.3}$$

which is strictly increasing in c, strictly decreasing in δ, and strictly increasing in p_R. In particular, $\underline{p}_S = p_S(\underline{c}) < p_S(\overline{c}) = \overline{p}_S$.

Proof. The supervisor compares w with $\Pi_S^C(c)$, so she sanctions with probability $\Pr\{w \geq \Pi_S^C(c)\} = 1 - F(\Pi_S^C(c))$, which is (A.3). Since $f > 0$, $\partial p_S/\partial c = f(\Pi_S^C(c)) > 0$, $\partial p_S/\partial \delta = -f(\Pi_S^C(c))\,p_R\,\mu < 0$, and $\partial p_S/\partial p_R = f(\Pi_S^C(c))\,[b + (1-\delta)\mu] > 0$. The last claim follows from $\underline{c} < \overline{c}$ and monotonicity in c. □

The lemma delivers the complementarity between patronage ties and collusion described in

Section 2: because patronage pairs bargain at lower cost, their supervisors are more easily bought in and sanction less often. It also shows that masking makes supervisors more lenient, since a bought-in supervisor expects a smaller sanction from the principal, and that principal oversight makes them stricter. I write $p_S(c)$ for $p_S(c;\delta,p_R)$ when the dependence on (δ,p_R) is not needed.

A.4 The agent's collusion decision

Anticipating the supervisor's behavior, the agent's expected payoff from collusion is

$$\begin{aligned}\Pi_A(c) &= \underbrace{[1-p_S(c)][1-p_R]\,b}_{\text{agent not caught}} - \underbrace{p_S(c)\,[1-\delta]\,\mu}_{\text{agent sanctioned by supervisor}} - \underbrace{[1-p_S(c)]\,p_R\,[1-\delta]\,\mu}_{\text{agent caught by principal}} \\ &= [1-p_S(c)][1-p_R]\,b - \mu[1-\delta]\{p_S(c)+[1-p_S(c)]\,p_R\}, \end{aligned} \tag{A.4}$$

and she colludes if and only if $\Pi_A(c) \geq \Pi_A^H$. Let

$$\Delta(c) \equiv \Pi_A(c) - \Pi_A^H \tag{A.5}$$

denote the gain from collusion, written $\Delta(c;\, p_F,\delta,p_R)$ when the other arguments matter. Since Π_A depends on c only through $p_S(c)$, with

$$\frac{\partial \Pi_A}{\partial p_S} = -(1-p_R)\,[b+\mu(1-\delta)] < 0, \tag{A.6}$$

Lemma 1 yields the following monotonicity properties.

Lemma 2 (Monotonicity of the gain from collusion)**.** Δ *is strictly decreasing in c, strictly decreas-*

ing in p_F, strictly increasing in δ, and strictly decreasing in p_R:

$$\frac{\partial \Delta}{\partial c} = -(1-p_R)\left[b+\mu(1-\delta)\right]\frac{\partial p_S}{\partial c} \;<\; 0, \tag{A.7}$$

$$\frac{\partial \Delta}{\partial p_F} = -\alpha\phi \;<\; 0, \tag{A.8}$$

$$\frac{\partial \Delta}{\partial \delta} = \mu\left\{p_S(c)+\left[1-p_S(c)\right]p_R\right\} \;-\; (1-p_R)\left[b+\mu(1-\delta)\right]\frac{\partial p_S}{\partial \delta} \;>\; 0, \tag{A.9}$$

$$\frac{\partial \Delta}{\partial p_R} = -\left[1-p_S(c)\right]\left[b+\mu(1-\delta)\right] \;-\; (1-p_R)\left[b+\mu(1-\delta)\right]\frac{\partial p_S}{\partial p_R} \;<\; 0. \tag{A.10}$$

Proof. Differentiate (A.4) and (A.1), using (A.6) for the terms that operate through p_S. In (A.7), $\partial p_S/\partial c > 0$ by Lemma 1. In (A.9), the first term is the direct effect of a smaller expected sanction and is positive. The second is positive because $\partial p_S/\partial \delta < 0$ (a more lenient supervisor). In (A.10), the first term combines the direct effect of a more likely principal sanction with the loss of the bribe, and the second is negative because $\partial p_S/\partial p_R > 0$ (a stricter supervisor). □

Lemma 2 implies that the agent colludes if and only if the bargaining cost is low enough. For the cutoff to be interior I assume:

Assumption 1 (Interior cutoff)**.** $\Delta(0) > 0 > \lim_{c\to\infty}\Delta(c)$.

The first inequality says that collusion pays when the supervisor is as lenient as possible. The second says that honest effort pays when the supervisor sanctions for sure ($p_S \to 1$), that is, $-\mu(1-\delta) < p_F\alpha\phi - e$, which holds whenever honest effort has a non-negative net payoff.

Proposition 1 (Collusion cutoff and comparative statics)**.** *Under Assumption 1 there is a unique cutoff $c^* \in (0,\infty)$ with $\Delta(c^*) = 0$, and the agent colludes if and only if $c \le c^*$. Writing $q \equiv p_S(c^*)$ for the supervisor's sanction probability at which the agent is indifferent,*

$$q \;=\; \frac{(1-p_R)b - \mu(1-\delta)p_R - p_F\alpha\phi + e}{(1-p_R)\left[b+\mu(1-\delta)\right]}, \tag{A.11}$$

and, inverting (A.3) *in c,*

$$c^* \;=\; (1-p_R)\,b - p_R(1-\delta)\,\mu - F^{-1}(1-q). \tag{A.12}$$

The cutoff satisfies:

1. $\dfrac{\partial c^*}{\partial \delta} > 0$*: a higher masking factor lowers the expected sanction, so more agent–supervisor pairs collude.*

2. $\dfrac{\partial c^*}{\partial p_R} < 0$*: more intense principal oversight raises the expected cost of collusion, so fewer pairs collude.*

3. $\dfrac{\partial c^*}{\partial p_F} < 0$*: an agent with higher detection ability has a better outside option, so fewer pairs collude.*

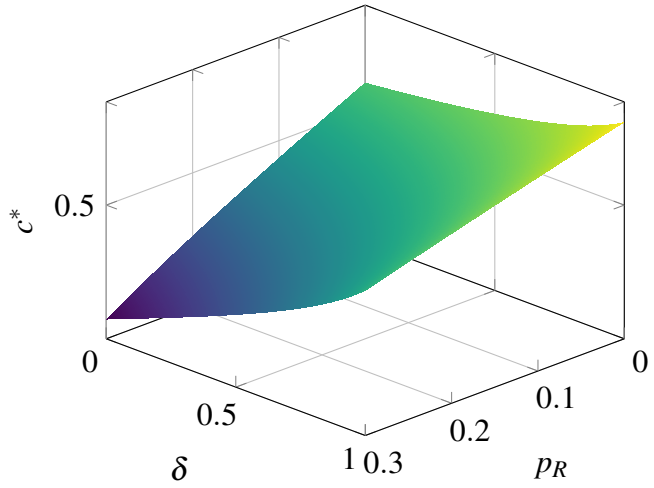
Proof. Δ is continuous and strictly decreasing in c (Lemma 2), so under Assumption 1 it has a unique root $c^* \in (0,\infty)$, with $\Delta(c) \geq 0$ if and only if $c \leq c^*$. Setting (A.5) to zero and solving for $p_S(c^*)$ gives (A.11). Substituting q into (A.3) and solving for c gives (A.12). Since c^* is defined implicitly by $\Delta(c^*; p_F, \delta, p_R) = 0$ with $\partial\Delta/\partial c < 0$, the implicit function theorem gives, for $x \in \{\delta, p_R, p_F\}$,

$$\frac{\partial c^*}{\partial x} = -\frac{\partial\Delta/\partial x}{\partial\Delta/\partial c},$$

whose sign is that of $\partial\Delta/\partial x$: positive for δ and negative for p_R and p_F by Lemma 2. □

Proposition 1 characterizes the perfect Bayesian equilibrium of the game: in Period 2 the supervisor sanctions if and only if $w \geq \Pi_S^C(c)$ (Lemma 1), and in Period 1 the agent, who knows F but not w, colludes if and only if $c \leq c^*$. More agent–supervisor pairs collude when principal oversight is weak (p_R low) or when misconduct is easier to mask (δ high), and patronage pairs are the first to do so because of their lower bargaining cost. Figure A3 plots the cutoff for a parameterized example.

Figure A3: Collusion cutoff c^* as a function of the principal's detection probability p_R and the masking factor δ



Notes: The surface plots c^* from equations (A.11)–(A.12) for a logistic F with location 0 and scale 0.1, so that $F^{-1}(1-q) = -0.1\ln[q/(1-q)]$, with $b = 0.6$, $\mu = 0.5$, and $p_F\alpha\phi - e = 0.1$, over $p_R \in [0, 0.3]$ and $\delta \in [0,1]$. Assumption 1 holds throughout this range. Consistent with Proposition 1, the cutoff is decreasing in p_R and increasing in δ. Lighter colors indicate a higher cutoff, that is, a larger set of agent–supervisor pairs that collude.

A.5 Predictions

There are four types of agent–supervisor dyads. Only the patronage pair (*PP*) has the low bargaining cost $\underline{c}$. The other three—a patronage agent with a non-patronage supervisor (*PN*), a non-patronage agent with a patronage supervisor (*NP*), and a non-patronage pair (*NN*)—have $\overline{c}$. Detection ability is $\underline{p}_F$ in *PP* and *PN* and $\overline{p}_F$ in *NP* and *NN*. Let $\Delta_{PP} \equiv \Delta(\underline{c}; \underline{p}_F)$, $\Delta_{PN} \equiv \Delta(\overline{c}; \underline{p}_F)$, and $\Delta_{NP} = \Delta_{NN} \equiv \Delta(\overline{c}; \overline{p}_F)$ denote the gains from collusion, all evaluated at the same (δ, p_R).

Corollary 1 (Prediction I: collusion is most attractive within patronage pairs)**.** *For every* (δ, p_R)*,*

$$\Delta_{PP} > \Delta_{PN} > \Delta_{NN} = \Delta_{NP},$$

and the gap between the two homogeneous dyads decomposes into two positive channels,

$$\Delta_{PP} - \Delta_{NN} \;=\; \underbrace{\Pi_A(\underline{c}) - \Pi_A(\overline{c})}_{\text{weaker oversight: } \underline{p}_S < \overline{p}_S} \;+\; \underbrace{(\overline{p}_F - \underline{p}_F)\,\alpha\phi}_{\text{lower human capital}} \;>\; 0.$$

Hence, whenever a dyad with at least one non-patronage member colludes, so does the patronage pair. Moreover, writing $\ell \equiv \underline{p}_F \alpha\phi - \Pi_A(\underline{c})$ *and* $u \equiv \underline{p}_F \alpha\phi - \Pi_A(\overline{c})$*, the set of effort costs for which the patronage pair colludes while every other dyad exerts effort is* $\{e : \Delta_{PN} < 0 \le \Delta_{PP}\} = [\ell, u) \cap (0,1)$*, where* $u - \ell = \Pi_A(\underline{c}) - \Pi_A(\overline{c}) > 0$*. It is non-empty whenever* $[\ell, u)$ *meets the admissible range of e.*

Proof. Δ is strictly decreasing in c and in p_F (Lemma 2). *PP* and *PN* differ only in c, *PN* and *NN* only in p_F, and *NP* and *NN* in neither. The decomposition follows from $\Pi_A^H = p_F \alpha\phi - e$. Since $\Delta = \Pi_A(c) - p_F \alpha\phi + e$ increases one-for-one in e, $\Delta_{PP} \ge 0$ if and only if $e \ge \ell$ and $\Delta_{PN} < 0$ if and only if $e < u$. Intersecting with $e \in (0,1)$ gives the stated set. □

The empirical specifications estimate the patronage-pair effect on fraud detection net of the inspector's and the administrator's own types. In the model, let H_{ij} be the probability that dyad ij's agent exerts effort (an indicator in the binary model, a share of shipments under Remark 1) and $D_{ij} = p_{F_i} H_{ij}$ her detection probability. Because $\Delta_{PP} > \Delta_{PN}$ and $H_{NP} = H_{NN}$, the detection interaction is

$$D_{PP} - D_{PN} - D_{NP} + D_{NN} \;=\; \underline{p}_F\,(H_{PP} - H_{PN}) \;=\; -\,\underline{p}_F \Pr\{\Delta_{PN} < 0 \le \Delta_{PP}\} \;\le\; 0.$$

It is nonzero only because $\underline{c} < \overline{c}$: with equal bargaining costs, $H_{PP} = H_{PN}$. Its size, however, also depends on the patronage agent's detection ability $\underline{p}_F$, which scales the detection lost per shipment, and on the share of shipments on which pairing with a patronage supervisor flips the agent's choice. Inspector fixed effects absorb the level difference in detection between agent types, not these two factors. The model also implies that a non-patronage agent behaves identically under either type

of supervisor ($D_{NP} = D_{NN}$), since only the both-patronage dyad enjoys the low bargaining cost.

Corollary 2 (Prediction II: masking lowers detection within patronage pairs)**.** *Suppose that (a) the patronage pair is marginal, $\Delta_{PP}|_{\delta=0} < 0 < \Delta_{PP}|_{\delta=1}$, and (b) dyads with bargaining cost $\overline{c}$ are not: $\Delta(\overline{c}; \underline{p}_F)|_{\delta=1} \leq 0$, that is,*

$$[1 - p_S(\overline{c})|_{\delta=1}]\,(1 - p_R)\,b \;\leq\; \underline{p}_F\,\alpha\phi - e,$$

so that even with no expected sanction, the bribe a patronage agent can keep under a strict supervisor is worth no more than her net payoff from honest effort. Then there is a unique $\delta^*_{PP} \in (0,1)$ such that the patronage pair colludes if and only if $\delta \geq \delta^*_{PP}$, while every other dyad exerts effort for all $\delta \in (0,1)$. Fraud detection therefore falls with the masking factor within patronage pairs, from $\underline{p}_F$ to 0 as δ crosses δ^*_{PP}, and is invariant to δ for patronage agents supervised by non-patronage supervisors and for non-patronage agents.[70]

Proof. Δ_{PP} is continuous and strictly increasing in δ (Lemma 2), so (a) gives a unique root $\delta^*_{PP} \in (0,1)$ with $\Delta_{PP} \geq 0$ if and only if $\delta \geq \delta^*_{PP}$. For the other dyads, the same monotonicity gives $\Delta(\overline{c}; p_F) < \Delta(\overline{c}; p_F)|_{\delta=1} \leq \Delta(\overline{c}; \underline{p}_F)|_{\delta=1} \leq 0$ for every $\delta < 1$ and $p_F \in \{\underline{p}_F, \overline{p}_F\}$, where the middle inequality uses that Δ is decreasing in p_F. At $\delta = 1$ the expected sanction vanishes and (A.4) reduces to $[1 - p_S(\overline{c})](1 - p_R)\,b$, which gives the displayed form of (b). □

Remark 1 (From thresholds to frequencies). Predictions I and II are stated in the paper as differences in the frequency of collusion. They follow from the threshold results once shipments are heterogeneous. Suppose the bribe a shipment can generate, b, is drawn for each shipment, independently of w, from a distribution G on $(0,1)$ with continuous density g, common to the dyads

[70] Prediction II is a statement about which dyads cross the collusion margin as δ rises, not about the slope of the collusion payoff in δ. That slope is not ranked across dyads in general: by (A.9) and Lemma 1, $\partial^2\Delta/\partial c\,\partial\delta = \mu(1 - p_R)\,f(\Pi^C_S(c)) - \mu\,p_R(1 - p_R)\,[b + \mu(1 - \delta)]\,f'(\Pi^C_S(c))$ when f is differentiable, and the second term can dominate. Masking changes behavior only where it flips the sign of Δ, and conditions (a) and (b) say that this happens for the patronage pair and for no other dyad.

being compared, and is observed by both officials before they act. Δ is strictly increasing in b, directly and because a larger bribe makes the supervisor easier to buy in:

$$\frac{\partial \Delta}{\partial b} = (1-p_R)\Big\{1 - p_S(c) + (1-p_R)\left[b + \mu(1-\delta)\right] f(\Pi_S^C(c))\Big\} > 0.$$

Dyad ij therefore colludes on the shipments with $b \geq b_{ij}^*$, where b_{ij}^* is the root of $\Delta = 0$ in b when one exists in $(0,1)$, $b_{ij}^* = 0$ when $\Delta \geq 0$ at $b = 0$ (collusion on every shipment), and $b_{ij}^* = 1$ when $\Delta \leq 0$ at $b = 1$ (collusion on none), with the values of Δ at the endpoints understood as limits. Its collusion frequency is $1 - G(b_{ij}^*)$ and its honesty probability is $H_{ij} = G(b_{ij}^*)$. By Lemma 2 and the implicit function theorem, an interior b_{ij}^* is increasing in c and in p_F, decreasing in δ, and increasing in p_R. The frequency ordering across dyads is therefore the weak version of Corollary 1, and it is strict between the patronage pair and the mixed dyad if and only if $\Pr_G\{\Delta_{PN} < 0 \leq \Delta_{PP}\} > 0$, that is, if some shipments lie between the two thresholds. At an interior threshold, the effect of δ on dyad ij's collusion frequency is $-g(b_{ij}^*)\,\partial b_{ij}^*/\partial \delta \geq 0$. It is zero while the threshold stays at a corner, and at a transition between a corner and the interior the frequency need not be differentiable. This is the smooth counterpart of condition (b) in Corollary 2, which keeps the other dyads honest when it holds for every feasible bribe. Where thresholds are interior, H_{ij} is twice continuously differentiable in the model's parameters, including the audit intensity of Section A.6, whenever F and G are. Its curvature in audit intensity is not pinned down by this construction.

A.6 Principal's audit allocation problem under scarce capacity

In the baseline model, I took the principal's monitoring technology as given. Here I extend the setup to clarify how a higher-level principal (e.g., a post-clearance audit unit) would allocate scarce audits endogenously across agent–supervisor dyads. None of the results in Sections A.1–A.5 depends on this extension. The principal commits to the audit rule before Period 1 and both officials observe it, so the honesty response defined below is anticipatory. This extension also resolves

a timing issue in the customs mapping: local inspector detection generates revenue immediately at the port, while principal audits are re-inspections that primarily target violations that were not detected (or were concealed) locally.

Two distinct detection channels. Consider a dyad ij (agent i, supervisor j). Let r_{ij} denote an observable risk index (composed of, for example, patronage pair indicator, a higher masking factor δ, etc.). Let $H_{ij}(a_{ij}; r_{ij}) \in [0,1]$ denote the probability that the agent behaves honestly (exerts effort) in dyad ij when the principal assigns audit intensity a_{ij}. Treating r_{ij} as a continuous index, assume H_{ij} is twice continuously differentiable in (a,r). The discrete case is covered at the end of the proof of Proposition 2. In the binary version of the model, $H_{ij}(a_{ij}; r_{ij}) = \mathbb{1}\{\Pi_A^H > \Pi_A(c_{ij}, a_{ij}, \delta)\}$. With heterogeneity in bribes across shipments, $H_{ij}(\cdot)$ is the share of shipments on which the agent exerts effort, which is smooth where the bribe threshold of Remark 1 is interior. The shape restrictions on H_{ij} imposed in Proposition 2 are additional assumptions on this response, not implications of the baseline.

Conditional on exerting effort, the violation is detected locally with probability p_{F_i}. Thus, the unconditional probability of local detection is $H_{ij}(a_{ij}; r_{ij})\, p_{F_i}$, generating net revenue to the principal equal to $R^L(\phi) = (2-\alpha)\phi$, since the principal recoups the evaded amount ϕ and applies a penalty fine of ϕ, while the detecting official receives a bonus share $\alpha\phi$ (as defined in the baseline model).

Let $a_{ij} \in [0,1]$ denote the probability that a shipment handled by dyad ij is audited. Audits are drawn with this probability after the officials act, independently of the local inspection outcome and of the officials' private draws, so a missed violation is audited with probability a_{ij}. The principal cannot tell ex ante which cleared shipments hide a violation, and an audit of a shipment whose violation was already detected locally yields nothing further. If the violation is not detected locally (with probability $1 - H_{ij}(a_{ij}; r_{ij}) p_{F_i}$) and the shipment is audited, the audit detects the underlying violation with probability $\pi \in (0,1)$ and yields recoverable revenue $R^A(\phi) = \kappa \cdot 2\phi$, where

$\kappa \in (0,1)$ captures imperfect conversion of assessed amounts into collected revenue (e.g., due to appeals). Any fine-sharing with auditors can be absorbed into κ.

In practice, audit intensity is constrained not only by a hard staffing cap, but also by increasing marginal administrative and political costs of intensifying audits on a given dyad (case complexity, legal follow-up, and disruption). A convex cost is a parsimonious reduced form for these increasing marginal costs and allows interior audit intensities (smooth targeting) rather than knife-edge cutoffs, although corner solutions remain possible. Accordingly, I let the per-dyad cost of audit intensity be quadratic:

$$C(a_{ij}) = \frac{\lambda}{2} a_{ij}^2, \qquad \lambda > 0.$$

Principal's objective and audit-capacity constraint. Let n_{ij} be the number of shipments handled by dyad ij, n the total number of shipments, and $s_{ij} = n_{ij}/n$ dyad ij's share of them. The principal chooses $\{a_{ij}\}$ to maximize expected net revenue per shipment:

$$U_P = \sum_{ij} s_{ij} \Big[H_{ij}(a_{ij}; r_{ij})\, p_{F_i} R^L(\phi) + (1 - H_{ij}(a_{ij}; r_{ij}) p_{F_i})\, a_{ij}\, \pi R^A(\phi) - \frac{\lambda}{2} a_{ij}^2 \Big], \tag{A.13}$$

subject to a scarce audit-capacity constraint:

$$\sum_{ij} s_{ij}\, a_{ij} \;\leq\; \bar{A}, \qquad \bar{A} \in (0,1), \tag{A.14}$$

where the left-hand side is the share of all shipments audited. The constraint captures that only a small fraction of shipments can be audited (empirically, I find this to be on the order of 1–2% of all shipments per year).

Link to collusion parameters. Audit intensity enters the agent's collusion payoff through the probability of being caught by the principal: the baseline detection probability becomes dyad-specific, $p_R^{ij} = a_{ij}\,\pi$, so that the agent's expected sanction from the principal, conditional on the

supervisor being bought in, is $a_{ij}\,\pi\,(1-\delta_{ij})\,\mu$, and masking (higher δ_{ij}) attenuates the direct sanction component of audit deterrence (the forfeiture of the bribe and the supervisor's response in Lemma 1 are its other components). Thus, dyads with lower collusion costs c_{ij} (e.g., patronage pairs) or a higher masking factor δ are more likely to collude absent audits. Whether they are also more responsive to targeted oversight is hypothesis (ii) below, which the baseline does not imply.

Proposition 2 (Targeted audits under scarce capacity)**.** *Assume $H_{ij}(a;r_{ij}) \in [0,1]$ is the probability that the local dyad ij behaves honestly (i.e., the agent exerts effort in inspecting), and H_{ij} is weakly increasing in audit intensity (a). Let r_{ij} be an observable risk index (e.g., patronage-pair indicator, higher masking factor δ, suspect assignments), and let p_{F_i} denote agent i's detection ability conditional on effort. Consider dyads that share the same detection ability p_F and the same response function $H(a;r)$ and differ only in r. Suppose:*

(i) $H_{ij}(a;r_{ij})$ is weakly decreasing in r_{ij}, so the stock of locally-missed cases $1-H_{ij}(a;r_{ij})p_{F_i}$ is weakly increasing in r_{ij}.

(ii) Audits are (weakly) more deterrent for higher-risk dyads, i.e., $\frac{\partial^2 H_{ij}(a;r_{ij})}{\partial a\,\partial r_{ij}} \geq 0$.

(iii) A violation caught locally yields at least as much revenue as one caught by an audit, $R^L(\phi) \geq \pi R^A(\phi)$, i.e., $2-\alpha \geq 2\pi\kappa$.

(iv) $H_{ij}(a;r_{ij})$ is weakly concave in a.

For a discrete risk index, (i) and (ii) are read as $H(a;r_2) \leq H(a;r_1)$ and $H_a(a;r_2) \geq H_a(a;r_1)$ for every $a \in [0,1]$ whenever $r_2 \geq r_1$. Then, at any optimum of (A.13)–(A.14)*, the optimal audit intensity $a^*(r)$ is weakly increasing in r among these dyads, including at the bounds $a^* \in \{0,1\}$. The result does not rank dyads that differ in detection ability or in their response function.*

Proof. Let $L \equiv R^L(\phi) = (2-\alpha)\phi$ and $A \equiv \pi R^A(\phi) = \pi\kappa 2\phi$. Fix any dyad ij and suppress sub-

scripts for ease of notation. Let the per-dyad expected gross revenue be

$$V(a;r) \;\equiv\; LH(a;r)\,p_F \;+\; aA(1-H(a;r)\,p_F),$$

so that the marginal benefit of audit intensity is:

$$V_a(a;r) = A(1-H(a;r)p_F) + H_a(a;r)\,p_F(L-aA) \;\equiv\; MB(a;r),$$

where H_a is the derivative of H with respect to audit intensity a. Consider the principal's Lagrangian (including the quadratic audit cost term and a linear capacity constraint),

$$\mathscr{L} = \sum_{ij} s_{ij}\Big(V_{ij}(a_{ij};r_{ij}) - \frac{\lambda}{2}a_{ij}^2\Big) \;+\; \nu\Big(\bar{A} - \sum_{ij} s_{ij}\,a_{ij}\Big) \;+\; \sum_{ij}\underline{\eta}_{ij}\,a_{ij} \;+\; \sum_{ij}\overline{\eta}_{ij}\,(1-a_{ij}),$$

where $\underline{\eta}_{ij} \geq 0$ and $\overline{\eta}_{ij} \geq 0$ are the Karush–Kuhn–Tucker (KKT) multipliers on the bounds $a_{ij} \geq 0$ and $a_{ij} \leq 1$. Let a_{ij}^* be an interior optimum, so the KKT bound multipliers are zero. Because $s_{ij} > 0$ multiplies both the dyad's marginal revenue and its marginal use of capacity, it drops out of the first-order condition, which is

$$MB_{ij}(a_{ij}^*;r_{ij}) \;=\; \lambda\,a_{ij}^* \;+\; \nu. \tag{A.15}$$

Define the net marginal gain (Γ) from increasing audit intensity for dyad ij as

$$\Gamma(a;r) \equiv MB(a;r) - \lambda a - \nu,$$

so that (A.15) is $\Gamma(a^*(r);r) = 0$.

Step 1: $\Gamma_r(a;r) \geq 0$**.** Differentiate $MB(a;r)$ with respect to the risk index r (holding p_F fixed):

$$\frac{\partial MB(a;r)}{\partial r} = -A\, p_F\, H_r(a;r) \;+\; p_F\,(L - aA)\, H_{ar}(a;r).$$

Assumption (i) gives $H_r(a;r) \leq 0$, so the first term is weakly nonnegative. Assumption (ii) gives $H_{ar}(a;r) \geq 0$, and assumption (iii) gives $L \geq A$, hence $L - aA \geq 0$ for every $a \in [0,1]$: raising honesty shifts cases from "missed, then audited with probability a" to "caught locally" without lowering expected revenue, because an audit recovers less than a local detection once its detection probability π and the conversion friction κ are taken into account. The second term is therefore weakly nonnegative. Hence,

$$\Gamma_r(a;r) = \frac{\partial MB(a;r)}{\partial r} \geq 0.$$

Step 2: $\Gamma_a(a;r) < 0$**.** Differentiating MB once more,

$$\frac{\partial MB(a;r)}{\partial a} = V_{aa}(a;r) = -2A\, p_F\, H_a(a;r) + p_F\,(L - aA)\, H_{aa}(a;r) \;\leq\; 0,$$

because $H_a \geq 0$, $L - aA \geq 0$ by assumption (iii), and $H_{aa} \leq 0$ by assumption (iv). Hence $\Gamma_a(a;r) = MB_a(a;r) - \lambda \leq -\lambda < 0$ for every a: the per-dyad objective is strictly concave in a, so the interior optimum $a^*(r)$ is unique. Recall that λa is the marginal audit cost and ν is the shadow value of scarce audit capacity.

Step 3: Monotone comparative statics. Fix the common multiplier $\nu \geq 0$. Because $\Gamma(a;r)$ is strictly decreasing in a (Step 2), the per-dyad objective is strictly concave on $[0,1]$ and its maximizer is unique: $a^*(r) = 0$ if $\Gamma(0;r) \leq 0$, $a^*(r) = 1$ if $\Gamma(1;r) \geq 0$, and otherwise $a^*(r)$ is the unique root of $\Gamma(a;r) = 0$, that is, of (A.15). Because Γ is weakly increasing in r (Step 1), $a^*(r)$ is weakly increasing in r. If the higher-risk dyad is at zero, so is the lower-risk one. If the lower-risk dyad is

at one, so is the higher-risk one. At an interior root the implicit function theorem gives

$$\frac{da^*(r)}{dr} = -\frac{\Gamma_r(a^*(r);r)}{\Gamma_a(a^*(r);r)} \;\geq\; 0.$$

For a discrete risk index, such as a patronage-pair indicator, the finite-difference reading of (i) and (ii) gives, for $r_2 \geq r_1$,

$$MB(a;r_2) - MB(a;r_1) = -A\,p_F\,[H(a;r_2) - H(a;r_1)] + p_F\,(L - aA)\,[H_a(a;r_2) - H_a(a;r_1)] \;\geq\; 0,$$

so Γ is again weakly increasing in r (the increasing-differences condition of Milgrom and Shannon (1994)) and the bounded-choice argument applies unchanged. The ranking does not extend to dyads with different p_F: a lower p_F raises the stock of missed cases, which increases MB, but lowers the return to deterrence $p_F H_a(L - aA)$, so its net effect on a^* is ambiguous.

Finally, the multiplier satisfies $\nu \geq 0$, $\sum_{ij} s_{ij} a^*_{ij} \leq \bar{A}$, and $\nu\,[\bar{A} - \sum_{ij} s_{ij} a^*_{ij}] = 0$. Because ν is a scalar common to all dyads (the shadow value of the *aggregate* capacity constraint), it shifts Γ uniformly across dyads, so choosing ν to satisfy these conditions preserves the monotone ranking of a^*_{ij} in r_{ij}. □

Therefore, among otherwise comparable dyads, oversight is more effective when audit intensity is targeted at the high-risk ones. Uniform audit rules may underperform, especially under vertical patronage alignment.

Mapping to the empirical setting. In Paraguayan customs, the fine share kept by the detecting agent (α) is institutionally fixed (set to 0.5) and the PCA team's capacity is limited, making (A.14) an empirically relevant constraint. The proposition above motivates targeted audits toward high-risk dyads (e.g., vertically aligned patronage pairs and periods with high masking δ), holding the inspector's detection ability fixed, consistent with the paper's interpretation of post-clearance audits as a closer "ground truth" signal for locally missed fraud.

B Methodology for uncovering *suspect assignments*

Following Chalendard et al. (2023), to determine whether an observed inspection share for a given inspector (with a given broker) is significantly distinct from the expected inspection share based on random assignment I construct standard errors for the expected inspection shares using simulation methods that take five steps.[71] First, I obtain 99% confidence intervals for inspectors' productivities (how many inspections each does in a semester at each port) using the method for constructing confidence intervals for multinomial proportions by Sison and Glaz (1995). Second, I simulate the productivity distribution across inspectors 100 times, drawing from the 99% confidence interval of observed productivities (obtained in the previous step), conservatively assuming the productivities are uniformly distributed. Next, for each productivity simulation, I take the total number of shipments as given and simulate which inspectors are assigned to his shipments 100 times assuming multinomial assignment. Afterward, I test whether the observed number of shipments of a given broker handled by a given inspector is larger than the 99th percentile of the respective simulated multinomial assignment. This allows us to quantify the percentage of times an inspector-broker pair in a given semester-port has a significant excess inspection share. Finally, I generate a binary indicator for inspector-broker pairs for which at least 90% of the productivity simulations reject the null hypothesis of random assignment.

[71] As I estimate excess inspections between inspectors and brokers by port and semester, the five steps to calculate standard errors are repeated for the 31 customs ports and 10 semesters in my sample covering the years 2017 to 2021.

C Additional tables and figures

Figure C1: Image of a physical inspection, which opens shipment containers



Notes: This is a stock photography of a physical inspection, physical inspections in Paraguayan customs are done by a single inspector.

Figure C2: Fraud detection rate for visual inspectors on red channel shipments

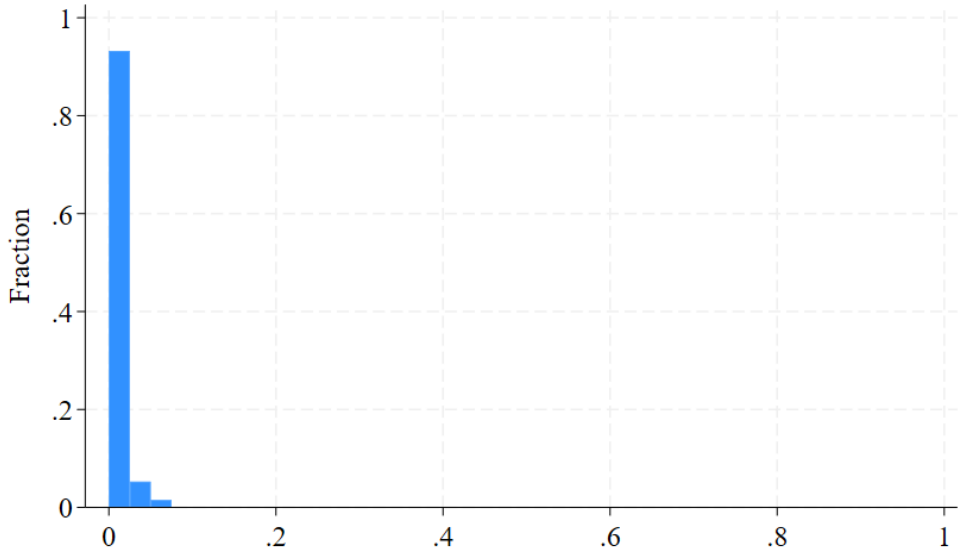
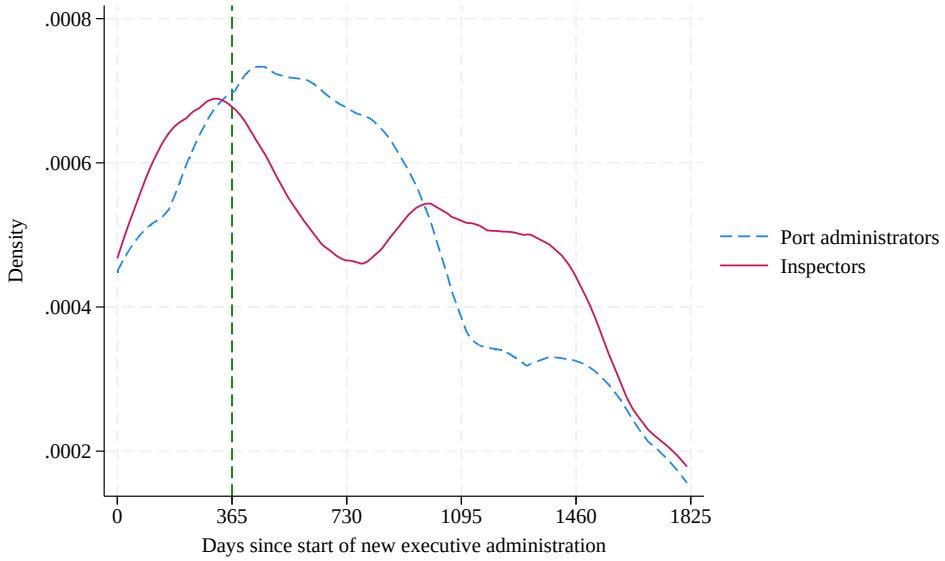
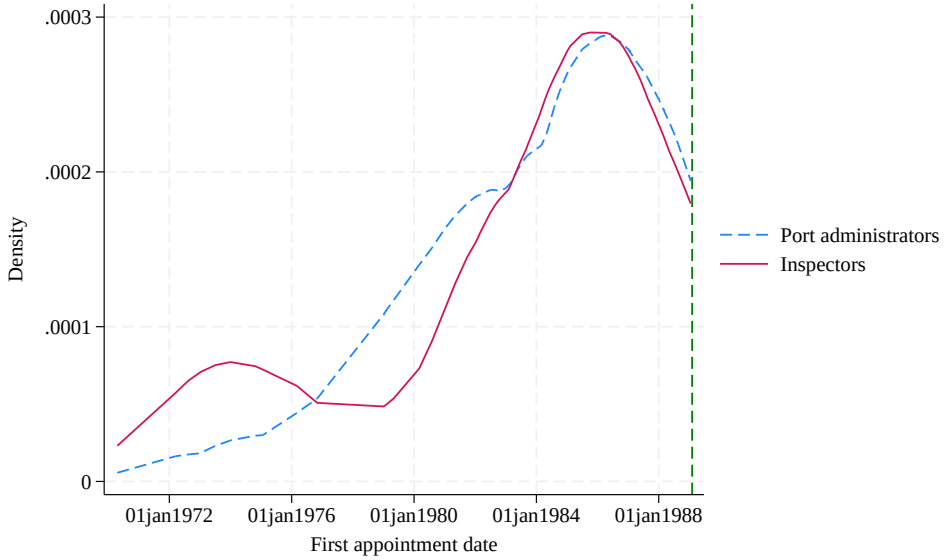


Figure C3: Kernel density plots of first appointment date within executive administrations during democracy for customs officials



Notes: Each plot shows the Epanechnikov kernel density estimates of the first appointment date within executive administrations during democracy for customs officials in the data. A vertical line is drawn at the first year cutoff.

Figure C4: Kernel density plots of first appointment date for customs officials hired during dictatorship



Notes: Each plot shows the Epanechnikov kernel density estimates of the first appointment date during dictatorship for customs officials in the data. A vertical line is drawn at the end of the dictatorship

Table C1: Comparison of patronage and non-patronage officials (excluding dictatorship hires)

(a) Valuation inspectors

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Education level	28	3.43	62	4.47	90	-1.04***
		(0.32)		(0.19)		
College degree	28	0.43	62	0.76	90	-0.33***
		(0.10)		(0.05)		
Age	28	48.79	62	45.08	90	3.71*
		(1.80)		(1.00)		
Political engagement index	28	3.57	62	3.76	90	-0.19
		(0.22)		(0.13)		

(b) Port administrators

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Education level	14	4.57	32	4.53	46	0.04
		(0.40)		(0.28)		
College degree	14	0.79	32	0.75	46	0.04
		(0.11)		(0.08)		
Age	14	51.14	32	50.84	46	0.30
		(1.57)		(1.51)		
Political engagement index	14	5.14	32	4.12	46	1.02**
		(0.52)		(0.19)		

Notes: The education level variable goes from no education mentioned (0), primary (1), secondary (2), some professional/vocational/technician (non-academic) training (3), some (academic) college courses (4), college degree (5), master's degree (6), and doctorate (7). The "political engagement index" aggregates indicators for various types of political involvement that the customs official has taken part in, which are: (i) being affiliated with the incumbent party, (ii) having been a poll worker in recent elections (2013 or 2018), (iii) having been a poll watcher or party representative in recent elections (2013 or 2018), (iv) having a leadership role in the incumbent Colorado party since 2006, (v) being a local party headquarter (*seccional*) leader (Dosek, 2019; Bandiera et al., 2025) sometime between 2006 and 2022, (vi) being a local representative to party authorities sometime between 2006 and 2022, and (vii) having voted in the most recent general elections for my data's time period (2008, 2013, or 2018), (viii) having been a political candidate, and (ix) being a politician's relative.

Table C2: Comparison of patronage and non-patronage valuation inspectors' survey responses - inspector characteristics, effort, and motivation

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Relatives at customs	27	0.370	34	0.176	61	0.194*
		(0.095)		(0.066)		
Has been promoted	28	0.964	34	0.941	62	0.023
		(0.036)		(0.041)		
Has received a bonus	28	0.821	34	0.853	62	-0.032
		(0.074)		(0.062)		
Has been investigated for misconduct	27	0.259	33	0.182	60	0.077
		(0.086)		(0.068)		
Hours worked	27	43.222	34	44.500	61	-1.278
		(3.085)		(1.892)		
Extra hours worked	28	8.143	32	9.156	60	-1.013
		(0.776)		(0.761)		
Overall satisfaction	29	3.724	34	3.882	63	-0.158
		(0.216)		(0.242)		
Pay satisfaction	29	3.103	34	3.176	63	-0.073
		(0.201)		(0.166)		
Stress	29	2.103	33	1.939	62	0.164
		(0.167)		(0.189)		
Overall motivation	29	3.724	34	3.647	63	0.077
		(0.171)		(0.197)		
Intrinsic motivation for learning	29	3.655	34	3.529	63	0.126
		(0.103)		(0.165)		
Intrisic motivation for challenges	29	4.793	34	4.441	63	0.352
		(0.077)		(0.185)		
Extrinsic motivation for income	29	4.276	34	4.147	63	0.129
		(0.192)		(0.194)		
Extrinsic motivation for financial security	29	2.966	34	3.206	63	-0.240
		(0.182)		(0.132)		
Organizational pride	29	4.448	34	4.382	63	0.066
		(0.117)		(0.104)		

Notes: The questions about satisfaction, stress, motivation, and organizational pride were on a scale from 1 to 5, with 5 indicating greater satisfaction/stress/motivation/pride.

Table C3: Comparison of patronage and non-patronage valuation inspectors' survey responses - inspector promotion and institutional environment (I)

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
How easy to obtain a similar outside option	29	2.517	34	2.676	63	-0.159
		(0.190)		(0.162)		
Promotion prospects	28	4.464	32	4.750	60	-0.286
		(0.233)		(0.168)		
Shipment assignment diversity	29	4.310	34	4.471	63	-0.160
		(0.205)		(0.154)		
Discretion in shipment assigned	29	3.103	33	3.091	62	0.013
		(0.255)		(0.251)		
Belief in inspection channel accuracy	29	3.759	34	3.882	63	-0.124
		(0.246)		(0.214)		
Believe IT support is good	29	4.103	34	4.088	63	0.015
		(0.213)		(0.186)		
Believe training is good	29	3.793	34	3.824	63	-0.030
		(0.125)		(0.099)		
Believe management is good	29	4.621	34	4.647	63	-0.026
		(0.126)		(0.157)		
Believes inspectors lack key information	29	3.690	34	3.176	63	0.513
		(0.217)		(0.217)		
Believes it's useful to speak with brokers	29	3.207	34	3.235	63	-0.028
		(0.182)		(0.174)		
Level of frequency speaking with brokers	29	2.828	34	2.941	63	-0.114
		(0.186)		(0.179)		
Knows risky brokers	29	3.207	33	3.242	62	-0.036
		(0.240)		(0.250)		
Knows risky importers	29	3.379	33	3.364	62	0.016
		(0.224)		(0.249)		

Notes: All questions except the promotion prospects one were on a scale from 1 to 5. The promotion prospects question was on a scale from 1 to 6.

Table C4: Comparison of patronage and non-patronage valuation inspectors' survey responses - institutional environment (II)

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Believes colleagues are honest	29	4.828	34	4.676	63	0.151
		(0.071)		(0.156)		
Believes supervisor is honest	29	3.793	34	3.735	63	0.058
		(0.115)		(0.136)		
Believes documentary inspections are accurate	29	4.276	34	4.029	63	0.246
		(0.156)		(0.209)		
Believes physical inspections are accurate	29	4.276	31	3.935	60	0.340
		(0.156)		(0.222)		
Believes brokers pay bribes	29	2.103	33	1.697	62	0.406
		(0.194)		(0.202)		
Believes importers pay bribes	29	1.862	34	1.588	63	0.274
		(0.190)		(0.185)		
Believes brokers pay bribes more than importers	28	1.964	34	1.706	62	0.258
		(0.202)		(0.209)		
Believes they receive threats	28	2.714	34	2.118	62	0.597*
		(0.229)		(0.206)		
Believes inspector misconduct is penalized	28	4.607	34	4.353	62	0.254
		(0.139)		(0.206)		
Believes they can report misconduct freely to supervisors	28	3.500	34	3.559	62	-0.059
		(0.167)		(0.153)		
Believes they can report misconduct freely anonymously online	27	4.556	33	4.636	60	-0.081
		(0.163)		(0.156)		
Knows customs' risk score system	28	0.571	32	0.719	60	-0.147
		(0.095)		(0.081)		

Notes: All questions except the question about knowing customs' risk score system were on a scale from 1 to 5. The question about knowing customs' risk score is either 1 if they know it and 0 if they do not.

Table C5: Comparison of patronage and non-patronage valuation inspectors' survey responses - risk management system

Variable	(1) Patronage N	(1) Patronage Mean/(SE)	(2) Non-patronage N	(2) Non-patronage Mean/(SE)	(1)-(2) Pairwise t-test N	(1)-(2) Pairwise t-test Mean difference
Believes risk score information would help inspectors	27	2.778	30	2.700	57	0.078
		(0.111)		(0.119)		
Believes importers' fraud history would help inspectors	28	2.643	32	2.594	60	0.049
		(0.117)		(0.126)		
Believes brokers' fraud history would help inspectors	27	2.667	33	2.485	60	0.182
		(0.119)		(0.131)		
Believes shipment financial transaction info would help inspectors	27	2.926	34	2.765	61	0.161
		(0.074)		(0.095)		
Believes unit price info would help inspectors	25	2.520	34	2.588	59	-0.068
		(0.154)		(0.120)		
Believes origin country declaration would help inspectors	26	2.731	33	2.788	59	-0.057
		(0.118)		(0.104)		
Would feel supported with info provision	28	2.857	32	2.688	60	0.170
		(0.099)		(0.114)		
Would feel monitored with info provision	28	2.679	31	2.161	59	0.517**
		(0.137)		(0.168)		
Believes they need more time often	29	2.862	32	2.406	61	0.456*
		(0.184)		(0.173)		
Believes trade facilitation is more important than combating fraud	29	1.586	33	1.545	62	0.041
		(0.153)		(0.145)		
Believes irregular fine exemptions occur	24	0.083	32	0.156	56	-0.073
		(0.058)		(0.065)		

Notes: All questions except the question about needing time more often and whether irregular fine exemptions occur were on a scale from 1 to 3. The question about needing time more often was on a scale from 1 to 4 and the question about irregular fine exemptions was either 1 if they believe they occur or 0 if they do not.

Table C6: Comparison of patronage and non-patronage valuation inspectors' survey responses - varied questions

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Extent of time discounting	26	1.423	31	1.484	57	-0.061
		(0.347)		(0.314)		
How much money they would donate hypothetically	25	132,200.000	30	165,000.000	55	-32,800.000
		(23,169.808)		(28,882.441)		
Importance of their political party community	24	1.458	30	1.600	54	-0.142
		(0.269)		(0.247)		
Importance of their union/labor community	24	3.417	30	3.333	54	0.083
		(0.199)		(0.221)		
Prioritizes revenue collection over own income	25	1.600	29	1.690	54	-0.090
		(0.173)		(0.165)		

Notes: The question about the extent of time discounting went from 1 to 10. The question about how much money they would donate was out of 500,000 Paraguayan Guaranis (PYG), which is around $66 at the exchange rate for August 2024. The question about the importance of their community went from 1 to 5. The question about prioritizing revenue collection was on a scale from 1 to 3.

C.1 Tables with patronage hires using a 1.5-year window for election-cycle hires

Table C7: Patronage hires and customs fraud detected

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector × Patronage administrator	-0.0477	-0.0925**	-0.0369**	-0.0366**	0.0008	-0.0002	-0.0013	-0.0013
	(0.0321)	(0.0362)	(0.0151)	(0.0150)	(0.0039)	(0.0055)	(0.0055)	(0.0054)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2603	0.2618	0.5752	0.5766	0.0216	0.0218	0.0316	0.0317

Notes: All regressions include inspector and administrator fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year and a half of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C8: Patronage pairs and suspect assignments

	Binary significant excess inspection share (1)
Patronage inspector	0.0024
× Patronage administrator	(0.0018)
Mean of Dependent Variable	0.0082
Observations	77,955
R^2	0.0394

Notes: The regression includes inspector, administrator, and customs broker fixed effects. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector combinations which are significantly excessive for 90% or more of the simulations. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year and a half of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. The sample covers all possible non-singleton pairings of valuation inspectors and customs brokers at all semesters and ports.

Table C9: Customs fraud detected, patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Significant excess	0.0089	-0.1213**	-0.0847*	-0.0848*	0.0173**	0.0663***	0.0664***	0.0663***
insp. share	(0.0145)	(0.0538)	(0.0471)	(0.0471)	(0.0077)	(0.0252)	(0.0249)	(0.0250)
Patronage inspector ×	-0.0104	-0.0244**	-0.0170**	-0.0170**	-0.0013	-0.0034	-0.0036	-0.0036
Patronage administrator	(0.0094)	(0.0107)	(0.0084)	(0.0084)	(0.0040)	(0.0057)	(0.0057)	(0.0056)
Patronage inspector ×	-0.0167	-0.0143	-0.0134	-0.0133	-0.0141	-0.0189**	-0.0187**	-0.0184*
Significant excess insp. share	(0.0205)	(0.0217)	(0.0172)	(0.0172)	(0.0095)	(0.0095)	(0.0094)	(0.0094)
Patronage administrator ×	0.0139	0.0075	0.0059	0.0061	-0.0053	-0.0083	-0.0080	-0.0077
Significant excess insp. share	(0.0190)	(0.0199)	(0.0172)	(0.0173)	(0.0089)	(0.0089)	(0.0088)	(0.0087)
Patronage inspector × Patronage	-0.0112	-0.0127	-0.0081	-0.0082	0.0224	0.0274**	0.0267**	0.0265**
administrator × Significant excess insp. share	(0.0273)	(0.0274)	(0.0231)	(0.0231)	(0.0140)	(0.0136)	(0.0134)	(0.0134)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,041	292,033	292,033	292,033	294,041	292,033	292,033	292,033
R^2	0.6667	0.6677	0.7118	0.7118	0.0670	0.0677	0.0745	0.0748

Notes: All regressions include inspector, administrator, and customs broker fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C10: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Workload deviations	-0.0004	-0.0842**	-0.0230	-0.0227	-0.0012	0.0126*	0.0097	0.0097
	(0.0079)	(0.0386)	(0.0187)	(0.0185)	(0.0010)	(0.0071)	(0.0069)	(0.0070)
Patronage inspector ×	-0.0571*	-0.1035***	-0.0414***	-0.0412***	0.0044	0.0037	0.0025	0.0025
Patronage administrator	(0.0321)	(0.0362)	(0.0152)	(0.0151)	(0.0038)	(0.0054)	(0.0053)	(0.0053)
Patronage inspector ×	0.0161	0.0222*	0.0109*	0.0112*	-0.0070***	-0.0092***	-0.0088***	-0.0087***
Workload deviations	(0.0123)	(0.0127)	(0.0061)	(0.0061)	(0.0026)	(0.0031)	(0.0031)	(0.0031)
Patronage administrator ×	0.0232	0.0206	0.0099	0.0109*	-0.0016	-0.0015	-0.0009	-0.0008
Workload deviations	(0.0147)	(0.0141)	(0.0065)	(0.0063)	(0.0013)	(0.0014)	(0.0014)	(0.0014)
Patronage inspector × Patronage	-0.0531***	-0.0461***	-0.0194**	-0.0202***	0.0218***	0.0209***	0.0199***	0.0199***
administrator × Workload deviations	(0.0178)	(0.0167)	(0.0076)	(0.0076)	(0.0048)	(0.0040)	(0.0040)	(0.0040)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2613	0.2633	0.5754	0.5768	0.0261	0.0266	0.0361	0.0363

Notes: All regressions include inspector and administrator fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year and a half of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C11: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected		
	(1)	(2)	(3)
Patronage inspector × Patronage administrator	-0.1109***	-0.1043*	-0.0714
	(0.0422)	(0.0555)	(0.0520)
Mean of Dependent Variable	13,511,141	13,505,530	13,505,530
HS2-product FE	X	X	X
Interactions w/ insp. char.		X	X
Evasion risk index			X
Observations	79,709	79,589	79,589
Pseudo R^2	0.1659	0.1662	0.3099

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects. Regressions in columns (2)-(3) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year and a half of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

C.2 Balance tables on shipment values

In this Appendix subsection I examine whether the declared shipment values handled by patronage and non-patronage valuation inspectors differ, and whether the baseline regression specification (1) mitigates these imbalances for patronage pairs. Balance table C12 compares the declared values of shipments handled by patronage and non-patronage valuation inspectors. It shows that patronage inspectors tend to inspect goods with higher value and tax liability, which plausibly lead to higher incentives for tax evasion. This is consistent with the results from Chalendard et al. (2023) who find that the most corrupt inspectors (according to their measure of suspect assignments from excess inspections) are assigned to the most lucrative shipments with highest expected tax yield. On the other hand, Table C13 demonstrates that once we run our baseline regression specification (1) with inspector and administrator fixed effects we can observe that patronage pairs do not have shipments with significantly higher value or tax liability.

Table C12: Balance table of shipment values and taxes paid handled by patronage and non-patronage valuation inspectors

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Total taxes paid on shipment (in PYG)	111445	54,863,108 (410,554)	182635	51,624,434 (315,119)	294080	3,238,673***
Total declared CIF value of shipment (in USD)	111445	64,187 (707)	182635	53,285 (431)	294080	10,901***
Total declared FOB value of shipment (in USD)	111445	60,015 (685)	182635	49,321 (415)	294080	10,695***
Total declared freight cost of shipment (in USD)	111445	3,716 (34)	182635	3,621 (23)	294080	95**

Notes:"Taxes paid" refers to the total taxes paid for the shipment in the local currency, Paraguayan Guaranis (PYG). The US Dollar (USD)-PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. "CIF value" refers to the CIF (cost, insurance, and freight) value for the shipment in US dollars.

Table C13: Shipment values and taxes paid on shipments handled by patronage pairs with same baseline regression specification

	Total taxes paid on shipment (in PYG)	Total declared CIF value of shipment (in USD)	Total declared FOB value of shipment (in USD)	Total declared freight cost of shipment (in USD)
	(1)	(2)	(3)	(4)
Patronage inspector × Patronage administrator	-3,630,680 (3,642,550)	-4,259 (7,222)	-4,465 (6,950)	242 (303)
Mean of Dependent Variable	52,774,075	57,248	53,259	3,606
Observations	292,069	292,069	292,069	292,069
R^2	0.0698	0.0699	0.0688	0.0496

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. The sample covers red channel shipments.

C.3 Tables controlling for high-value ports

In this Appendix subsection, I additionally control for ports that are highlighted by customs officials as "high-value" ports given the value of goods handled at these ports, and hence opportunities for corruption. These are the *Ciudad del Este* ports (*Ciudad del Este*, *Campestre*, and ALGESA), the airports (Pettirossi, CODESA, and Guarani), and the three main river ports at the time period of my data (*Puerto Seguro Fluvial*, TERPORT, and TERPORT Villeta). The latter three are upstream river ports close to the capital. Tables C14 to C17 show the main results are robust to controlling for these high-value ports.

Table C14: Patronage hires and customs fraud detected

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector × Patronage administrator	-0.0724**	-0.1303***	-0.0363**	-0.0363**	0.0036	0.0045	0.0028	0.0028
	(0.0316)	(0.0352)	(0.0155)	(0.0154)	(0.0038)	(0.0056)	(0.0056)	(0.0056)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2628	0.2648	0.5759	0.5773	0.0217	0.0218	0.0316	0.0318

Notes: All regressions include inspector and administrator fixed effects and control for high-value ports. These are the *Ciudad del Este* ports (*Ciudad del Este*, *Campestre*, and ALGESA), the airports (Pettirossi, CODESA, and Guarani), and the three main river ports at the time period of my data (*Puerto Seguro Fluvial*, TERPORT, and TERPORT Villeta). Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C15: Customs fraud detected, patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Significant excess	0.0153	-0.1186**	-0.0848*	-0.0849*	0.0123**	0.0522**	0.0525**	0.0524**
insp. share	(0.0143)	(0.0487)	(0.0440)	(0.0440)	(0.0059)	(0.0238)	(0.0236)	(0.0236)
Patronage inspector ×	-0.0081	-0.0246**	-0.0110	-0.0110	0.0008	0.0006	0.0004	0.0003
Patronage administrator	(0.0097)	(0.0116)	(0.0091)	(0.0091)	(0.0039)	(0.0057)	(0.0057)	(0.0057)
Patronage inspector ×	-0.0295	-0.0435*	-0.0313*	-0.0312*	-0.0081	-0.0127	-0.0129	-0.0128
Significant excess insp. share	(0.0214)	(0.0236)	(0.0185)	(0.0185)	(0.0099)	(0.0107)	(0.0108)	(0.0108)
Patronage administrator ×	0.0031	-0.0097	-0.0062	-0.0060	-0.0028	-0.0051	-0.0049	-0.0046
Significant excess insp. share	(0.0181)	(0.0185)	(0.0171)	(0.0171)	(0.0075)	(0.0071)	(0.0071)	(0.0070)
Patronage inspector × Patronage	-0.0021	0.0128	0.0060	0.0060	0.0279*	0.0304**	0.0297**	0.0297**
administrator × Significant excess insp. share	(0.0280)	(0.0285)	(0.0244)	(0.0244)	(0.0155)	(0.0151)	(0.0150)	(0.0151)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,041	292,033	292,033	292,033	294,041	292,033	292,033	292,033
R^2	0.6667	0.6677	0.7118	0.7118	0.0671	0.0678	0.0746	0.0749

Notes: All regressions include inspector, administrator, and customs broker fixed effects and also control for high-value ports. These are the *Ciudad del Este* ports (*Ciudad del Este*, *Campestre*, and ALGESA), the airports (Pettirossi, CODESA, and Guarani), and the three main river ports at the time period of my data (*Puerto Seguro Fluvial*, TERPORT, and TERPORT Villeta). Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C16: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Workload deviations	0.0055	-0.0557	-0.0117	-0.0114	-0.0029**	0.0050	0.0024	0.0025
	(0.0095)	(0.0398)	(0.0194)	(0.0192)	(0.0012)	(0.0058)	(0.0058)	(0.0058)
Patronage inspector ×	-0.0811**	-0.1370***	-0.0389**	-0.0390**	0.0073**	0.0080	0.0063	0.0063
Patronage administrator	(0.0315)	(0.0353)	(0.0158)	(0.0157)	(0.0037)	(0.0055)	(0.0055)	(0.0055)
Patronage inspector ×	0.0078	0.0100	0.0052	0.0058	-0.0054*	-0.0080**	-0.0078**	-0.0077**
Workload deviations	(0.0139)	(0.0142)	(0.0069)	(0.0068)	(0.0029)	(0.0035)	(0.0035)	(0.0035)
Patronage administrator ×	0.0172	0.0140	0.0078	0.0083	0.0008	0.0007	0.0012	0.0013
Workload deviations	(0.0144)	(0.0138)	(0.0062)	(0.0061)	(0.0014)	(0.0013)	(0.0013)	(0.0013)
Patronage inspector × Patronage	-0.0523***	-0.0436**	-0.0176**	-0.0182**	0.0207***	0.0206***	0.0199***	0.0198***
administrator × Workload deviations	(0.0181)	(0.0173)	(0.0079)	(0.0079)	(0.0045)	(0.0043)	(0.0042)	(0.0042)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2642	0.2663	0.5760	0.5775	0.0266	0.0269	0.0364	0.0366

Notes: All regressions include inspector and administrator fixed effects and also control for high-value ports. These are the *Ciudad del Este* ports (*Ciudad del Este*, *Campestre*, and ALGESA), the airports (Pettirossi, CODESA, and Guarani), and the three main river ports at the time period of my data (*Puerto Seguro Fluvial*, TERPORT, and TERPORT Villeta). Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C17: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected		
	(1)	(2)	(3)
Patronage inspector × Patronage administrator	-0.1064**	-0.1074*	-0.0972*
	(0.0415)	(0.0607)	(0.0571)
Mean of Dependent Variable	13,511,141	13,505,530	13,505,530
HS2-product FE	X	X	X
Interactions w/ insp. char.		X	X
Evasion risk index			X
Observations	79,709	79,589	79,589
Pseudo R^2	0.1672	0.1677	0.3105

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects and also control for high-value ports. These are the *Ciudad del Este* ports (*Ciudad del Este*, *Campestre*, and ALGESA), the airports (Pettirossi, CODESA, and Guarani), and the three main river ports at the time period of my data (*Puerto Seguro Fluvial*, TERPORT, and TERPORT Villeta). Regressions in columns (2)-(3) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

C.4 Additional tables on workload deviations

Table C18: Inspectors' workload tally and total shipments per inspector each day

	Inspector workload tally (1)
Total shipments in a day per inspector	0.4397***
	(0.0694)
Mean of Dependent Variable	4.7945
Observations	292,627
R^2	0.4433

Notes: Standard errors are clustered at the day level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the estimated running tally of each inspector's workload: how many shipments inspectors have pending to inspect. "Total shipments in a day per inspector" refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). The sample covers red channel shipments.

Table C19: Balance table of workload handled by patronage and non-patronage valuation inspectors

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Total shipments in a day	111445	9.4957	182635	10.0499	294080	-0.5542***
		(0.0269)		(0.0149)		
Workload deviations	111445	-0.0461	182635	0.0281	294080	-0.0742***
		(0.0036)		(0.0020)		

Notes: "Total shipments in a day" refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Workload deviations" is the standardized version (with mean 0 and standard deviation of 1) of "Total shipments in a day."

Table C20: Balance table of workload handled by patronage and non-patronage valuation inspectors (controlling for port administrator fixed effects)

	(1) Patronage		(2) Non-patronage		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Total shipments in a day	111445	9.4957	182635	10.0499	294080	-0.5542***
		(0.0269)		(0.0149)		
Workload deviations	111445	-0.0461	182635	0.0281	294080	-0.0742***
		(0.0036)		(0.0020)		

Notes: This table is similar to Table C19 except it also controls for port administrator fixed effects on the pairwise regressions. "Total shipments in a day" refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Workload deviations" is the standardized version (with mean 0 and standard deviation of 1) of "Total shipments in a day."

C.5 Composition of shipments handled by patronage pairs and coarser product controls

The tables in this subsection support the distinction, drawn in Section 5, between a compositional channel (patronage pairs handle shipments whose product mix makes fraud less likely to be detected) and a treatment channel (a patronage pair is less likely to detect fraud on a given shipment). Table C21 compares the product sectors handled by patronage and non-patronage pairs: patronage pairs handle fewer transportation shipments (HS chapters 86-89), which account for about four-fifths of the shipments on which inspectors detect fraud, and more chemicals, metals, and machinery. Table C22 replaces the product chapter fixed effects of Table 2 with coarser product sector fixed effects (15 groupings of chapters) and yields virtually the same within-product estimate.

Table C21: Types of product sectors handled by patronage pairs

	(1) Patronage pair		(2) Non-patronage pair		(1)-(2) Pairwise t-test	
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
01-05 Animal	61265	0.003	232815	0.003	294080	-0.000
		(0.000)		(0.000)		
06-15 Vegetable	61265	0.095	232815	0.104	294080	-0.009***
		(0.001)		(0.001)		
16-24 Foodstuffs	61265	0.041	232815	0.037	294080	0.004***
		(0.001)		(0.000)		
25-27 Minerals	61265	0.021	232815	0.020	294080	0.001**
		(0.001)		(0.000)		
28-38 Chemicals	61265	0.178	232815	0.160	294080	0.018***
		(0.002)		(0.001)		
39-40 Plastic / Rubber	61265	0.173	232815	0.173	294080	-0.000
		(0.002)		(0.001)		
41-43 Hides, Skins	61265	0.044	232815	0.043	294080	0.000
		(0.001)		(0.000)		
44-49 Wood	61265	0.060	232815	0.059	294080	0.002
		(0.001)		(0.000)		
50-63 Textiles, Clothing	61265	0.079	232815	0.084	294080	-0.005***
		(0.001)		(0.001)		
64-67 Footwear	61265	0.046	232815	0.045	294080	0.001
		(0.001)		(0.000)		
68-71 Stone/Glass	61265	0.071	232815	0.068	294080	0.002**
		(0.001)		(0.001)		
72-83 Metals	61265	0.142	232815	0.133	294080	0.009***
		(0.001)		(0.001)		
84-85 Mach/Elec	61265	0.278	232815	0.270	294080	0.007***
		(0.002)		(0.001)		
86-89 Transportation	61265	0.282	232815	0.309	294080	-0.026***
		(0.002)		(0.001)		
90-97 Miscellaneous	61265	0.181	232815	0.177	294080	0.005***
		(0.002)		(0.001)		
Differentiated share	61265	0.783	232815	0.797	294080	-0.014***
		(0.002)		(0.001)		
Differentiated indicator	61265	0.808	232815	0.821	294080	-0.013***
		(0.002)		(0.001)		

Notes: The table compares the product sectors (comprised of the product code chapters listed before each sector title) and differentiated characteristics (share of differentiated items and whether the shipment had differentiated items) of red channel shipments handled by patronage pairs and by non-patronage pairs, without fixed effects. Each sector row is the share of shipments with at least one item in that sector. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table C22: Patronage hires and customs fraud detected, product sector fixed effects

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Patronage inspector × Patronage administrator	-0.1294***	-0.0363**	0.0045	0.0032
	(0.0349)	(0.0165)	(0.0056)	(0.0056)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Product sector FE		X		X
Observations	292,069	292,069	292,069	292,069
R^2	0.2628	0.5546	0.0218	0.0288

Notes: All regressions include inspector and administrator fixed effects. "Product sector FE" corresponds to product sector (groupings of similar product chapters, as listed in Table C21) fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

C.6 Patronage pairs and the scope for evasion

This subsection asks whether patronage pairs' lower fraud detection depends on how much tax is at stake on a shipment. If the gap reflects lower effort where evasion pays, it should be concentrated on shipments with high tax rates and absent where there is little to evade. A capacity shortfall would lower detection regardless. The measure of tax at stake must be fixed before the inspection, so I use the *statutory* rate: for each declared item, the customs duty and internal taxes implied by its declared tariff classification and country of origin under the 2017 tariff schedule (following the rules that assign the extra-zone, Mercosur, auto-parts, and used-goods rates), averaged across the shipment's items with declared CIF weights.[72] The resulting rate has a mean of 17.5% (terciles at 10.4% and 21%). Two further indicators mark shipments with little scope for evasion: all items carry a zero statutory customs duty (34% of shipments, which then owe only internal taxes), and all items are declared from Mercosur countries (29%).

Table C23 estimates the most stringent specification of Table 2 (with inspector and administrator fixed effects, interactions with inspector characteristics, product chapter fixed effects, and the evasion risk index) separately by tercile of the statutory rate. Patronage pairs detect less fraud only in the top tercile (-0.036, significant at 5%), where inspectors detect fraud on 61% of shipments, in the bottom and middle terciles the coefficient is positive and insignificant. Table C24 instead interacts the patronage pair (and each patronage indicator) with each low-scope indicator on the full sample. The pair effect on the remaining shipments is -0.047 to -0.049, the interaction offsets it, and the implied effect on low-scope shipments is between 0.008 and 0.013 with standard errors of about 0.021. The PCA outcome shows no pattern in either table. One caveat applies: inspectors detect fraud on only 2–3% of low-scope shipments, so there is little detection to lose there, and the tables cannot distinguish a pair effect that is truly zero from one that has no room to appear. Within the shipments where detection is common, however, the effect of patronage pairs is as large as in

[72] The taxes actually paid on a shipment cannot be used for this purpose, since they include the inspector's adjustment and any fines, and therefore rise mechanically with detection.

the full sample.

Table C23: Patronage hires and customs fraud detected, by tercile of the statutory tax rate

	Fraud detected by inspector			Fraud detected by PCA, but not inspector		
Statutory tax rate tercile:	Bottom	Middle	Top	Bottom	Middle	Top
	(1)	(2)	(3)	(4)	(5)	(6)
Patronage inspector ×	0.0080	0.0188	-0.0357**	-0.0004	0.0060	-0.0034
Patronage administrator	(0.0065)	(0.0187)	(0.0171)	(0.0055)	(0.0071)	(0.0078)
Mean of Dependent Variable	0.0296	0.1904	0.6052	0.0185	0.0177	0.0079
Mean statutory tax rate (%)	8.7	15.9	28.2	8.7	15.9	28.2
Interactions w/ insp. char.	X	X	X	X	X	X
HS2-product FE	X	X	X	X	X	X
Evasion risk index	X	X	X	X	X	X
Observations	96,084	98,959	95,430	96,084	98,959	95,430
R^2	0.0833	0.3159	0.6957	0.0601	0.0371	0.0320

Notes: Each column estimates column (4) of Table 2 on the red channel shipments in one tercile of the statutory tax rate. The statutory tax rate is the customs duty plus internal taxes implied by the declared tariff classification and country of origin of each item under the 2017 tariff schedule, averaged across the shipment's items with declared CIF weights. All regressions include inspector and administrator fixed effects, control for interactions of the patronage administrator indicator with inspector education and age, for product chapter (HS 2-digit code) fixed effects, and for the evasion risk index. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table C24: Patronage hires and customs fraud detected on shipments with low scope for evasion

	Fraud detected by inspector			Fraud detected by PCA, but not inspector		
Low scope for evasion:	Bottom tercile tax rate	Zero customs duty	Mercosur origin	Bottom tercile tax rate	Zero customs duty	Mercosur origin
	(1)	(2)	(3)	(4)	(5)	(6)
Patronage inspector ×	-0.0487**	-0.0488***	-0.0467***	0.0025	0.0023	0.0015
Patronage administrator	(0.0189)	(0.0188)	(0.0173)	(0.0064)	(0.0064)	(0.0068)
Patronage inspector × Patronage	0.0565*	0.0578**	0.0596**	0.0018	0.0027	0.0067
administrator × Low scope	(0.0295)	(0.0293)	(0.0280)	(0.0068)	(0.0070)	(0.0088)
Patronage pair effect, low-scope shipments	0.0078	0.0090	0.0129	0.0043	0.0050	0.0082
	(0.0214)	(0.0213)	(0.0216)	(0.0068)	(0.0068)	(0.0072)
Mean of Dep. Var., other shipments	0.3940	0.3964	0.3754	0.0129	0.0132	0.0160
Mean of Dep. Var., low-scope shipments	0.0296	0.0277	0.0203	0.0185	0.0178	0.0117
Share of low-scope shipments	0.331	0.333	0.290	0.331	0.333	0.290
Interactions w/ insp. char.	X	X	X	X	X	X
HS2-product FE	X	X	X	X	X	X
Evasion risk index	X	X	X	X	X	X
Observations	290,477	290,477	292,069	290,477	290,477	292,069
R^2	0.5915	0.5921	0.5961	0.0323	0.0323	0.0328

Notes: Each column estimates column (4) of Table 2 on all red channel shipments, adding a low-scope indicator and its interactions with the patronage inspector indicator, the patronage administrator indicator, and their product. The table reports the patronage pair coefficient (the effect on shipments outside the low-scope group) and its interaction with the indicator, followed by their sum, the pair effect on low-scope shipments, with its standard error. Low scope for evasion is defined, in turn, as a statutory tax rate in the bottom tercile (at most 10.4%), a zero statutory customs duty on every item of the shipment, and a declared Mercosur origin (Brazil, Argentina, or Uruguay) for every item. The statutory tax rate is defined in Table C23. All regressions include inspector and administrator fixed effects, control for interactions of the patronage administrator indicator with inspector education and age, for product chapter (HS 2-digit code) fixed effects, and for the evasion risk index. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

C.7 Seasonal fixed effects

Trade flows, staffing, and the composition of shipments vary over the year, and the baseline specification (1) leaves such seasonal variation in the error term. Tables C25 to C28 add fixed effects for the calendar quarter (four dummies) or the calendar month (twelve dummies) of the shipment's inspection date to columns (2), (4), (6), and (8) of Tables 2, 4, and 5, and to columns (2) and (3) of Table 6. Every coefficient moves by less than a tenth of a standard error. The patronage pair coefficient on fraud detected is -0.131 with quarter and -0.134 with month fixed effects in the column (2) specification, against -0.129 in Table 2, and -0.037 and -0.038 in the column (4) specification, against -0.036. The interactions of the patronage pair with suspect assignments and with workload deviations, and the fraud value estimates, are unchanged. The suspect-assignment regression of Table 3 is estimated at the semester level and adding a calendar-semester dummy leaves its coefficient at 0.0031, with the same standard error of 0.0018. Seasonality therefore does not account for the results.

Table C25: Patronage hires and customs fraud detected, with seasonal fixed effects

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector × Patronage administrator	-0.1310***	-0.0365**	-0.1337***	-0.0380**	0.0046	0.0029	0.0047	0.0030
	(0.0349)	(0.0153)	(0.0348)	(0.0153)	(0.0056)	(0.0056)	(0.0056)	(0.0055)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Seasonal FE	Quarter	Quarter	Month	Month	Quarter	Quarter	Month	Month
Interactions w/ insp. char.	X	X	X	X	X	X	X	X
HS2-product FE		X		X		X		X
Evasion risk index		X		X		X		X
Observations	292,069	292,069	292,069	292,069	292,069	292,069	292,069	292,069
R^2	0.2633	0.5767	0.2640	0.5769	0.0219	0.0318	0.0222	0.0321

Notes: Columns (1)-(2) and (5)-(6) add calendar quarter fixed effects, and columns (3)-(4) and (7)-(8) calendar month fixed effects, of the shipment's presentation date to columns (2), (4), (6), and (8) of Table 2. All regressions include inspector and administrator fixed effects and control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table C26: Excess inspection shares, patronage hires, and customs fraud detected, with seasonal fixed effects

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Significant excess	-0.1147**	-0.0809*	-0.1156**	-0.0817*	0.0516**	0.0518**	0.0516**	0.0518**
insp. share	(0.0487)	(0.0443)	(0.0482)	(0.0439)	(0.0237)	(0.0235)	(0.0236)	(0.0234)
Patronage inspector ×	-0.0252**	-0.0113	-0.0264**	-0.0122	0.0006	0.0004	0.0007	0.0004
Patronage administrator	(0.0115)	(0.0090)	(0.0115)	(0.0091)	(0.0057)	(0.0057)	(0.0057)	(0.0057)
Patronage inspector ×	-0.0443*	-0.0312*	-0.0442*	-0.0311*	-0.0122	-0.0122	-0.0122	-0.0121
Significant excess insp. share	(0.0235)	(0.0185)	(0.0235)	(0.0184)	(0.0106)	(0.0108)	(0.0106)	(0.0108)
Patronage administrator ×	-0.0092	-0.0050	-0.0099	-0.0057	-0.0051	-0.0045	-0.0050	-0.0044
Significant excess insp. share	(0.0186)	(0.0173)	(0.0185)	(0.0172)	(0.0072)	(0.0071)	(0.0072)	(0.0071)
Patronage inspector × Patronage	0.0124	0.0049	0.0124	0.0050	0.0304**	0.0296**	0.0303**	0.0296**
administrator × Significant excess insp. share	(0.0284)	(0.0246)	(0.0283)	(0.0244)	(0.0152)	(0.0151)	(0.0151)	(0.0151)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Seasonal FE	Quarter	Quarter	Month	Month	Quarter	Quarter	Month	Month
Interactions w/ insp. char.	X	X	X	X	X	X	X	X
HS2-product FE		X		X		X		X
Evasion risk index		X		X		X		X
Observations	292,033	292,033	292,033	292,033	292,033	292,033	292,033	292,033
R^2	0.6679	0.7119	0.6680	0.7120	0.0678	0.0749	0.0680	0.0752

Notes: Columns (1)-(2) and (5)-(6) add calendar quarter fixed effects, and columns (3)-(4) and (7)-(8) calendar month fixed effects, of the shipment's presentation date to columns (2), (4), (6), and (8) of Table 4. All regressions follow equation (3) and include inspector, administrator, and customs broker fixed effects, and control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table C27: Workload deviations, patronage hires, and customs fraud detected, with seasonal fixed effects

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Workload deviations	-0.0597	-0.0142	-0.0597	-0.0144	0.0050	0.0024	0.0046	0.0020
	(0.0389)	(0.0185)	(0.0387)	(0.0185)	(0.0058)	(0.0058)	(0.0058)	(0.0058)
Patronage inspector ×	-0.1379***	-0.0392**	-0.1407***	-0.0407***	0.0081	0.0064	0.0082	0.0065
Patronage administrator	(0.0351)	(0.0156)	(0.0350)	(0.0157)	(0.0055)	(0.0054)	(0.0055)	(0.0054)
Patronage inspector ×	0.0086	0.0047	0.0091	0.0048	-0.0079**	-0.0076**	-0.0078**	-0.0074**
Workload deviations	(0.0141)	(0.0069)	(0.0140)	(0.0069)	(0.0035)	(0.0034)	(0.0035)	(0.0034)
Patronage administrator ×	0.0168	0.0101	0.0176	0.0106*	0.0006	0.0013	0.0007	0.0014
Workload deviations	(0.0140)	(0.0062)	(0.0139)	(0.0062)	(0.0014)	(0.0013)	(0.0014)	(0.0014)
Patronage inspector × Patronage	-0.0452**	-0.0188**	-0.0463***	-0.0193**	0.0205***	0.0197***	0.0203***	0.0195***
administrator × Workload deviations	(0.0175)	(0.0080)	(0.0174)	(0.0080)	(0.0042)	(0.0042)	(0.0042)	(0.0041)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Seasonal FE	Quarter	Quarter	Month	Month	Quarter	Quarter	Month	Month
Interactions w/ insp. char.	X	X	X	X	X	X	X	X
HS2-product FE		X		X		X		X
Evasion risk index		X		X		X		X
Observations	292,069	292,069	292,069	292,069	292,069	292,069	292,069	292,069
R^2	0.2650	0.5769	0.2657	0.5771	0.0270	0.0367	0.0272	0.0370

Notes: Columns (1)-(2) and (5)-(6) add calendar quarter fixed effects, and columns (3)-(4) and (7)-(8) calendar month fixed effects, of the shipment's presentation date to columns (2), (4), (6), and (8) of Table 5. All regressions include inspector and administrator fixed effects and control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

Table C28: Patronage hires and customs fraud value detected (PPML), with seasonal fixed effects

	Fraud value detected			
	(1)	(2)	(3)	(4)
Patronage inspector × Patronage administrator	-0.1081*	-0.0978*	-0.1072*	-0.0949*
	(0.0622)	(0.0577)	(0.0623)	(0.0577)
Mean of Dependent Variable	13,505,530	13,505,530	13,505,530	13,505,530
Seasonal FE	Quarter	Quarter	Month	Month
HS2-product FE	X	X	X	X
Interactions w/ insp. char.	X	X	X	X
Evasion risk index		X		X
Observations	79,589	79,589	79,589	79,589
Pseudo R^2	0.1667	0.3099	0.1671	0.3103

Notes: Columns (1)-(2) add calendar quarter fixed effects, and columns (3)-(4) calendar month fixed effects, of the shipment's presentation date to columns (2) and (3) of Table 6. The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006) following equation (4) on shipments where the inspector detected fraud. All regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects and control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

C.8 Tables showing balance among shipments' evasion risk and robustness to the imbalances

Table C29: Tax evasion risk of shipments handled by patronage pairs with same baseline regression specification

	SAS/SGIR-SBR risk score (1)	SAS/SGIR risk score (2)	SBR risk score (3)	Log initial CIF value (4)	Mixed shipment dummy (5)	Differentiated share (6)	Log initial weight (7)	Log weighted avg. initial unit price diff. (8)	Time-sensitive share (9)	Registered during extra hours (10)
Patronage inspector × Patronage administrator	0.4320	3.2016	0.7668	0.0842	-0.0188	-0.0822***	0.1079	0.0560***	0.0271**	0.0084*
	(1.9399)	(3.6260)	(1.8322)	(0.0725)	(0.0174)	(0.0287)	(0.1681)	(0.0202)	(0.0117)	(0.0050)
Mean of Dependent Variable	74.2499	14.9511	80.6459	9.8486	0.4967	0.7972	8.2326	0.0885	0.0949	0.0787
Observations	152,561	152,545	152,545	292,069	292,069	292,069	292,069	292,069	292,069	292,069
R^2	0.1195	0.3985	0.0831	0.1859	0.0579	0.2404	0.5047	0.0156	0.0901	0.0087

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. The SAS/SGIR-SBR risk score is the final risk score assigned to the shipment, which takes a weighted average of the SAS/SGIR and the SBR risk score. The SAS/SGIR and SBR risk scores are two different types of evasion risk scores the customs authority uses. The initial CIF value refers to the sum of the product cost, insurance, and freight value as initially declared by the customs broker, which determines the tax base. The mixed shipment dummy takes into account whether the shipment has items belong to different product subheadings (six-digit HS codes). The differentiated share is the share of shipment items that have differentiated products. The initial unit price difference is first determined at the shipment-item level by taking the difference between the initially declared unit price by the customs broker and the median unit price for the same product code among all imports in the Paraguayan customs data. The weighted average at the shipment level is determined by using the shipment items' physical weights when taking the weighted average. The time-sensitive share corresponds to the proportion of items within a shipment that are time-sensitive according to the classification of Hummels and Schaur (2013). Registered during extra hours indicates that the shipment was registered at customs outside of normal port hours. The sample for Columns (1)-(3) covers red channel shipments which had a customs risk score assigned. Shipments that are assigned to the red channel based on legislation do not get assigned a risk score as this motive suffices for an inspection. The sample for the remaining columns includes all red channel shipments.

Table C30: Patronage hires and customs fraud detected

	Fraud detected by inspector			Fraud detected by PCA, but not inspector		
	(1)	(2)	(3)	(4)	(5)	(6)
Patronage inspector × Patronage administrator	-0.1294***	-0.0356**	-0.0288**	0.0045	0.0028	0.0028
	(0.0349)	(0.0155)	(0.0144)	(0.0056)	(0.0056)	(0.0056)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148
HS2-product FE		X	X		X	X
Evasion risk covariates			X			X
Observations	292,069	292,069	292,069	292,069	292,069	292,069
R^2	0.2628	0.5751	0.5885	0.0218	0.0316	0.0348

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk covariates" consist of tax evasion risk variables that are imbalanced between patronage and non-patronage pairs, they are described as follows. The differentiated share is the share of shipment items that have differentiated products. The initial unit price difference is first determined at the shipment-item level by taking the difference between the initially declared unit price by the customs broker and the median unit price for the same product code among all imports in the Paraguayan customs data. The weighted average at the shipment level is determined by using the shipment items' physical weights when taking the weighted average. The time-sensitive share corresponds to the proportion of items within a shipment that are time-sensitive according to the classification of Hummels and Schaur (2013). Registered during extra hours indicates that the shipment was registered at customs outside of normal port hours. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (4)-(6) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C31: Customs fraud detected, patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Significant excess	-0.1187**	-0.1179**	-0.0833*	-0.0952**	0.0539**	0.0525**	0.0530**	0.0541**
insp. share	(0.0489)	(0.0488)	(0.0442)	(0.0432)	(0.0240)	(0.0238)	(0.0236)	(0.0232)
Patronage inspector ×	-0.0237**	-0.0245**	-0.0106	-0.0092	0.0027	0.0006	0.0005	0.0005
Patronage administrator	(0.0114)	(0.0115)	(0.0091)	(0.0088)	(0.0056)	(0.0057)	(0.0057)	(0.0057)
Patronage inspector ×	-0.0351**	-0.0429*	-0.0301	-0.0321*	0.0058	-0.0124	-0.0126	-0.0125
Significant excess insp. share	(0.0166)	(0.0237)	(0.0185)	(0.0181)	(0.0082)	(0.0107)	(0.0108)	(0.0107)
Patronage administrator ×		-0.0092	-0.0050	-0.0057		-0.0048	-0.0045	-0.0046
Significant excess insp. share		(0.0185)	(0.0172)	(0.0170)		(0.0071)	(0.0071)	(0.0070)
Patronage inspector × Patronage		0.0123	0.0048	0.0104		0.0301**	0.0293*	0.0285*
administrator × Significant excess insp. share		(0.0286)	(0.0246)	(0.0240)		(0.0151)	(0.0150)	(0.0147)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
HS2-product FE			X	X			X	X
Evasion risk covariates				X				X
Observations	292,033	292,033	292,033	292,033	292,033	292,033	292,033	292,033
R^2	0.6677	0.6677	0.7118	0.7151	0.0675	0.0677	0.0745	0.0771

Notes: All regressions include inspector, administrator, and customs broker fixed effects. All regressions control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk covariates" consist of tax evasion risk variables that are imbalanced between patronage and non-patronage pairs, they are described as follows. The differentiated share is the share of shipment items that have differentiated products. The initial unit price difference is first determined at the shipment-item level by taking the difference between the initially declared unit price by the customs broker and the median unit price for the same product code among all imports in the Paraguayan customs data. The weighted average at the shipment level is determined by using the shipment items' physical weights when taking the weighted average. The time-sensitive share corresponds to the proportion of items within a shipment that are time-sensitive according to the classification of Hummels and Schaur (2013). Registered during extra hours indicates that the shipment was registered at customs outside of normal port hours. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (4)-(6) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C32: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Workload deviations	-0.0661*	-0.0608	-0.0146	-0.0160	0.0144*	0.0049	0.0022	0.0021
	(0.0373)	(0.0385)	(0.0186)	(0.0180)	(0.0080)	(0.0058)	(0.0058)	(0.0057)
Patronage inspector ×	-0.1287***	-0.1365***	-0.0384**	-0.0318**	0.0036	0.0080	0.0064	0.0063
Patronage administrator	(0.0350)	(0.0351)	(0.0158)	(0.0148)	(0.0056)	(0.0055)	(0.0055)	(0.0055)
Patronage inspector ×	-0.0241**	0.0096	0.0049	0.0054	0.0091***	-0.0080**	-0.0078**	-0.0075**
Workload deviations	(0.0101)	(0.0139)	(0.0069)	(0.0066)	(0.0032)	(0.0036)	(0.0035)	(0.0035)
Patronage administrator ×		0.0160	0.0089	0.0098		0.0008	0.0013	0.0015
Workload deviations		(0.0139)	(0.0062)	(0.0060)		(0.0013)	(0.0013)	(0.0013)
Patronage inspector × Patronage		-0.0451***	-0.0184**	-0.0197***		0.0206***	0.0198***	0.0193***
administrator × Workload deviations		(0.0173)	(0.0079)	(0.0075)		(0.0043)	(0.0042)	(0.0041)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
HS2-product FE			X	X			X	X
Evasion risk covariates				X				X
Observations	292,069	292,069	292,069	292,069	292,069	292,069	292,069	292,069
R^2	0.2641	0.2645	0.5754	0.5887	0.0250	0.0269	0.0364	0.0394

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk covariates" consist of tax evasion risk variables that are imbalanced between patronage and non-patronage pairs, they are described as follows. The differentiated share is the share of shipment items that have differentiated products. The initial unit price difference is first determined at the shipment-item level by taking the difference between the initially declared unit price by the customs broker and the median unit price for the same product code among all imports in the Paraguayan customs data. The weighted average at the shipment level is determined by using the shipment items' physical weights when taking the weighted average. The time-sensitive share corresponds to the proportion of items within a shipment that are time-sensitive according to the classification of Hummels and Schaur (2013). Registered during extra hours indicates that the shipment was registered at customs outside of normal port hours. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (4)-(6) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C33: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected	
	(1)	(2)
Patronage inspector × Patronage administrator	-0.1080*	-0.1079*
	(0.0618)	(0.0618)
Mean of Dependent Variable	13,505,530	13,505,530
HS2-product FE	X	X
Evasion risk covariates		X
Observations	79,589	79,589
Pseudo R^2	0.1662	0.1679

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects. The regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk covariates" consist of tax evasion risk variables that are imbalanced between patronage and non-patronage pairs, they are described as follows. The differentiated share is the share of shipment items that have differentiated products. The initial unit price difference is first determined at the shipment-item level by taking the difference between the initially declared unit price by the customs broker and the median unit price for the same product code among all imports in the Paraguayan customs data. The weighted average at the shipment level is determined by using the shipment items' physical weights when taking the weighted average. The time-sensitive share corresponds to the proportion of items within a shipment that are time-sensitive according to the classification of Hummels and Schaur (2013). Registered during extra hours indicates that the shipment was registered at customs outside of normal port hours. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

C.9 Tables looking at fraud value on all shipments

Table C34: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected		
	(1)	(2)	(3)
Patronage inspector × Patronage administrator	-0.0979	-0.1369	-0.1456*
	(0.0637)	(0.0903)	(0.0804)
Mean of Dependent Variable	3,790,575	3,796,102	3,796,102
HS2-product FE	X	X	X
Interactions w/ insp. char.		X	X
Evasion risk index			X
Observations	293,753	291,794	291,794
Pseudo R^2	0.4504	0.4507	0.4977

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects. Regressions in columns (2)-(3) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments.

C.10 Tables controlling for low-capacity interactions

In this Appendix subsection, I additionally control for inspectors and administrators having low capacity with other independent variables. In particular, I proxy for low capacity as having a below-median education, which is a college degree for both inspectors and administrators. Table C35 runs the baseline specification while also controlling for a "low-capacity pair." Table C36 controls for low-capacity pairs, interactions between significant excess inspection shares and low-capacity inspectors and administrators respectively, and their triple interaction. Table C37 has a similar specification with workload deviations instead. Finally, Table C38 also controls for low-capacity pairs. The patronage pair coefficients on fraud detection, suspect assignments, and fraud value retain their magnitude and significance. In Table C37, the interaction of patronage pairs with workload deviations on inspector-detected fraud is -0.037 (standard error 0.018) in column (2) and -0.018 (standard error 0.008) in the most stringent specification of column (4), with product chapter FEs and the evasion risk index, both significant at the 5% level, whereas the interaction on PCA-detected fraud remains 0.014 and significant at the 1% level in every column. Low-capacity pairs themselves do not detect less fraud as workload rises (their interaction is 0.015 in column (2) and 0.011 in columns (3)-(4), not significant), which is not what a human-capital channel would predict. Appendix C.11 discusses these results, and Appendix C.13 reports the low-capacity pair coefficient side by side with the patronage pair coefficient, alone and jointly (Tables C50 and C52). Overall, the effect of patronage pairs goes beyond their lower qualifications.

Table C35: Patronage hires and customs fraud detected

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector × Patronage administrator	-0.0760**	-0.1280***	-0.0346**	-0.0345**	0.0042	0.0043	0.0026	0.0027
	(0.0318)	(0.0350)	(0.0153)	(0.0152)	(0.0038)	(0.0056)	(0.0056)	(0.0056)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Low-capacity interactions	X	X	X	X	X	X	X	X
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2612	0.2631	0.5753	0.5767	0.0217	0.0218	0.0316	0.0318

Notes: All regressions include inspector and administrator fixed effects and control for low-capacity pairs (as proxied by both the inspector and administrator having below-median education). Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C36: Customs fraud detected, patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Significant excess	0.0172	-0.2255***	-0.1767**	-0.1765**	0.0102*	0.0796***	0.0827***	0.0830***
insp. share	(0.0173)	(0.0863)	(0.0841)	(0.0840)	(0.0061)	(0.0272)	(0.0271)	(0.0270)
Patronage inspector ×	-0.0103	-0.0240**	-0.0101	-0.0102	0.0012	0.0006	0.0004	0.0003
Patronage administrator	(0.0097)	(0.0114)	(0.0090)	(0.0090)	(0.0038)	(0.0058)	(0.0058)	(0.0057)
Patronage inspector ×	-0.0243	-0.0460*	-0.0328*	-0.0327*	-0.0137	-0.0117	-0.0118	-0.0117
Significant excess insp. share	(0.0227)	(0.0240)	(0.0190)	(0.0190)	(0.0100)	(0.0107)	(0.0108)	(0.0108)
Patronage administrator ×	0.0018	-0.0108	-0.0065	-0.0064	-0.0026	-0.0044	-0.0041	-0.0038
Significant excess insp. share	(0.0198)	(0.0187)	(0.0175)	(0.0175)	(0.0074)	(0.0071)	(0.0070)	(0.0070)
Patronage inspector × Patronage	-0.0014	0.0143	0.0066	0.0066	0.0282*	0.0296*	0.0288*	0.0288*
administrator × Significant excess insp. share	(0.0284)	(0.0287)	(0.0249)	(0.0249)	(0.0152)	(0.0151)	(0.0150)	(0.0151)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Low-capacity interactions	X	X	X	X	X	X	X	X
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,041	292,033	292,033	292,033	294,041	292,033	292,033	292,033
R^2	0.6668	0.6678	0.7118	0.7118	0.0672	0.0677	0.0746	0.0749

Notes: All regressions include inspector, administrator, and customs broker fixed effects. They also control for low-capacity pairs (as proxied by both the inspector and administrator having below-median education), interactions between significant excess inspection shares and low-capacity inspectors and administrators respectively, and their triple interaction. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C37: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Workload deviations	0.0165	-0.2525***	-0.1151***	-0.1164***	-0.0046***	0.0085	0.0056	0.0055
	(0.0114)	(0.0869)	(0.0384)	(0.0384)	(0.0016)	(0.0106)	(0.0103)	(0.0102)
Patronage inspector ×	-0.0854***	-0.1341***	-0.0375**	-0.0376**	0.0072**	0.0066	0.0049	0.0049
Patronage administrator	(0.0317)	(0.0353)	(0.0158)	(0.0156)	(0.0035)	(0.0054)	(0.0054)	(0.0054)
Patronage inspector ×	0.0097	0.0052	0.0033	0.0040	-0.0067*	-0.0067*	-0.0065*	-0.0064*
Workload deviations	(0.0145)	(0.0133)	(0.0066)	(0.0066)	(0.0035)	(0.0036)	(0.0036)	(0.0036)
Patronage administrator ×	0.0124	0.0092	0.0061	0.0068	0.0027*	0.0028*	0.0032**	0.0033**
Workload deviations	(0.0149)	(0.0135)	(0.0061)	(0.0060)	(0.0016)	(0.0015)	(0.0015)	(0.0015)
Patronage inspector × Patronage	-0.0477**	-0.0373**	-0.0171**	-0.0180**	0.0145***	0.0144***	0.0138***	0.0137***
administrator × Workload deviations	(0.0189)	(0.0178)	(0.0079)	(0.0078)	(0.0037)	(0.0039)	(0.0039)	(0.0039)
Low-capacity inspector ×	-0.0586*	-0.0504	-0.0369**	-0.0382**	0.0044	0.0043	0.0045	0.0044
Low-capacity administrator	(0.0335)	(0.0338)	(0.0157)	(0.0155)	(0.0041)	(0.0044)	(0.0043)	(0.0043)
Low-capacity inspector ×	-0.0040	0.1302**	0.0650***	0.0657***	-0.0021	-0.0095	-0.0093	-0.0092
Workload deviations	(0.0122)	(0.0528)	(0.0234)	(0.0234)	(0.0019)	(0.0063)	(0.0062)	(0.0061)
Low-capacity administrator ×	-0.0282*	-0.0269*	-0.0123*	-0.0120*	0.0032	0.0031	0.0027	0.0027
Workload deviations	(0.0144)	(0.0138)	(0.0066)	(0.0066)	(0.0022)	(0.0022)	(0.0021)	(0.0021)
Low-capacity inspector × Low-capacity	0.0175	0.0146	0.0112	0.0112	0.0097***	0.0098***	0.0101***	0.0101***
administrator × Workload deviations	(0.0207)	(0.0202)	(0.0100)	(0.0099)	(0.0032)	(0.0031)	(0.0031)	(0.0031)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Low-capacity interactions	X	X	X	X	X	X	X	X
Interactions w/ insp. char.		X	X	X		X	X	X
HS2-product FE			X	X			X	X
Evasion risk index				X				X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2630	0.2654	0.5757	0.5771	0.0276	0.0278	0.0372	0.0374

Notes: All regressions include inspector and administrator fixed effects. They also control for low-capacity pairs (as proxied by both the inspector and administrator having below-median education), interactions between workload deviations and low-capacity inspectors and administrators respectively, and their triple interaction. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C38: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected		
	(1)	(2)	(3)
Patronage inspector × Patronage administrator	-0.1112***	-0.1089*	-0.0983*
	(0.0425)	(0.0622)	(0.0578)
Mean of Dependent Variable	13,511,141	13,505,530	13,505,530
Low-capacity interactions	X	X	X
HS2-product FE	X	X	X
Interactions w/ insp. char.		X	X
Evasion risk index			X
Observations	79,709	79,589	79,589
Pseudo R^2	0.1659	0.1662	0.3099

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects and control for low-capacity pairs (as proxied by both the inspector and administrator having below-median education). Regressions in columns (2)-(3) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

C.11 Capacity constraints versus masking under high workload

Section 5 reads the additional decline in patronage pairs' fraud detection on high-workload days as strategic: workload spikes give plausible deniability for lower effort. A mechanical reading is also possible. Patronage inspectors have about one education level less than non-patronage inspectors (Table 1), and officials with less human capital may suffer steeper cognitive overload when many shipments arrive, so their detection would fall faster under load without any intent to collude. The interactions of workload deviations with inspector education and age in Table 5 hold observed human capital fixed but cannot rule out capacity deficits that are unobserved and specific to patronage hires. This subsection sets out why the specification already separates the two mechanisms and draws on the low-capacity table above.

The two mechanisms differ in *where* the workload response should appear. A capacity constraint operates at the inspector level: an inspector's ability to process shipments does not depend on who supervises her. Strategic masking, in the framework of Section 2, operates at the pair level: the masking factor δ lowers the expected penalty from collusion, which only pays when collusion costs are low, that is, within patronage pairs. The specification in Section 5 includes the interaction of workload deviations with the patronage inspector indicator, which is the workload slope of a patronage inspector *when supervised by a non-patronage administrator*. In Appendix Table C67 this slope is 0.010 (standard error 0.014) for inspector-detected fraud and -0.008 for fraud later uncovered by the PCA team. The additional decline remains entirely in the triple interaction, which requires the administrator to be a patronage hire as well. For a capacity story to generate this pattern it would have to run through the administrator's capacity, but patronage administrators do not differ from non-patronage administrators in education (Panel B of Table 1 shows 81 versus 75 percent hold a college degree, a difference that is not statistically significant).

Table C37 makes the same point with a direct proxy for capacity. If lower human capital produced a steeper decline under load, pairs of two below-college officials should detect less fraud as workload rises. Their interaction with workload deviations is instead positive and not statisti-

cally significant (0.015 in column (2) and 0.011 with product chapter fixed effects, with standard errors of 0.020 and 0.010). On PCA-detected fraud, low-capacity pairs show a positive interaction (0.010) comparable to that of patronage pairs (0.014), so this outcome does not separate the two groups. Once the low-capacity interactions are included, the patronage pair interaction on inspector-detected fraud is -0.037 (standard error 0.018) in column (2) and -0.018 (standard error 0.008) in the most stringent specification of column (4), with product chapter fixed effects and the evasion risk index, both significant at the 5 percent level, while the interaction on PCA-detected fraud remains 0.014 and significant at the 1 percent level in every column. Nor do low-capacity pairs detect less fraud on average: when the low-capacity pair replaces the patronage pair indicator, its coefficient on inspector-detected fraud is -0.055 and not significant (Appendix Table C50), and entering it alongside the patronage pair leaves the patronage pair coefficient unchanged (Appendix Table C52).

C.12 Workload deviations with outcomes aggregated to the inspector-port-day

Table C39 asks whether the workload results hold when the two outcomes, fraud detected by the inspector and fraud detected by the PCA team, are aggregated. It collapses the data to the inspector-port-day and uses as outcomes the share of the inspector's red channel shipments that day with fraud detected by the inspector and by the PCA team, weighted by the number of shipments. With the joint specification the triple interaction is -0.015 (5 percent level) for inspector-detected fraud and 0.019 (1 percent level) for PCA-detected fraud. The reallocation of detection from the pair to the PCA team under high workload therefore survives aggregation to the inspector-port-day.

Table C39: Patronage hires, workload deviations and the share of an inspector's shipments with fraud detected, inspector-port-day level

	Share with fraud detected by inspector (1)	Share with fraud detected by PCA (2)
Workload deviations	-0.0059	0.0017
	(0.0148)	(0.0058)
Patronage inspector ×	-0.0179	0.0056
Patronage administrator	(0.0123)	(0.0053)
Patronage inspector ×	0.0049	-0.0075**
Workload deviations	(0.0059)	(0.0034)
Patronage administrator ×	0.0096**	0.0014
Workload deviations	(0.0049)	(0.0013)
Patronage inspector × Patronage	-0.0146**	0.0193***
administrator × Workload deviations	(0.0067)	(0.0041)
Mean of Dependent Variable	0.2725	0.0148
Interactions w/ insp. char.	X	X
HS2-product shares	X	X
Evasion risk index	X	X
Observations (inspector-port-days)	46,847	46,847
R^2	0.7972	0.1140

Notes: One observation per valuation inspector, port and day. The outcomes are the shares of the inspector's red channel shipments that day with fraud detected by the inspector (column (1)) and by the post-clearance audit (PCA) team but not the inspector (column (2)). Regressions use the joint specification of column (4) of Table 5, with the product chapter dummies and the evasion risk index entered as their inspector-day means, are weighted by the number of shipments, and include inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively.

C.13 Tables using other types of affinity matches, alone and jointly with patronage pairs

This Appendix subsection addresses the concern, raised in Section 5, that the patronage pair coefficient may capture homogeneity of the inspector-administrator pair on a dimension other than patronage (Marx et al., 2021). Tables C40 to C49 first *replace* the patronage pair indicator with two alternative affinity measures, shared gender and age similarity (an age gap of at most 10 years), in the main specifications. Same-gender pairs account for 59.1% and age-similar pairs for 52% of red channel shipments. Same-gender pairs detect more fraud than mixed-gender pairs, the opposite of patronage pairs. Similar-age pairs, in contrast, detect less fraud and have more suspect assignments, a pattern that resembles the one for patronage pairs but with coefficients about half as large.

Tables C51 to C56 therefore enter the alternative matches *jointly* with the patronage pair, following the approach in Marx et al. (2021). I consider five matches: same gender, similar age, same education level, concurrent hires (inspector and administrator appointed under the same executive administration, a proxy for a shared hiring cohort and faction), and low-capacity pairs, in which neither official holds a college degree, the median education level. The last match is not a homophily measure but a direct check on the competence explanation: since patronage inspectors are less educated, the patronage pair coefficient could reflect two low-human-capital officials working together. In the data every low-capacity pair is also a same-education pair, so the low-capacity match isolates the least-educated subset of the same-education pairs. Table C51 regresses each match on the patronage pair using the baseline specification (1): conditional on the fixed effects, patronage pairs are no more likely to be same-gender, similar-age, or low-capacity pairs (2.7 percentage points, not significant), but are 22 percentage points more likely to share an education level and 30 percentage points more likely to be concurrent hires. Tables C52 and C53 add the matches one at a time and jointly to the fraud detection regressions of Table 2. The horse race with

low-capacity pairs is in column (6): the patronage pair coefficient on inspector fraud detection is unchanged at -0.129, while the low-capacity pair coefficient is -0.051, not significant, and less than half the patronage effect. Table C50 shows the low-capacity pair on its own, replacing the patronage pair indicator as in the same-gender and similar-age tables above. With all five matches included, the patronage pair coefficient falls from -0.129 to -0.095 and remains significant at the 1% level, while similar-age pairs and concurrent hires have independent negative effects, same-gender pairs a positive one, and the same-education and low-capacity matches none. Tables C54 and C55 repeat the exercise with the four affinity matches for fraud detection interacted with suspect assignments and with workload deviations. The corresponding low-capacity interactions are in Tables C36 and C37. The interactions of patronage pairs with suspect assignments and workload deviations on PCA-detected fraud are unchanged, while the interaction with workload deviations on inspector-detected fraud becomes -0.033 and while still remaining significant once the other matches are included. On the intensive margin (Table C56), the patronage pair coefficient on the fraud value detected is unaffected by the alternative matches (-0.108 without them, -0.109 with the low-capacity pair, and -0.112 with all five), none of which has a significant effect of its own.

Overall, the alternative matches absorb part, but not all, of the patronage pair effect, and the part they absorb is concentrated in the two dimensions (similar age and concurrent hiring) that are themselves products of patronage hiring waves rather than in generic homophily on gender or education, or in the two officials' low human capital. The results should therefore be seen as showing that patronage ties have explanatory power beyond generic similarity between officials and beyond their lower qualifications, while acknowledging that the data cannot fully separate the effect of shared patronage status from that of the shared cohort and age structure that patronage hiring creates.

Table C40: Same-gender pairs and customs fraud detected

	Fraud detected by inspector (1)	Fraud detected by PCA, but not inspector (2)
Same-gender pair	0.1177***	0.0028
	(0.0389)	(0.0023)
Mean of Dependent Variable	0.2725	0.0148
Observations	292,069	292,069
R^2	0.2641	0.0218

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on column (2) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. "Same-gender pairs" identifies inspectors and administrators working on a shipment with the same gender. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C41: Same-gender pairs and suspect assignments

	Binary significant excess inspection share (1)
Same-gender pair	0.0021*
	(0.0012)
Mean of Dependent Variable	0.0082
Observations	77,955
R^2	0.0394

Notes: The regression includes inspector, administrator, and customs broker fixed effects. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector combinations which are significantly excessive for 90% or more of the simulations. "Same-gender pairs" identifies inspectors and administrators working on a shipment with the same gender. The sample covers all possible non-singleton pairings of valuation inspectors and customs brokers at all semesters and ports.

Table C42: Customs fraud detected, same-gender pairs, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Significant excess insp. share	-0.1205**	-0.1170**	0.0534**	0.0436*
	(0.0489)	(0.0476)	(0.0240)	(0.0233)
Same-gender pair	0.0213**	0.0217**	0.0049**	0.0036*
	(0.0091)	(0.0092)	(0.0022)	(0.0022)
Patronage inspector × Significant excess insp. share	-0.0358**	-0.0367**	0.0060	0.0087
	(0.0167)	(0.0168)	(0.0082)	(0.0086)
Patronage administrator × Significant excess insp. share		-0.0050		0.0053
		(0.0131)		(0.0066)
Same-gender pair × Significant excess insp. share		-0.0048		0.0173***
		(0.0138)		(0.0062)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,033	292,033	292,033	292,033
R^2	0.6677	0.6677	0.0676	0.0678

Notes: All regressions include inspector, administrator, and customs broker fixed effects. All regressions control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C43: Same-gender pairs, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Workload deviations	-0.0634*	-0.0604	0.0142*	0.0003
	(0.0364)	(0.0391)	(0.0079)	(0.0061)
Same-gender pair	0.1160***	0.1155***	0.0034	0.0045*
	(0.0388)	(0.0391)	(0.0022)	(0.0024)
Patronage inspector ×	-0.0276***	-0.0279**	0.0092***	0.0087***
Workload deviations	(0.0104)	(0.0109)	(0.0032)	(0.0025)
Patronage administrator ×		-0.0008		0.0084***
Workload deviations		(0.0096)		(0.0019)
Same-gender pair ×		-0.0027		0.0076***
Workload deviations		(0.0123)		(0.0019)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,069	292,069	292,069	292,069
R^2	0.2653	0.2653	0.0250	0.0264

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. "Same-gender pairs" identifies inspectors and administrators working on a shipment with the same gender. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C44: Same-gender pairs and customs fraud value detected by valuation inspectors

	Fraud value detected
	(1)
Same-gender pair	0.0555
	(0.0374)
Mean of Dependent Variable	13,505,530
HS2-product FE	X
Observations	79,589
Pseudo R^2	0.1662

Notes: The table runs a Poisson pseudo-maximum likelihood (PPML) regression (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regression includes inspector, administrator, and product chapter (HS 2-digit code) fixed effects. The regression controls for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made. "Same-gender pairs" identifies inspectors and administrators working on a shipment with the same gender. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

Table C45: Similar age pairs and customs fraud detected

	Fraud detected by inspector (1)	Fraud detected by PCA, but not inspector (2)
Similar age pair	-0.0736***	-0.0017
	(0.0185)	(0.0022)
Mean of Dependent Variable	0.2725	0.0148
Observations	292,069	292,069
R^2	0.2635	0.0218

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on column (2) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. "Similar age pairs" identifies inspectors and administrators working on a shipment with a similar age, defined as having an age gap of at most 10 years. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C46: Similar age pairs and suspect assignments

	Binary significant excess inspection share (1)
Similar age pair	0.0016*
	(0.0009)
Mean of Dependent Variable	0.0082
Observations	77,955
R^2	0.0394

Notes: The regression includes inspector, administrator, and customs broker fixed effects. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector combinations which are significantly excessive for 90% or more of the simulations. "Similar age pairs" identifies inspectors and administrators working on a shipment with a similar age, defined as having an age gap of at most 10 years. The sample covers all possible non-singleton pairings of valuation inspectors and customs brokers at all semesters and ports.

Table C47: Customs fraud detected, similar age pairs, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Significant excess insp. share	-0.1163**	-0.1171**	0.0542**	0.0564**
	(0.0489)	(0.0483)	(0.0241)	(0.0225)
Similar age pair	-0.0185***	-0.0178***	-0.0025	-0.0037
	(0.0047)	(0.0048)	(0.0022)	(0.0024)
Patronage inspector × Significant excess insp. share	-0.0359**	-0.0362**	0.0060	0.0063
	(0.0167)	(0.0167)	(0.0082)	(0.0080)
Patronage administrator × Significant excess insp. share		-0.0066		0.0067
		(0.0133)		(0.0066)
Similar age pair × Significant excess insp. share		-0.0106		0.0185***
		(0.0135)		(0.0061)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,033	292,033	292,033	292,033
R^2	0.6678	0.6678	0.0676	0.0678

Notes: All regressions include inspector, administrator, and customs broker fixed effects. All regressions control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C48: Similar age pairs, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Workload deviations	-0.0675*	-0.0711**	0.0141*	0.0062
	(0.0358)	(0.0347)	(0.0079)	(0.0062)
Similar age pair	-0.0742***	-0.0788***	-0.0023	-0.0018
	(0.0184)	(0.0186)	(0.0022)	(0.0021)
Patronage inspector ×	-0.0298***	-0.0271**	0.0092***	0.0068***
Workload deviations	(0.0107)	(0.0111)	(0.0032)	(0.0025)
Patronage administrator ×		0.0028		0.0091***
Workload deviations		(0.0100)		(0.0020)
Similar age pair ×		-0.0313**		0.0047**
Workload deviations		(0.0132)		(0.0020)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,069	292,069	292,069	292,069
R^2	0.2649	0.2655	0.0250	0.0261

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. "Similar age pairs" identifies inspectors and administrators working on a shipment with a similar age, defined as having an age gap of at most 10 years. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C49: Similar age pairs and customs fraud value detected by valuation inspectors

	Fraud value detected
	(1)
Similar age pair	0.0081
	(0.0249)
Mean of Dependent Variable	13,505,530
HS2-product FE	X
Observations	79,589
Pseudo R^2	0.1660

Notes: The table runs a Poisson pseudo-maximum likelihood (PPML) regression (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regression includes inspector, administrator, and product chapter (HS 2-digit code) fixed effects. The regression controls for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made. "Similar age pairs" identifies inspectors and administrators working on a shipment with a similar age, defined as having an age gap of at most 10 years. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

Table C50: Low-capacity pairs and customs fraud detected

	Fraud detected by inspector (1)	Fraud detected by PCA, but not inspector (2)	Fraud value detected (3)
Low-capacity pair	-0.0548	0.0066	-0.0004
	(0.0340)	(0.0051)	(0.0590)
Mean of Dependent Variable	0.2725	0.0148	13,507,186
Interactions w/ insp. char.	X	X	X
HS2-product FE			X
Observations	291,913	291,913	79,558

Notes: The table replaces the patronage pair indicator with a "low-capacity pair" indicator, equal to 1 when neither the inspector nor the administrator holds a college degree (the median education level), in the fraud detection specifications of Table 2 (columns (1)-(2)) and the fraud value PPML specification of Table 6 (column (3)). All regressions include inspector and administrator fixed effects and control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The low-capacity inspector and administrator indicators are absorbed by the fixed effects. Column (3) also includes product chapter (HS 2-digit code) fixed effects and is estimated on red channel shipments on which the valuation inspector detected fraud, its coefficient is a semi-elasticity of the fraud value adjustment in Paraguayan Guaranis. The outcome variable in column (2) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C51: Patronage pairs and matches on other dimensions

	Same-gender pair (1)	Similar age pair (2)	Age gap (years) (3)	Same education pair (4)	Concurrent hires (5)	Low-capacity pair (6)
Patronage inspector × Patronage administrator	-0.1004	0.0586	-0.5832	0.2151**	0.2982***	0.0274
	(0.0618)	(0.0998)	(1.3117)	(0.1000)	(0.0814)	(0.0495)
Mean of Dependent Variable	0.5958	0.5247	10.8235	0.3525	0.1765	0.0660
Observations	291,913	291,913	291,913	291,913	292,069	291,913
R^2	0.7952	0.5757	0.6955	0.5690	0.5237	0.6424

Notes: Each column regresses an indicator for the inspector-administrator pair matching on the column's dimension on the patronage pair indicator, using the baseline specification of equation (1). All regressions include inspector and administrator fixed effects and control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. "Same-gender pair" equals 1 when the inspector and administrator have the same gender. "Similar age pair" equals 1 when their ages differ by at most 10 years. "Age gap" is the absolute age difference in years. "Same education pair" equals 1 when both hold the same education level. "Concurrent hires" equals 1 when both were hired during the same executive administration. "Low-capacity pair" equals 1 when neither official holds a college degree (the median education level), so that in the data every low-capacity pair is also a same-education pair. Matches are missing when the relevant trait is not observed for one of the officials. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C52: Patronage pairs and customs fraud detected by inspectors, controlling for matches on other dimensions

	Fraud detected by inspector						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Patronage inspector × Patronage administrator	-0.1294***	-0.1193***	-0.1264***	-0.1253***	-0.1063***	-0.1292***	-0.0951***
	(0.0349)	(0.0339)	(0.0345)	(0.0349)	(0.0335)	(0.0350)	(0.0326)
Same-gender pair		0.1120***					0.1115***
		(0.0385)					(0.0367)
Similar age pair			-0.0719***				-0.0598***
			(0.0184)				(0.0178)
Same education pair				-0.0247			-0.0104
				(0.0180)			(0.0264)
Concurrent hires					-0.0774***		-0.0600***
					(0.0203)		(0.0201)
Low-capacity pair						-0.0507	-0.0228
						(0.0339)	(0.0489)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.2725	0.2725	0.2725
Interactions w/ insp. char.	X	X	X	X	X	X	X
Observations	292,069	291,913	291,913	291,913	292,069	291,913	291,913
R^2	0.2628	0.2660	0.2656	0.2632	0.2649	0.2632	0.2702

Notes: All regressions include inspector and administrator fixed effects and control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. Column (1) reproduces column (2) of Table 2. Columns (2)-(6) add one alternative match at a time and column (7) adds all five jointly. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Alternative matches are defined as in Table C51. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C53: Patronage pairs and customs fraud detected by the PCA team, controlling for matches on other dimensions

	Fraud detected by PCA, but not inspector						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Patronage inspector ×	0.0045	0.0048	0.0047	0.0038	0.0042	0.0044	0.0039
Patronage administrator	(0.0056)	(0.0057)	(0.0056)	(0.0059)	(0.0056)	(0.0057)	(0.0060)
Same-gender pair		0.0030					0.0029
		(0.0023)					(0.0023)
Similar age pair			-0.0018				-0.0022
			(0.0021)				(0.0022)
Same education pair				0.0033			0.0026
				(0.0026)			(0.0033)
Concurrent hires					0.0010		0.0014
					(0.0019)		(0.0019)
Low-capacity pair						0.0065	0.0024
						(0.0051)	(0.0062)
Mean of Dependent Variable	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148	0.0148
Interactions w/ insp. char.	X	X	X	X	X	X	X
Observations	292,069	291,913	291,913	291,913	292,069	291,913	291,913
R^2	0.0218	0.0218	0.0218	0.0219	0.0218	0.0218	0.0219

Notes: The outcome variable is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Column (1) reproduces column (6) of Table 2, the remaining columns follow Table C52. All regressions include inspector and administrator fixed effects and control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. Alternative matches are defined as in Table C51. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C54: Customs fraud detected, patronage pairs, and suspect assignments, controlling for matches on other dimensions

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Significant excess insp. share	-0.1179**	-0.1122**	0.0525**	0.0458**
	(0.0488)	(0.0490)	(0.0238)	(0.0207)
Patronage inspector × Patronage administrator	-0.0245**	-0.0125	0.0006	0.0008
	(0.0115)	(0.0113)	(0.0057)	(0.0061)
Patronage inspector × Significant excess insp. share	-0.0429*	-0.0474*	-0.0124	-0.0106
	(0.0237)	(0.0247)	(0.0107)	(0.0104)
Patronage administrator × Significant excess insp. share	-0.0092	-0.0097	-0.0048	-0.0082
	(0.0185)	(0.0179)	(0.0071)	(0.0076)
Patronage inspector × Patronage administrator × Significant excess insp. share	0.0123	0.0236	0.0301**	0.0301**
	(0.0286)	(0.0291)	(0.0151)	(0.0145)
Same-gender pair		0.0201**		0.0042*
		(0.0088)		(0.0023)
Same-gender pair × Significant excess insp. share		-0.0006		0.0164***
		(0.0136)		(0.0061)
Similar age pair		-0.0128***		-0.0036
		(0.0049)		(0.0024)
Similar age pair × Significant excess insp. share		-0.0006		0.0143**
		(0.0147)		(0.0061)
Same education pair		-0.0071		0.0013
		(0.0055)		(0.0024)
Same education pair × Significant excess insp. share		-0.0330**		0.0157**
		(0.0145)		(0.0067)
Concurrent hires		-0.0266***		-0.0006
		(0.0067)		(0.0018)
Concurrent hires × Significant excess insp. share		-0.0335*		0.0073
		(0.0195)		(0.0095)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,033	291,877	292,033	291,877
R^2	0.6677	0.6683	0.0677	0.0685

Notes: All regressions follow equation (3) and include inspector, administrator, and customs broker fixed effects. All regressions control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. Columns (1) and (3) reproduce columns (2) and (6) of Table 4. Columns (2) and (4) add the four alternative matches and their interactions with significant excess inspection shares. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Alternative matches are defined as in Table C51. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C55: Patronage pairs, customs fraud detected and workload deviations, controlling for matches on other dimensions

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Workload deviations	-0.0608	-0.0662*	0.0049	-0.0026
	(0.0385)	(0.0371)	(0.0058)	(0.0049)
Patronage inspector ×	-0.1365***	-0.0969***	0.0080	0.0077
Patronage administrator	(0.0351)	(0.0322)	(0.0055)	(0.0057)
Patronage inspector ×	0.0096	0.0048	-0.0080**	-0.0063*
Workload deviations	(0.0139)	(0.0134)	(0.0036)	(0.0034)
Patronage administrator ×	0.0160	0.0153	0.0008	0.0002
Workload deviations	(0.0139)	(0.0141)	(0.0013)	(0.0013)
Patronage inspector × Patronage	-0.0451***	-0.0326*	0.0206***	0.0173***
administrator × Workload deviations	(0.0173)	(0.0188)	(0.0043)	(0.0044)
Same-gender pair		0.1088***		0.0049**
		(0.0361)		(0.0023)
Same-gender pair ×		-0.0004		0.0067***
Workload deviations		(0.0114)		(0.0016)
Similar age pair		-0.0656***		-0.0019
		(0.0176)		(0.0022)
Similar age pair ×		-0.0279**		0.0016
Workload deviations		(0.0136)		(0.0015)
Same education pair		-0.0196		0.0024
		(0.0166)		(0.0023)
Same education pair ×		-0.0273***		0.0043***
Workload deviations		(0.0097)		(0.0013)
Concurrent hires		-0.0581***		0.0001
		(0.0197)		(0.0017)
Concurrent hires ×		0.0128		0.0011
Workload deviations		(0.0105)		(0.0025)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,069	291,913	292,069	291,913
R^2	0.2645	0.2727	0.0269	0.0278

Notes: All regressions include inspector and administrator fixed effects and control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. Columns (1) and (3) reproduce columns (2) and (6) of Table 5. Columns (2) and (4) add the four alternative matches and their interactions with workload deviations. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Alternative matches are defined as in Table C51. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of inspectors present at the port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C56: Patronage pairs and customs fraud value detected by valuation inspectors, controlling for matches on other dimensions

	Fraud value detected						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Patronage inspector ×	-0.1080*	-0.1046*	-0.1084*	-0.1151*	-0.1075*	-0.1093*	-0.1123*
Patronage administrator	(0.0618)	(0.0615)	(0.0623)	(0.0619)	(0.0633)	(0.0623)	(0.0629)
Same-gender pair		0.0528					0.0561
		(0.0373)					(0.0377)
Similar age pair			0.0081				0.0112
			(0.0249)				(0.0257)
Same education pair				0.0166			0.0318
				(0.0271)			(0.0437)
Concurrent hires					-0.0017		-0.0032
					(0.0293)		(0.0303)
Low-capacity pair						0.0098	-0.0427
						(0.0594)	(0.0964)
Mean of Dependent Variable	13,505,530	13,507,186	13,507,186	13,507,186	13,505,530	13,507,186	13,507,186
Interactions w/ insp. char.	X	X	X	X	X	X	X
HS2-product FE	X	X	X	X	X	X	X
Observations	79,589	79,558	79,558	79,558	79,589	79,558	79,558
Pseudo R^2	0.1662	0.1664	0.1662	0.1662	0.1662	0.1662	0.1664

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006) following equation (4), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is the fraud value adjustment (additional tax and penalty payment) in Paraguayan Guaranis. All regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects and control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. Column (1) reproduces column (2) of Table 6. Columns (2)-(6) add one alternative match at a time and column (7) adds all five jointly. Alternative matches are defined as in Table C51. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments on which the valuation inspector detected fraud.

C.14 Tables analyzing tenure heterogeneity

Table C57 evaluates the fraud detection for pairings of distinct patronage types. Panel A focuses on pairings where the inspector is an electoral patronage hire and the administrator is a dictatorship hire, while Panel B examines the reverse pairing, with the inspector as a dictatorship hire and the administrator as an electoral patronage hire. The results in both panels align with those in Table 2. Specifically, in column (1), both panels show that these pairings are associated with significantly lower customs fraud detection. Similarly, in column (2), the pairings exhibit positive coefficients implying fraud detected by the PCA team (but not the port inspector), yet the wide standard errors render these coefficients statistically insignificant.

Table C57: Distinct patronage types and customs fraud detected

	Fraud detected by inspector (1)	Fraud detected by PCA, but not inspector (2)
Panel A: Electoral patronage inspector & Dictatorship administrator		
Electoral patronage inspector × Dictatorship administrator	-0.0774*	0.0006
	(0.0397)	(0.0045)
Mean of Dependent Variable	0.2725	0.0148
Observations	292,069	292,069
R^2	0.2620	0.0218
	(1)	(2)
Panel B: Dictatorship inspector & Electoral patronage administrator		
Dictatorship inspector × Electoral patronage administrator	-0.1363**	0.0087
	(0.0560)	(0.0079)
Mean of Dependent Variable	0.2725	0.0148
Observations	292,069	292,069
R^2	0.2651	0.0218

Notes: All regressions include inspector and administrator fixed effects. All regressions in Panel A control for interactions of the dictatorship administrator indicator with: (1) inspector education and (2) inspector age. All regressions in Panel B control for interactions of the electoral patronage administrator indicator with: (1) inspector education and (2) inspector age. The electoral patronage and dictatorship inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on column (2) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Electoral patronage hires are those inspectors and administrators that joined the customs authority within a year of a new executive administration's start. Dictatorship hires are those that joined during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Next, I examine whether officials' tenure (their employment duration) is significantly associated with fraud detection. Table C58 analyzes the interaction between inspectors' tenure and patronage inspectors and/or administrators in relation to fraud detection. In column (2), only the coefficient for patronage pairs is significant, while none of the coefficients involving inspector tenure are. In column (4), two coefficients involving inspector tenure are significant, but they have opposite signs, such that patronage pairs with inspectors having longer tenure tend to have slightly more undetected fraud that is later uncovered by the PCA team.

Table C58: Patronage hires, customs fraud detected and inspector tenure

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Inspector tenure	-0.0863	-0.1117	0.0574	0.0615
	(0.4169)	(0.4145)	(0.0531)	(0.0534)
Patronage inspector ×	-0.1316***	-0.1446***	0.0028	0.0036
Patronage administrator	(0.0350)	(0.0378)	(0.0052)	(0.0052)
Patronage inspector ×	0.0054	0.0167	-0.0067	-0.0125
Inspector tenure	(0.1507)	(0.1468)	(0.0236)	(0.0241)
Patronage administrator ×		0.0362		-0.0090**
Inspector tenure		(0.0339)		(0.0041)
Patronage inspector × Patronage		-0.0089		0.0098**
administrator × Inspector tenure		(0.0335)		(0.0049)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,069	292,069	292,069	292,069
R^2	0.2643	0.2645	0.0235	0.0237

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and inspector tenure with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Inspector tenure' is standardized and refers to how long the inspector has been at customs by taking into account the date difference between the inspectors' first appointment and the date of the shipment inspection. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

C.15 Table analyzing concurrent hires

I examine whether the performance patterns observed for patronage hires are amplified when inspectors and port administrators are concurrent hires (i.e., if both were appointed by same executive administration). As individuals hired under the same administration likely belong to the same political faction, this measure serves as a rough proxy for shared factional affiliation within their party.[73] Identifying concurrent hires captures cohort dynamics (Bandiera et al., 2009). The regressions use the fixed effects of equation (1), so the table is directly comparable to Table 2. Columns (1)-(2) show that concurrent hires are associated with lower customs fraud detection, although the estimate is imprecise (significant at the 10% level in column (1) only), while the patronage pair coefficient is essentially the same as in Table 2 and its interactions with concurrent hires are not statistically significant. Columns (3)-(4) show no significant relationship between concurrent hires and fraud detection by the PCA team.

[73] While I have data on Paraguayans' partisanship, no comparable data exist on intra-party faction membership. Factional alignment is also fluid in Paraguay and shifts with changes in power (Última Hora, 2007, 2017, 2022).

Table C59: Patronage hires, customs fraud detected and concurrent hires

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Concurrent hires	-0.2497*	-0.2280	-0.0159	-0.0093
	(0.1429)	(0.1457)	(0.0130)	(0.0148)
Patronage inspector ×	-0.1198***	-0.1254***	0.0033	0.0018
Patronage administrator	(0.0330)	(0.0367)	(0.0059)	(0.0066)
Patronage inspector ×	-0.0695	-0.0728	0.0024	-0.0075
Concurrent hires	(0.0463)	(0.0673)	(0.0045)	(0.0075)
Patronage administrator ×		-0.0524		0.0030
Concurrent hires		(0.0510)		(0.0051)
Patronage inspector × Patronage		0.0298		0.0125
administrator × Concurrent hires		(0.0923)		(0.0103)
Mean of Dependent Variable	0.2725	0.2725	0.0148	0.0148
Observations	292,069	292,069	292,069	292,069
R^2	0.2654	0.2656	0.0218	0.0219

Notes: All regressions include inspector and administrator fixed effects. "Concurrent hires" is a variable equal to 1 when the inspector and administrator were hired during the same executive administration. All regressions control for interactions of the patronage administrator indicator and concurrent hires with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

C.16 Additional tables showing columns without inspector and administrator FEs

Table C60: Patronage hires and customs fraud detected

	Fraud detected by inspector			Fraud detected by PCA, but not inspector		
	(1)	(2)	(3)	(4)	(5)	(6)
Patronage inspector	0.0559			0.0051		
	(0.0492)			(0.0053)		
Patronage administrator	-0.3641***			0.0092		
	(0.1322)			(0.0103)		
Patronage inspector × Patronage administrator	-0.1548**	-0.1294***	-0.0356**	-0.0011	0.0045	0.0028
	(0.0624)	(0.0349)	(0.0155)	(0.0061)	(0.0056)	(0.0056)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148
Inspector FE		X	X		X	X
Administrator FE		X	X		X	X
HS2-product FE			X			X
Observations	292,071	292,069	292,069	292,071	292,069	292,069
R^2	0.0253	0.2628	0.5751	0.0012	0.0218	0.0316

Notes: "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects in columns (2)-(3) and (5)-(6). Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (4)-(6) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C61: Patronage pairs and suspect assignments

	Binary significant excess inspection share	
	(1)	(2)
Patronage inspector	-0.0019	
	(0.0012)	
Patronage administrator	0.0006	
	(0.0014)	
Patronage inspector	0.0006	0.0031*
× Patronage administrator	(0.0018)	(0.0018)
Mean of Dependent Variable	0.0082	0.0082
Inspector FE		X
Administrator FE		X
Observations	77,958	77,955
R^2	0.0196	0.0394

Notes: The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects in column (2). Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector combinations which are significantly excessive for 90% or more of the simulations. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. The sample covers all possible non-singleton pairings of valuation inspectors and customs brokers at all semesters and ports.

Table C62: Customs fraud detected, patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector	-0.0063				0.0061			
	(0.0087)				(0.0050)			
Patronage administrator	-0.0161				-0.0097			
	(0.0290)				(0.0091)			
Significant excess	-0.1143**	-0.1187**	-0.1179**	-0.0833*	0.0430*	0.0539**	0.0525**	0.0530**
insp. share	(0.0548)	(0.0489)	(0.0488)	(0.0442)	(0.0223)	(0.0240)	(0.0238)	(0.0236)
Patronage inspector ×	-0.0013	-0.0237**	-0.0245**	-0.0106	-0.0061	0.0027	0.0006	0.0005
Patronage administrator	(0.0133)	(0.0114)	(0.0115)	(0.0091)	(0.0056)	(0.0056)	(0.0057)	(0.0057)
Patronage inspector ×	-0.0124	-0.0351**	-0.0429*	-0.0301	0.0080	0.0058	-0.0124	-0.0126
Significant excess insp. share	(0.0167)	(0.0166)	(0.0237)	(0.0185)	(0.0080)	(0.0082)	(0.0107)	(0.0108)
Patronage administrator ×			-0.0092	-0.0050			-0.0048	-0.0045
Significant excess insp. share			(0.0185)	(0.0172)			(0.0071)	(0.0071)
Patronage inspector × Patronage			0.0123	0.0048			0.0301**	0.0293*
administrator × Significant excess insp. share			(0.0286)	(0.0246)			(0.0151)	(0.0150)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Inspector FE		X	X	X		X	X	X
Administrator FE		X	X	X		X	X	X
HS2-product FE				X				X
Observations	292,035	292,033	292,033	292,033	292,035	292,033	292,033	292,033
R^2	0.6536	0.6677	0.6677	0.7118	0.0508	0.0675	0.0677	0.0745

Notes: All regressions include customs broker fixed effects. "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. All regressions control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects in columns (2)-(4) and (6)-(8). Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C63: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector	0.0550				0.0050			
	(0.0483)				(0.0055)			
Patronage administrator	-0.3350**				0.0045			
	(0.1324)				(0.0092)			
Workload deviations	-0.0863	-0.0661*	-0.0608	-0.0146	0.0005	0.0144*	0.0049	0.0022
	(0.0677)	(0.0373)	(0.0385)	(0.0186)	(0.0094)	(0.0080)	(0.0058)	(0.0058)
Patronage inspector ×	-0.1471**	-0.1287***	-0.1365***	-0.0384**	-0.0011	0.0036	0.0080	0.0064
Patronage administrator	(0.0613)	(0.0350)	(0.0351)	(0.0158)	(0.0059)	(0.0056)	(0.0055)	(0.0055)
Patronage inspector ×	-0.0277	-0.0241**	0.0096	0.0049	0.0095**	0.0091***	-0.0080**	-0.0078**
Workload deviations	(0.0191)	(0.0101)	(0.0139)	(0.0069)	(0.0037)	(0.0032)	(0.0036)	(0.0035)
Patronage administrator ×			0.0160	0.0089			0.0008	0.0013
Workload deviations			(0.0139)	(0.0062)			(0.0013)	(0.0013)
Patronage inspector × Patronage			-0.0451***	-0.0184**			0.0206***	0.0198***
administrator × Workload deviations			(0.0173)	(0.0079)			(0.0043)	(0.0042)
Mean of Dependent Variable	0.2725	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
Inspector FE		X	X	X		X	X	X
Administrator FE		X	X	X		X	X	X
HS2-product FE				X				X
Observations	292,071	292,069	292,069	292,069	292,071	292,069	292,069	292,069
R^2	0.0285	0.2641	0.2645	0.5754	0.0052	0.0250	0.0269	0.0364

Notes: "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. All regressions control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects in columns (2)-(4) and (6)-(8). Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C64: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected	
	(1)	(2)
Patronage inspector	0.0960	
	(0.1015)	
Patronage administrator	0.5938**	
	(0.2449)	
Patronage inspector ×	-0.1939	-0.1080*
Patronage administrator	(0.1211)	(0.0618)
Mean of Dependent Variable	13,505,465	13,505,530
Inspector FE		X
Administrator FE		X
HS2-product FE	X	X
Observations	79,590	79,589
Pseudo R^2	0.0494	0.1662

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include product chapter (HS 2-digit code) fixed effects. The regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects in column (2). Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

C.17 Main results showing all coefficients (including covariates)

Table C65: Patronage hires and customs fraud detected

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Patronage inspector ×	-0.0705**	-0.1294***	-0.0356**	-0.0356**	0.0036	0.0045	0.0028	0.0028
Patronage administrator	(0.0317)	(0.0349)	(0.0155)	(0.0153)	(0.0038)	(0.0056)	(0.0056)	(0.0055)
Patronage administrator ×		-0.0029	-0.0043	-0.0042		0.0006	0.0008	0.0008
Inspector education level		(0.0092)	(0.0039)	(0.0039)		(0.0014)	(0.0014)	(0.0014)
Patronage administrator ×		0.0058***	0.0020***	0.0020***		0.0000	0.0001	0.0001
Inspector age		(0.0016)	(0.0007)	(0.0007)		(0.0002)	(0.0002)	(0.0002)
Evasion risk index				0.0204***				0.0020***
				(0.0028)				(0.0006)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
HS2-product FE			X	X			X	X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2608	0.2628	0.5751	0.5766	0.0217	0.0218	0.0316	0.0317

Notes: All regressions include inspector and administrator fixed effects. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The education level variable goes from no education mentioned (0), primary (1), secondary (2), some professional/vocational/technician (non-academic) training (3), some (academic) college courses (4), college degree (5), master's degree (6), and doctorate (7). "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C66: Customs fraud detected, patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Significant excess	0.0149	-0.1179**	-0.0833*	-0.0833*	0.0121**	0.0525**	0.0530**	0.0529**
insp. share	(0.0143)	(0.0488)	(0.0442)	(0.0442)	(0.0059)	(0.0238)	(0.0236)	(0.0236)
Patronage inspector ×	-0.0079	-0.0245**	-0.0106	-0.0107	0.0009	0.0006	0.0005	0.0004
Patronage administrator	(0.0097)	(0.0115)	(0.0091)	(0.0091)	(0.0039)	(0.0057)	(0.0057)	(0.0057)
Patronage inspector ×	-0.0291	-0.0429*	-0.0301	-0.0300	-0.0079	-0.0124	-0.0126	-0.0124
Significant excess insp. share	(0.0214)	(0.0237)	(0.0185)	(0.0185)	(0.0099)	(0.0107)	(0.0108)	(0.0108)
Patronage administrator ×	0.0036	-0.0092	-0.0050	-0.0049	-0.0026	-0.0048	-0.0045	-0.0043
Significant excess insp. share	(0.0181)	(0.0185)	(0.0172)	(0.0172)	(0.0075)	(0.0071)	(0.0071)	(0.0070)
Patronage inspector × Patronage	-0.0026	0.0123	0.0048	0.0048	0.0276*	0.0301**	0.0293*	0.0293*
administrator × Significant excess insp. share	(0.0280)	(0.0286)	(0.0246)	(0.0246)	(0.0155)	(0.0151)	(0.0150)	(0.0151)
Evasion risk index				0.0014				0.0028***
				(0.0016)				(0.0006)
Patronage administrator ×		-0.0033	-0.0033	-0.0034		0.0012	0.0012	0.0012
Inspector education level		(0.0029)	(0.0022)	(0.0022)		(0.0014)	(0.0014)	(0.0014)
Patronage administrator ×		0.0012**	0.0008*	0.0008*		0.0002	0.0002	0.0002
Inspector age		(0.0006)	(0.0004)	(0.0004)		(0.0002)	(0.0002)	(0.0002)
Significant excess insp. share ×		0.0092*	0.0065	0.0065		-0.0051**	-0.0052**	-0.0052**
Inspector education level		(0.0052)	(0.0045)	(0.0045)		(0.0024)	(0.0024)	(0.0024)
Significant excess insp. share ×		0.0022***	0.0016**	0.0016**		-0.0004	-0.0004	-0.0004
Inspector age		(0.0008)	(0.0007)	(0.0007)		(0.0004)	(0.0004)	(0.0004)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
HS2-product FE			X	X			X	X
Observations	294,041	292,033	292,033	292,033	294,041	292,033	292,033	292,033
R^2	0.6667	0.6677	0.7118	0.7118	0.0671	0.0677	0.0745	0.0749

Notes: All regressions include inspector, administrator, and customs broker fixed effects. The patronage inspector and administrator variables are absorbed by the inspector and administrator fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The education level variable goes from no education mentioned (0), primary (1), secondary (2), some professional/vocational/technician (non-academic) training (3), some (academic) college courses (4), college degree (5), master's degree (6), and doctorate (7). "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share she would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C67: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector				Fraud detected by PCA, but not inspector			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Workload deviations	0.0070	-0.0608	-0.0146	-0.0144	-0.0029**	0.0049	0.0022	0.0023
	(0.0095)	(0.0385)	(0.0186)	(0.0184)	(0.0012)	(0.0058)	(0.0058)	(0.0058)
Patronage inspector ×	-0.0798**	-0.1365***	-0.0384**	-0.0385**	0.0073**	0.0080	0.0064	0.0064
Patronage administrator	(0.0316)	(0.0351)	(0.0158)	(0.0156)	(0.0037)	(0.0055)	(0.0055)	(0.0054)
Patronage inspector ×	0.0077	0.0096	0.0049	0.0055	-0.0054*	-0.0080**	-0.0078**	-0.0077**
Workload deviations	(0.0136)	(0.0139)	(0.0069)	(0.0068)	(0.0029)	(0.0036)	(0.0035)	(0.0035)
Patronage administrator ×	0.0193	0.0160	0.0089	0.0095	0.0009	0.0008	0.0013	0.0014
Workload deviations	(0.0145)	(0.0139)	(0.0062)	(0.0061)	(0.0014)	(0.0013)	(0.0013)	(0.0013)
Patronage inspector × Patronage	-0.0548***	-0.0451***	-0.0184**	-0.0191**	0.0207***	0.0206***	0.0198***	0.0197***
administrator × Workload deviations	(0.0180)	(0.0173)	(0.0079)	(0.0079)	(0.0045)	(0.0043)	(0.0042)	(0.0042)
Evasion risk index				0.0204***				0.0021***
				(0.0028)				(0.0006)
Patronage administrator ×		-0.0037	-0.0045	-0.0044		0.0011	0.0012	0.0012
Inspector education level		(0.0093)	(0.0040)	(0.0040)		(0.0014)	(0.0014)	(0.0014)
Patronage administrator ×		0.0055***	0.0019**	0.0019**		0.0002	0.0002	0.0002
Inspector age		(0.0017)	(0.0008)	(0.0007)		(0.0002)	(0.0002)	(0.0002)
Workload deviations ×		0.0061*	0.0021	0.0020		-0.0014*	-0.0012	-0.0012
Inspector education level		(0.0034)	(0.0018)	(0.0018)		(0.0008)	(0.0008)	(0.0008)
Workload deviations ×		0.0009	0.0001	0.0001		-0.0000	0.0000	0.0000
Inspector age		(0.0006)	(0.0003)	(0.0003)		(0.0001)	(0.0001)	(0.0001)
Mean of Dependent Variable	0.2711	0.2725	0.2725	0.2725	0.0148	0.0148	0.0148	0.0148
HS2-product FE			X	X			X	X
Observations	294,077	292,069	292,069	292,069	294,077	292,069	292,069	292,069
R^2	0.2623	0.2645	0.5754	0.5768	0.0266	0.0269	0.0364	0.0366

Notes: All regressions include inspector and administrator FEs. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Regressions in columns (2)-(4) and (6)-(8) control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The education level variable goes from no education mentioned (0), primary (1), secondary (2), some professional/vocational/technician (non-academic) training (3), some (academic) college courses (4), college degree (5), master's degree (6), and doctorate (7). "HS2-product FE" corresponds to product chapter (HS 2-digit code) fixed effects. "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. The outcome variable on columns (5)-(8) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments.

Table C68: Patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected		
	(1)	(2)	(3)
Patronage inspector ×	-0.1112***	-0.1080*	-0.0976*
Patronage administrator	(0.0425)	(0.0618)	(0.0576)
Patronage administrator ×		-0.0127	-0.0025
Inspector education level		(0.0144)	(0.0128)
Patronage administrator ×		-0.0017	-0.0017
Inspector age		(0.0027)	(0.0024)
Evasion risk index			0.4963***
			(0.0174)
Mean of Dependent Variable	13,511,141	13,505,530	13,505,530
HS2-product FE	X	X	X
Observations	79,709	79,589	79,589
Pseudo R^2	0.1659	0.1662	0.3099

Notes: The table runs Poisson pseudo-maximum likelihood (PPML) regressions (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regressions include inspector, administrator, and product chapter (HS 2-digit code) fixed effects. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Regressions in columns (2)-(3) control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The education level variable goes from no education mentioned (0), primary (1), secondary (2), some professional/vocational/technician (non-academic) training (3), some (academic) college courses (4), college degree (5), master's degree (6), and doctorate (7). "Evasion risk index" consists of an index for tax evasion risk variables. It consists of the unweighted average of standardized variables capturing potential evasion: three risk scores from customs' risk management unit (the SAS/SGIR score, the SBR risk score, and their weighted average used in practice), the declared CIF value (tax base), a mixed-shipment indicator (whether the shipment includes products from at least two six-digit HS codes), the share of differentiated items, shipment weight, a weighted undervaluation measure (difference between declared and median unit prices, aggregated across items using product weights), the share of time-sensitive products, and an indicator for registration outside normal operating hours at the relevant customs port. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the fraud value adjustment made, which is the additional tax and penalty fine payment needed by the customs broker due to customs fraud being uncovered on the broker's shipment. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments with some fraud value detected by the valuation inspector.

C.18 Tables for shipments with inspector-broker pairs without *suspect assignments*

Table C69: Patronage hires and customs fraud detected for shipments with inspector-broker pairs without suspect assignments

	Fraud detected by inspector (1)	Fraud detected by PCA, but not inspector (2)
Patronage inspector × Patronage administrator	-0.1274***	0.0031
	(0.0350)	(0.0056)
Mean of Dependent Variable	0.2739	0.0143
Observations	281,839	281,839
R^2	0.2649	0.0201

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on column (2) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments, excluding shipments with inspector-broker pairs with significant excess inspection shares.

Table C70: Patronage hires, customs fraud detected and workload deviations at customs ports each day for shipments with inspector-broker pairs without suspect assignments

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Workload deviations	-0.0477	-0.0518	-0.0028	-0.0039
	(0.0329)	(0.0336)	(0.0029)	(0.0028)
Patronage inspector ×	-0.1272***	-0.1335***	0.0030	0.0047
Patronage administrator	(0.0351)	(0.0354)	(0.0055)	(0.0054)
Patronage inspector ×	-0.0167*	0.0071	0.0007	-0.0042
Workload deviations	(0.0094)	(0.0120)	(0.0011)	(0.0029)
Patronage administrator ×		0.0143		0.0006
Workload deviations		(0.0120)		(0.0012)
Patronage inspector × Patronage		-0.0365**		0.0070**
administrator × Workload deviations		(0.0164)		(0.0031)
Mean of Dependent Variable	0.2739	0.2739	0.0143	0.0143
Observations	281,839	281,839	281,839	281,839
R^2	0.2658	0.2660	0.0205	0.0207

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments, excluding shipments with inspector-broker pairs with significant excess inspection shares.

Table C71: Patronage hires and customs fraud value detected by valuation inspectors for shipments with inspector-broker pairs without suspect assignments

	Fraud value detected
	(1)
Patronage inspector ×	-0.1177*
Patronage administrator	(0.0615)
Mean of Dependent Variable	13,560,117
HS2-product FE	X
Observations	77,197
Pseudo R^2	0.1643

Notes: The table runs a Poisson pseudo-maximum likelihood (PPML) regression (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regression includes inspector, administrator, and product chapter (HS 2-digit code) fixed effects. The regression controls for interactions of the patronage administrator indicator with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. The sample covers all red channel shipments with some fraud value detected by the valuation inspector. The sample excludes shipments with inspector-broker pairs with significant excess inspection shares.

C.19 Tables for electoral patronage hires

Table C72: Electoral patronage hires and customs fraud detected

	Fraud detected by inspector (1)	Fraud detected by PCA, but not inspector (2)
Electoral patronage inspector× Electoral patronage administrator	-0.0569	-0.0033
	(0.0366)	(0.0051)
Mean of Dependent Variable	0.2667	0.0153
Observations	242,970	242,970
R^2	0.2568	0.0239

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the electoral patronage administrator indicator with: (1) inspector education and (2) inspector age. The electoral patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on column (2) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Electoral patronage hires are those inspectors and administrators that joined the customs authority within a year of a new executive administration's start during Paraguay's democratic period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments, except those handled by dictatorship inspectors as these form the other treatment group.

Table C73: Electoral patronage pairs and suspect assignments

	Binary significant excess inspection share (1)
Electoral patronage inspector × Electoral patronage administrator	0.0006
	(0.0022)
Mean of Dependent Variable	0.0073
Observations	64,769
R^2	0.0244

Notes: All regressions include inspector, administrator, and customs broker fixed effects. The electoral patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector combinations which are significantly excessive for 90% or more of the simulations. Electoral patronage hires are those inspectors and administrators that joined the customs authority within a year of a new executive administration's start during Paraguay's democratic period. The sample covers all possible pairings of valuation inspectors and customs brokers at all semesters and ports in my data, except those involving dictatorship inspectors as these are the other part of the treatment group.

Table C74: Customs fraud detected, electoral patronage hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Significant excess insp. share	-0.0775	-0.0732	0.0694***	0.0648***
	(0.0501)	(0.0494)	(0.0209)	(0.0199)
Electoral patronage inspector × Electoral patronage administrator	-0.0032	0.0019	-0.0068	-0.0106**
	(0.0151)	(0.0149)	(0.0045)	(0.0048)
Electoral patronage inspector × Significant excess insp. share	-0.0454**	-0.0262	-0.0015	-0.0148*
	(0.0177)	(0.0186)	(0.0086)	(0.0089)
Electoral patronage administrator × Significant excess insp. share		0.0249		-0.0022
		(0.0184)		(0.0070)
Electoral patronage inspector × Electoral patronage administrator × Significant excess insp. share		-0.0689*		0.0512*
		(0.0399)		(0.0275)
Mean of Dependent Variable	0.2667	0.2667	0.0153	0.0153
Observations	242,930	242,930	242,930	242,930
R^2	0.6554	0.6554	0.0702	0.0706

Notes: All regressions include inspector, administrator, and customs broker fixed effects. All regressions control for interactions of the electoral patronage administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The electoral patronage inspector variable is absorbed by the inspector fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Excess inspection share is the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share he would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. Electoral patronage hires are those inspectors and administrators that joined the customs authority within a year of a new executive administration's start during Paraguay's democratic period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments, except those handled by dictatorship inspectors as these form the other treatment group.

Table C75: Electoral patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Workload deviations	-0.0358	-0.0364	0.0081	0.0043
	(0.0450)	(0.0449)	(0.0051)	(0.0041)
Electoral patronage inspector ×	-0.0539	-0.0569	-0.0056	-0.0009
Electoral patronage administrator	(0.0358)	(0.0362)	(0.0046)	(0.0046)
Electoral patronage inspector ×	-0.0342***	-0.0250*	0.0104***	-0.0047
Workload deviations	(0.0108)	(0.0144)	(0.0032)	(0.0031)
Electoral patronage administrator ×		0.0097		0.0008
Workload deviations		(0.0204)		(0.0014)
Electoral patronage inspector × Electoral		-0.0175		0.0200***
patronage administrator × Workload deviations		(0.0241)		(0.0033)
Mean of Dependent Variable	0.2667	0.2667	0.0153	0.0153
Observations	242,970	242,970	242,970	242,970
R^2	0.2587	0.2587	0.0282	0.0302

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the electoral patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The electoral patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Electoral patronage hires are those inspectors and administrators that joined the customs authority within a year of a new executive administration's start during Paraguay's democratic period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments, except those handled by dictatorship inspectors as these form the other treatment group.

Table C76: Electoral patronage hires and customs fraud value detected by valuation inspectors

	Fraud value detected
	(1)
Electoral patronage inspector ×	-0.1195*
Electoral patronage administrator	(0.0672)
Mean of Dependent Variable	13,505,200
HS2-product FE	X
Observations	64,789
Pseudo R^2	0.1791

Notes: The table runs a Poisson pseudo-maximum likelihood (PPML) regression (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regression includes inspector, administrator, and product chapter (HS 2-digit code) fixed effects. The regression controls for interactions of the electoral patronage administrator indicator with: (1) inspector education and (2) inspector age. The electoral patronage inspector indicator variable is absorbed by the inspector fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Electoral patronage hires are those inspectors and administrators that joined the customs authority within a year of a new executive administration's start during Paraguay's democratic period. The sample covers all red channel shipments with some fraud value detected by the valuation inspector, except those handled by dictatorship inspectors as these form the other treatment group.

C.20 Tables for dictatorship hires

Table C77: Dictatorship hires and customs fraud detected

	Fraud detected by inspector (1)	Fraud detected by PCA, but not inspector (2)
Dictatorship inspector × Dictatorship administrator	-0.0359	0.0158**
	(0.0656)	(0.0065)
Mean of Dependent Variable	0.2839	0.0126
Observations	229,722	229,722
R^2	0.2552	0.0144

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the dictatorship administrator indicator with: (1) inspector education and (2) inspector age. The dictatorship inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on column (2) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Dictatorship hires are those inspectors and administrators that joined the customs authority during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments, except those handled by electoral patronage inspectors as these form the other treatment group.

Table C78: Dictatorship pairs and suspect assignments

	Binary significant excess inspection share (1)
Dictatorship inspector × Dictatorship administrator	0.0026
	(0.0037)
Mean of Dependent Variable	0.0075
Observations	61,038
R^2	0.0276

Notes: All regressions include inspector, administrator, and customs broker fixed effects. The dictatorship inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector combinations which are significantly excessive for 90% or more of the simulations. Dictatorship hires are those inspectors and administrators that joined the customs authority during Paraguay's dictatorship period. The sample covers all possible pairings of valuation inspectors and customs brokers at all semesters and ports in my data, except those involving electoral patronage inspectors as those correspond to the other part of the treatment group.

Table C79: Customs fraud detected, dictatorship hires, and suspect assignments between customs brokers and inspectors

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Significant excess insp. share	-0.1284**	-0.1293**	0.0483*	0.0482*
	(0.0576)	(0.0560)	(0.0259)	(0.0264)
Dictatorship inspector × Dictatorship administrator	-0.0243	-0.0289	0.0157**	0.0151**
	(0.0171)	(0.0177)	(0.0066)	(0.0065)
Dictatorship inspector × Significant excess insp. share	-0.0222	-0.0396	0.0208	0.0187
	(0.0231)	(0.0244)	(0.0144)	(0.0165)
Dictatorship administrator × Significant excess insp. share		-0.0329**		-0.0039
		(0.0166)		(0.0064)
Dictatorship inspector × Dictatorship administrator × Significant excess insp. share		0.0572		0.0070
		(0.0405)		(0.0216)
Mean of Dependent Variable	0.2839	0.2839	0.0126	0.0126
Observations	229,688	229,688	229,688	229,688
R^2	0.6874	0.6875	0.0625	0.0625

Notes: All regressions include inspector, administrator, and customs broker fixed effects. All regressions control for interactions of the dictatorship administrator indicator and significant excess inspection shares with: (1) inspector education and (2) inspector age. The dictatorship inspector variable is absorbed by the inspector fixed effects. Standard errors in parentheses are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Excess inspection share is the difference between the share of a given broker's shipments handled by an inspector in a given semester and the hypothetical share he would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to the proportion of inspection shares that are larger than the 99th percentile of the respective simulated multinomial assignment shares. This latter variable varies between 0 and 1. Dictatorship hires are those inspectors and administrators that joined the customs authority during Paraguay's dictatorship period. "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments, except those handled by electoral patronage inspectors as these form the other treatment group.

Table C80: Dictatorship hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Workload deviations	-0.0456	-0.0540	-0.0013	-0.0011
	(0.0425)	(0.0422)	(0.0040)	(0.0041)
Dictatorship inspector ×	-0.0425	-0.0472	0.0156**	0.0157**
Dictatorship administrator	(0.0653)	(0.0646)	(0.0065)	(0.0065)
Dictatorship inspector ×	0.0055	0.0201	0.0034	0.0030
Workload deviations	(0.0170)	(0.0182)	(0.0022)	(0.0029)
Dictatorship administrator ×		0.0248		-0.0003
Workload deviations		(0.0155)		(0.0009)
Dictatorship inspector × Dictatorship		-0.0514*		0.0013
administrator × Workload deviations		(0.0287)		(0.0029)
Mean of Dependent Variable	0.2839	0.2839	0.0126	0.0126
Observations	229,722	229,722	229,722	229,722
R^2	0.2569	0.2574	0.0147	0.0147

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the dictatorship administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The dictatorship inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (3)-(4) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Dictatorship hires are those inspectors and administrators that joined the customs authority during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers all red channel shipments, except those handled by electoral patronage inspectors as these form the other treatment group.

Table C81: Dictatorship hires and customs fraud value detected by valuation inspectors

	Fraud value detected
	(1)
Dictatorship inspector ×	0.0421
Dictatorship administrator	(0.0785)
Mean of Dependent Variable	13,574,923
HS2-product FE	X
Observations	65,207
Pseudo R^2	0.1603

Notes: The table runs a Poisson pseudo-maximum likelihood (PPML) regression (Santos Silva and Tenreyro, 2006), where the coefficients are interpreted as semi-elasticities: a coefficient β implies a $100 \times (e^{\beta} - 1)\%$ change in the expected value of the dependent variable. The dependent variable is measured in the local currency (Paraguayan Guaranis, PYG). The US dollar (USD) to PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. The regression includes inspector, administrator, and product chapter (HS 2-digit code) fixed effects. The regression controls for interactions of the dictatorship administrator indicator with: (1) inspector education and (2) inspector age. The dictatorship inspector indicator variable is absorbed by the inspector fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. Dictatorship hires are those inspectors and administrators that joined the customs authority during Paraguay's dictatorship period. The sample covers all red channel shipments with some fraud value detected by the valuation inspector. The sample excludes those handled by electoral patronage inspectors as these form the other treatment group.

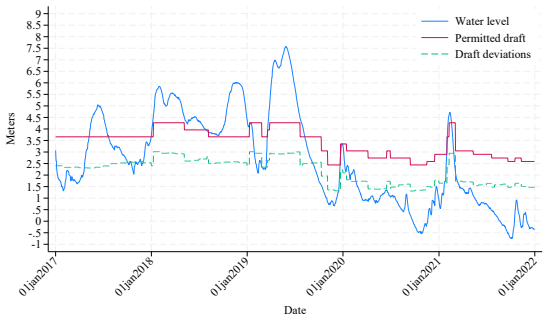
C.21 Results using water level deviations to proxy for workload deviations at river ports

To derive a more exogenous measure affecting the workload deviations discussed in the main text that cannot be influenced by customs officials, I use exogenous changes in the water level of the Paraguay River, the primary waterway in Paraguay. These changes influence the type and size of vessels that can navigate the river. In particular, the General Naval Prefecture of Paraguay regulates the maximum permitted vessel draft—the vertical distance between the waterline and a vessel's keel. Following Ignatenko (2024), I measure deviations in the water level and vessel draft from their monthly average across years to capture unexpected changes in them and I use these deviations to proxy for the total shipment volume at river ports. The relevant ports on the Paraguay River are Villeta, Pilar, Terport, Terport Villeta, *Puerto Seguro Fluvial*, Caacupemi, Caacupemi Pilar, *Puertos y Estibajes*, Empedril, and Concepción.

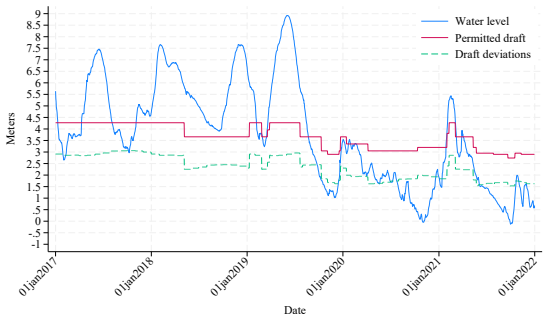
I obtained daily data on the permitted vessel draft from the General Naval Prefecture for four key upstream locations: Pilar, Asunción, Concepción, and Villeta. Similarly, I obtained data on the water level for these locations from the Directorate of Meteorology and Hydrology of Paraguay. I then linked these data to their corresponding river ports. Through this data I identify days when the vessel draft supports higher or lower shipment volumes, with a variable that subtracts the monthly average vessel draft across years with the current vessel draft to obtain the vessel draft deviations. Figure C5 below plots the daily water level, daily permitted vessel draft and draft deviations for the four upstream locations.

Figure C5: Daily permitted and expected vessel draft for upstream locations on the Paraguay river

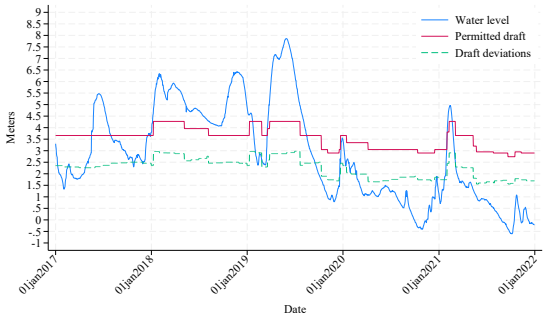
(a) Asunción



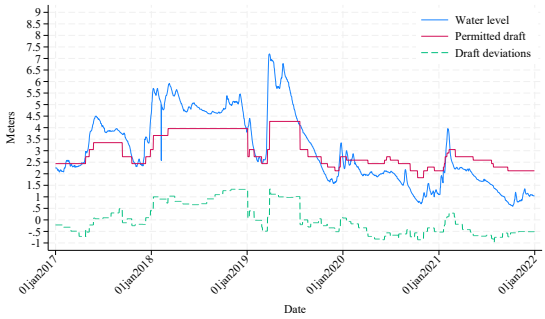
(b) Pilar



(c) Villeta



(d) Concepción



Notes: Each blue solid line plot shows the daily water level for four upstream locations on the Paraguay river: Pilar, Asunción, Villeta and Concepción. The water level is measured in meters and 0 is a reference point (elevation) above the mean sea level. Each solid red line plot shows the daily permitted vessel drafts and each dashed green line plots the draft deviations (from the monthly average across years) for four upstream locations on the Paraguay river: Pilar, Villeta, Asunción (particularly, *Remanso Castillo* in the greater Asunción area where several river ports are located), and Concepción. The vessel draft is measured in meters.

Table C82 explores the relationship between the river's water levels, vessel drafts, their deviations from monthly averages, and workload deviations. First, as a sanity check, we confirm that higher water levels are associated with higher vessel drafts, and similarly, deviations in water levels predict deviations in vessel drafts. We also find that both water level and vessel draft deviations are significant predictors of total red channel shipments per inspector per day at each customs port. Lastly, we confirm that workload deviations are influenced by both water level and vessel draft deviations. The stronger relationship between vessel drafts and shipment volumes likely stems from the fact that vessel drafts serve as a policy variable, capturing critical thresholds for vessel navigation. In contrast, beyond a certain point, higher water levels do not necessarily translate into proportionately higher shipment volumes. As noted by Ignatenko (2024), the river becomes unnavigable for standard barges upstream when water levels fall below three meters.

Table C82: Relationship between the river water level and the vessel draft, their deviations from monthly averages, and workload deviations

	Vessel draft	Vessel draft deviations	Total shipments in a day		Workload deviations	
	(1)	(2)	(3)	(4)	(5)	(6)
Water level	0.2224***					
	(0.0034)					
Water level deviations		0.8392***	0.4260***		0.0982***	
		(0.0137)	(0.0638)		(0.0147)	
Vessel draft deviations				0.2166***		0.0499***
				(0.0511)		(0.0118)
Mean of Dependent Variable	3.4447	-0.0000	5.6753	5.6753	-0.0000	-0.0000
Observations	8,103	8,103	8,103	8,103	8,103	8,103
R^2	0.7213	0.7043	0.0096	0.0025	0.0096	0.0025

Notes: Standard errors are clustered at the day level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Vessel draft deviations" is standardized and refers to the deviations in the vessel draft from its monthly average across years to capture unexpected changes in the vessel draft. "Water level deviations' is standardized and refers to the deviations in the water level from its monthly average across years to capture unexpected changes in the water level. The outcome variable for Columns (3)-(4) and (5)-(6) are "Total shipments in a day" and "Workload deviations" respectively. "Total shipments in a day" refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Workload deviations" is the standardized version (with mean 0 and standard deviation of 1) of "Total shipments in a day." "Observations" are at the port-day level for river ports.

Table C83 confirms that the water level, vessel draft, and workload deviations predict the total number of shipments assigned to inspectors each day, including patronage hires. Column (3) suggests that patronage inspectors get 3.5 more shipments per additional standard deviation in workload at the port as a whole. In contrast, Table C84 shows that water level deviations do not correlate with various measures of shipment value, including total taxes paid, the CIF (cost, insurance, and freight) value, and the FOB (free on board) value of the products. However, greater water level deviations are associated with lower freight costs. This aligns with Ignatenko (2024), who finds that higher water levels on the Paraguay River increase competition among transport companies, as more firms can operate when the water level is sufficiently high. Conversely, when water levels drop too low, some transport companies are unable to compete due to them lacking smaller vessels capable of navigating shallower waters.

We can also highlight that water level deviations do not correlate with suspect assignments or differentiated products. Table C85 indicates there is no significant correlation between excess inspection shares of inspectors towards customs brokers or importers and water level deviations. Finally, Table C86 shows that water level deviations do not correlate with the share of differentiated items or whether a shipment has differentiated items, which are products subject to greater tariff evasion due to the difficulties associated with assessing their quality and price (Javorcik and Narciso, 2008).

Table C83: Inspectors' total shipments assigned for the day, patronage hires, and deviations of the water level, vessel draft, and port workload

	Total shipments assigned to inspector on date		
	(1)	(2)	(3)
Water level deviations	0.3629***		
	(0.0541)		
Vessel draft deviations		0.3005***	
		(0.0539)	
Workload deviations			3.5486***
			(0.0144)
Patronage inspector × Water level deviations	0.2370***		
	(0.0666)		
Patronage inspector × Vessel draft deviations		0.1350**	
		(0.0627)	
Patronage inspector × Workload deviations			-0.0248
			(0.0327)
Mean of Dependent Variable	5.4294	5.4294	5.4294
Observations	17,577	17,577	17,577
R^2	0.1271	0.1222	0.7709

Notes: All regressions include inspector fixed effects. Standard errors are clustered at the day level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable is the total number of red channel shipments assigned to the inspector on a particular day. "Vessel draft deviations" is standardized and refers to the deviations in the vessel draft from its monthly average across years to capture unexpected changes in the vessel draft. "Water level deviations' is standardized and refers to the deviations in the water level from its monthly average across years to capture unexpected changes in the water level. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" are at the inspector-port-day level for river ports.

Table C84: Relationship between the water level deviations and shipment values

	Total taxes paid on shipment (in PYG) (1)	Total declared CIF value of shipment (in USD) (2)	Total declared FOB value of shipment (in USD) (3)	Total declared freight cost of shipment (in USD) (4)	Total final CIF value of shipment (in USD) (5)	Total final FOB value of shipment (in USD) (6)	Total final freight cost of shipment (in USD) (7)
Water level deviations	-592,701	816	1,067	-289***	296	644	-379***
	(930,148)	(900)	(848)	(71)	(901)	(849)	(73)
Mean of Dependent Variable	81,505,359	77,300	70,966	5,795	80,410	73,688	6,151
Observations	94,714	94,714	94,714	94,714	94,714	94,714	94,714
R^2	0.0000	0.0000	0.0000	0.0003	0.0000	0.0000	0.0006

Notes: Standard errors are clustered at the day level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Water level deviations' is standardized and refers to the deviations in the water level from its monthly average across years to capture unexpected changes in the water level. "Taxes paid" refers to the total taxes paid for the shipment in the local currency, Paraguayan Guaranis (PYG). The US Dollar (USD)-PYG exchange rate hovered between 5,500-7,100 PYG per USD during the time period of my data. "CIF value" refers to the CIF (cost, insurance, and freight) value for the shipment in US dollars. The difference between "declared" and "final" values is that declared values are those originally reported by customs brokers while final values are the values after shipments are inspected. The sample covers red channel shipments at the Paraguay river customs ports.

Table C85: Relationship between the water level deviations and suspect assignments

	Inspector-broker signif. excess inspection share (1)	Binary inspector-broker signif. excess inspection share (2)	Inspector-importer signif. excess inspection share (3)	Binary inspector-importer signif. excess inspection share (4)
Water level deviations	-0.0013	0.0006	-0.0002	0.0004
	(0.0012)	(0.0010)	(0.0013)	(0.0010)
Mean of Dependent Variable	0.0589	0.0264	0.0567	0.0232
Observations	94,714	94,714	94,714	94,714
R^2	0.0000	0.0000	0.0000	0.0000

Notes: Standard errors are clustered at the day level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Water level deviations' is standardized and refers to the deviations in the water level from its monthly average across years to capture unexpected changes in the water level. Suspect assignments are identified through excess inspection shares, which are the difference between the share of a given broker's (importer's) shipments handled by an inspector in a given semester and the hypothetical share he would be expected to handle if the allocation of shipments to inspectors were randomly assigned. Significant excess inspection shares refers to how often the inspection shares are larger than the 99th percentile of the respective simulated multinomial assignment shares. The binary significant excess inspection share refers to broker-inspector (importer-inspector) combinations which are significantly excessive for 90% or more of the simulations. The sample covers red channel shipments at the Paraguay river customs ports.

Table C86: Relationship between the water level deviations and differentiated products

	Share of differentiated items (1)	Any differentiated item (2)
Water level deviations	0.0014	0.0002
	(0.0015)	(0.0014)
Mean of Dependent Variable	0.8328	0.8563
Observations	94,714	94,714
R^2	0.0000	0.0000

Notes: Standard errors are clustered at the day level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. "Water level deviations' is standardized and refers to the deviations in the water level from its monthly average across years to capture unexpected changes in the water level. "Share of differentiated items" corresponds to the share (out of the total number of items in the shipment) that have a differentiated product, which is a product with a wide quality spectrum. "Any differentiated item" is a dummy variable equal to 1 when the shipment has at least one item with a differentiated product. The sample covers red channel shipments at the Paraguay river customs ports.

Table C87 presents regression results using unexpected water level deviations as a proxy for workload deviations, as these water level deviations affect the types of vessels able to access river ports and, consequently, the total shipment volume. The full regression on Column (2) with the triple interaction indicates that vertical patronage alignment magnifies underperformance: for each additional SD in workload, patronage pairs detect about 2.16 percentage points less fraud, equivalent to around 7% of the dependent variable mean. When focusing on fraud detected by the post-clearance audit team, we only find a negative significant relationship with water level deviations. Table C88 shows similar results when we focus on vessel draft deviations instead. The full regression on Column (2) with the triple interaction reveals that for each additional standard deviation in workload, patronage pairs detect approximately 4.31 percentage points less customs fraud, equivalent to around 14.9% of the dependent variable mean. Likewise, on Column (4) we observe a negative relationship between vessel draft deviations and fraud detected by the PCA team, though the interaction with patronage administrators partially offsets this. Table C89 runs the same analyses as Table 5, while restricting the sample to the Paraguay river customs ports, to make it comparable to the samples of Tables C87 and C88. It shows a similar pattern for Columns (1) and (2), and on Column (4) we find that patronage administrators handling shipments with an additional standard deviation in workload have 0.69 percentage points (29% more of the dependent variable mean) more fraud detected by the PCA team. This divergence likely arises from the fact that workload deviations directly affect inspectors' actions, whereas vessel draft and water level deviations influence inspectors and administrators more indirectly.

Table C87: Patronage hires, customs fraud detected and water level deviations at customs ports each day

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Water level deviations	0.0294	0.0005	-0.0353**	-0.0373**
	(0.0324)	(0.0339)	(0.0151)	(0.0155)
Patronage inspector ×	-0.0095	-0.0085	-0.0041	-0.0043
Patronage administrator	(0.0231)	(0.0225)	(0.0086)	(0.0086)
Patronage inspector ×	-0.0137	0.0095	-0.0015	-0.0004
Water level deviations	(0.0107)	(0.0121)	(0.0066)	(0.0078)
Patronage administrator ×		0.0291**		0.0084
Water level deviations		(0.0143)		(0.0054)
Patronage inspector × Patronage		-0.0507***		0.0002
administrator × Water level deviations		(0.0174)		(0.0074)
Mean of Dependent Variable	0.2890	0.2890	0.0238	0.0238
Observations	94,659	94,659	94,659	94,659
R^2	0.1100	0.1105	0.0335	0.0338

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and water level deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (4)-(6) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Water level deviations" is standardized and refers to the deviations in the water level from its monthly average across years to capture unexpected changes in the water level. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments at the Paraguay river customs ports.

Table C88: Patronage hires, customs fraud detected and vessel draft deviations at customs ports each day

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Vessel draft deviations	0.0097	-0.0129	-0.0311*	-0.0337**
	(0.0357)	(0.0373)	(0.0182)	(0.0166)
Patronage inspector ×	-0.0003	0.0040	-0.0032	-0.0043
Patronage administrator	(0.0225)	(0.0219)	(0.0086)	(0.0089)
Patronage inspector ×	-0.0287***	-0.0097	-0.0023	-0.0030
Vessel draft deviations	(0.0107)	(0.0115)	(0.0057)	(0.0067)
Patronage administrator ×		0.0207		0.0085*
Vessel draft deviations		(0.0141)		(0.0048)
Patronage inspector × Patronage		-0.0431**		0.0049
administrator × Vessel draft deviations		(0.0188)		(0.0068)
Mean of Dependent Variable	0.2890	0.2890	0.0238	0.0238
Observations	94,659	94,659	94,659	94,659
R^2	0.1104	0.1107	0.0336	0.0341

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and draft deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (4)-(6) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Vessel draft deviations" is standardized and refers to the deviations in the vessel draft from its monthly average across years to capture unexpected changes in the vessel draft. "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments at the Paraguay river customs ports.

Table C89: Patronage hires, customs fraud detected and workload deviations at customs ports each day

	Fraud detected by inspector		Fraud detected by PCA, but not inspector	
	(1)	(2)	(3)	(4)
Workload deviations	0.0518	0.0215	-0.0065	-0.0056
	(0.0454)	(0.0392)	(0.0094)	(0.0083)
Patronage inspector ×	-0.0200	-0.0229	-0.0007	0.0012
Patronage administrator	(0.0233)	(0.0230)	(0.0093)	(0.0091)
Patronage inspector ×	-0.0243*	0.0051	-0.0033	-0.0037
Workload deviations	(0.0127)	(0.0159)	(0.0046)	(0.0065)
Patronage administrator ×		0.0448***		0.0034
Workload deviations		(0.0131)		(0.0023)
Patronage inspector × Patronage		-0.0514***		0.0044
administrator × Workload deviations		(0.0162)		(0.0062)
Mean of Dependent Variable	0.2890	0.2890	0.0238	0.0238
Observations	94,659	94,659	94,659	94,659
R^2	0.1117	0.1127	0.0329	0.0330

Notes: All regressions include inspector and administrator fixed effects. All regressions control for interactions of the patronage administrator indicator and workload deviations with: (1) inspector education and (2) inspector age. The patronage inspector and administrator indicator variables are absorbed by the inspector and administrator fixed effects. Standard errors are clustered at the inspector-administrator pair level. ***, **, and * indicate significance at 1%, 5%, and 10% levels, respectively. The outcome variable on columns (4)-(6) is an indicator variable equal to 1 when a shipment is linked to fraud detected by the post-clearance audit (PCA) team, but not by the valuation inspector. Patronage hires are those inspectors and administrators that joined the customs authority: (1) within a year of a new executive administration's start during democracy, or (2) during Paraguay's dictatorship period. "Workload deviations" is standardized and refers to the total number of red channel shipments that arrived at the relevant customs port on the relevant day for the shipment, divided by the number of valuation inspectors present at the port that day (to take into account the differing capacities of customs ports). "Observations" refers to the number of non-singleton shipment observations. The sample covers red channel shipments at the Paraguay river customs ports.

D Counterfactual tax revenue in the absence of patronage

Following Chalendard et al. (2023), I define hypothetical tax revenue losses as the difference between the log of hypothetical tax yield (based on internal reference prices) and the log of actual tax yield: $loss = \log(T^H) - \log(T)$. Similarly, hypothetical losses in the absence of patronage hires are defined as: $loss^{NP} = \log(T^H) - \log(T^{NP})$, where T^{NP} is the counterfactual tax yield absent patronage, my estimate of interest. Rearranging these expressions yields:

$$\log(T^{NP}) = \log(T) - (loss^{NP} - loss) \tag{D.1}$$

I then use the β estimates obtained in Table 7 to quantify the predicted tax revenue loss from patronage hires as:

$$\widehat{loss} = \widehat{\beta}_1 PatronInsp + \widehat{\beta}_2 PatronAdm + \widehat{\beta}_3 PatronInsp \times PatronAdm + \widehat{\beta_X} X, \tag{D.2}$$

where the vector X includes all independent variables other than the patronage hire indicator variables. I use these estimates to predict the counterfactual tax revenue losses that would have taken place without the patronage hires as:

$$\widehat{loss^{NP}} = \widehat{\beta_X} X \tag{D.3}$$

Subtracting equation (D.2) from equation (D.3) gives:

$$\widehat{loss^{NP}} - \widehat{loss} = -\widehat{\beta}_1 PatronInsp - \widehat{\beta}_2 PatronAdm - \widehat{\beta}_3 PatronInsp \times PatronAdm$$

With this I calculate the counterfactual tax yield in the absence of patronage hires by inserting into equation (D.1) and taking the exponential:

$$\widehat{T^{NP}} = T \times \exp(\widehat{\beta}_1 PatronInsp + \widehat{\beta}_2 PatronAdm + \widehat{\beta}_3 PatronInsp \times PatronAdm) \tag{D.4}$$